Fpt University

TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REPORT

Report 4 – Software Design Description

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IV. Software Design Description

1. System Design

1.1 System Architecture

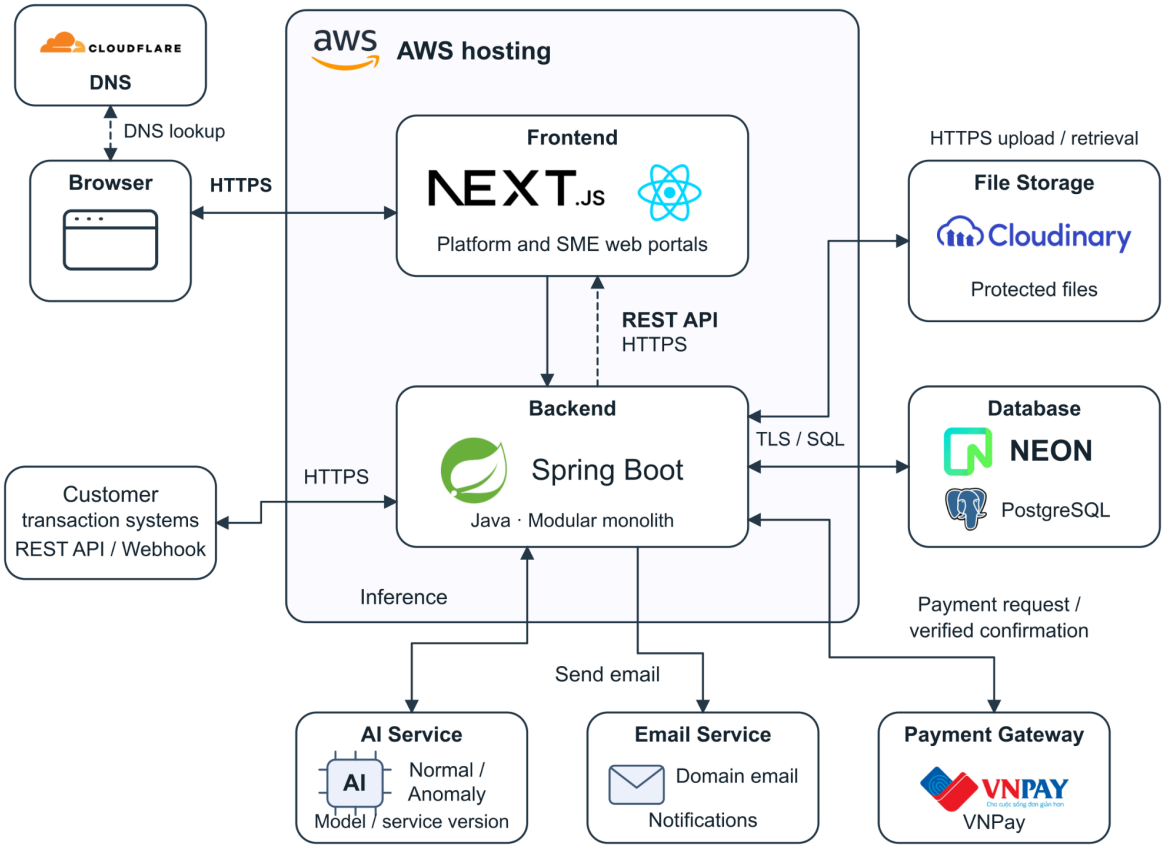


Figure 4.1. SME-FraudGuard System Architecture.

1.2 Package Diagram

The backend is organized as one Spring Boot monolith under com.fraudguard. Eleven business modules are accompanied by common infrastructure and external integration adapters. The packages below reflect the current codebase scaffold; business implementations are added in later development.

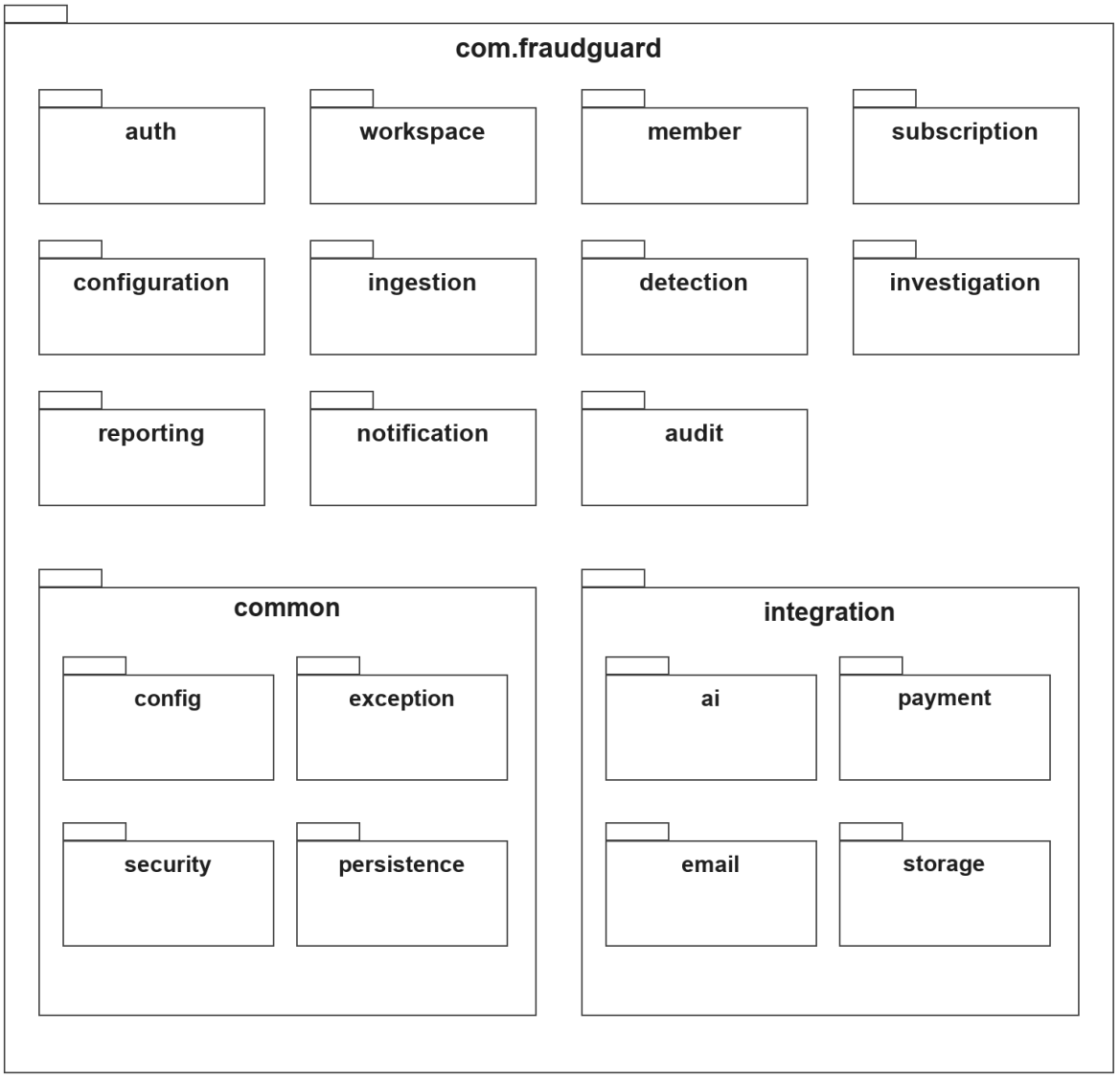


Figure 4.1.2.1. Backend Package Overview.

Each business module follows the layered structure below. Dashed arrows show permitted design dependencies, not implemented runtime calls. Other modules access service interfaces and DTOs; repositories, entities, mappers and service implementations remain private to their owning module. AI training remains outside this backend.

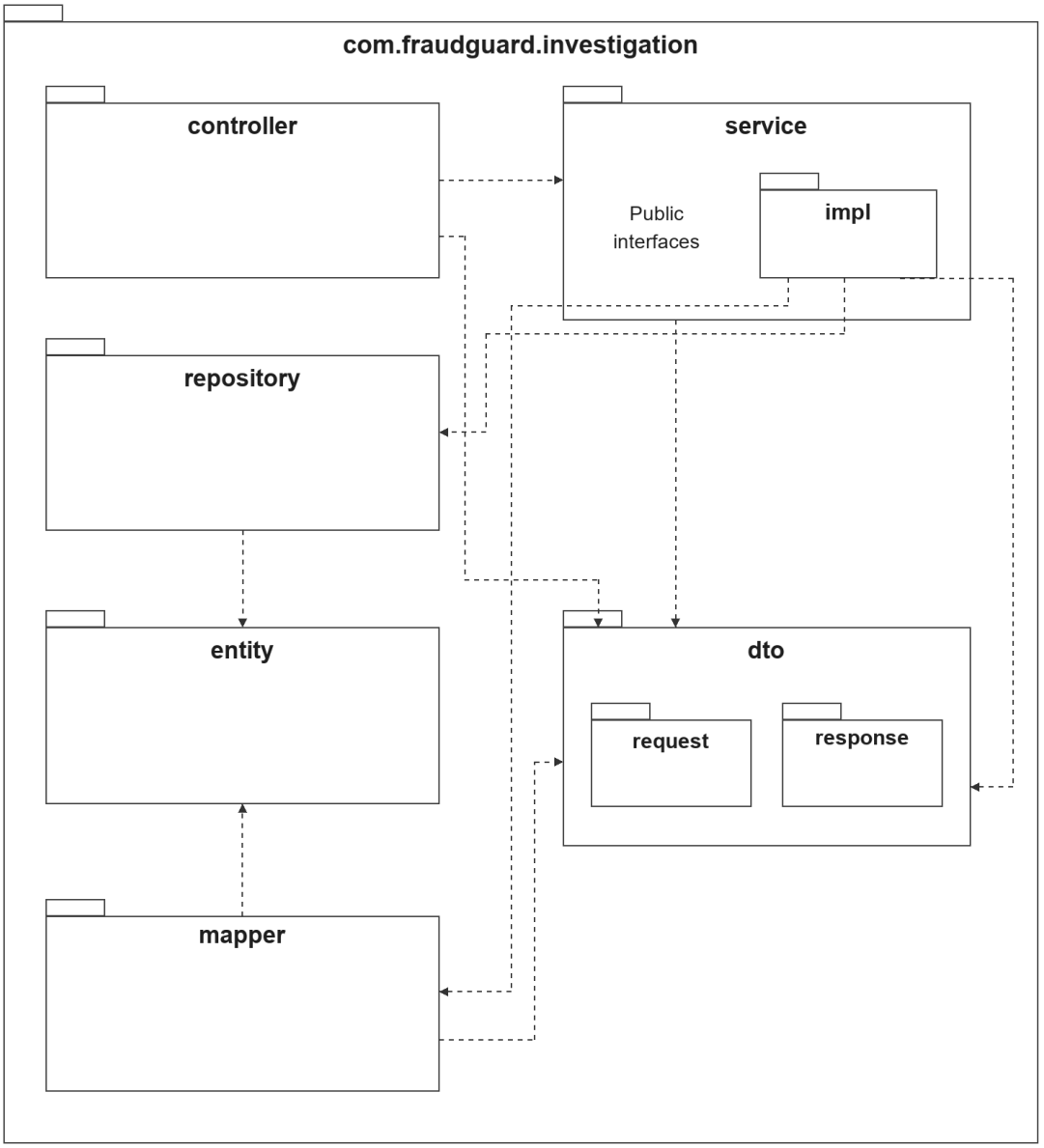


Figure 4.1.2.2. Layered Package Structure within a Business Module.

2. Database Design

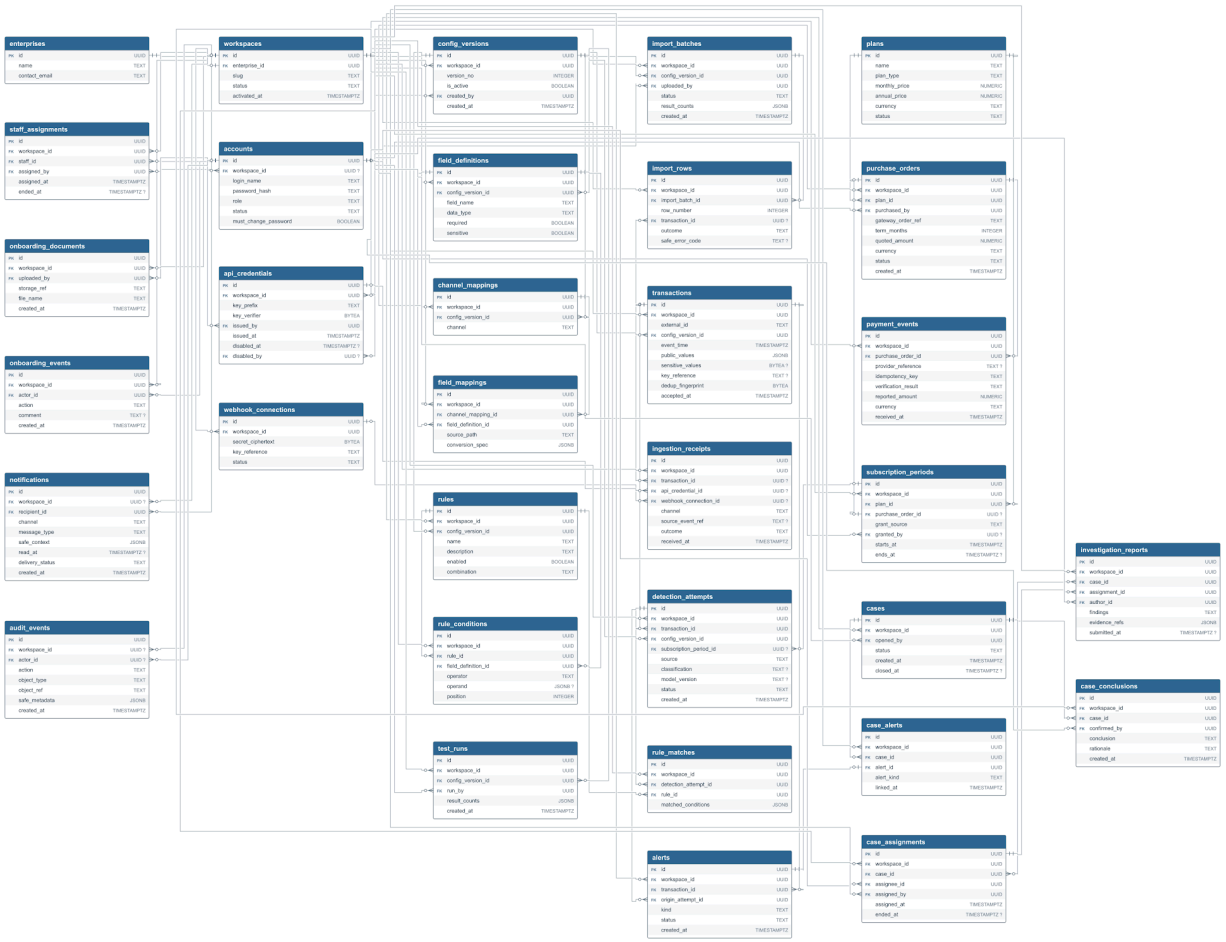


Figure 4.2. SME-FraudGuard Database Design.

3. Detailed Design

This section presents the target detailed design derived from Report 3 functional requirements, Sections 3.2-3.15. Class diagrams identify controllers/listeners, service interfaces and implementations, repositories, DTOs and domain types. Sequence diagrams use the same class names and operations, with explicit calls, returns, activation bars and repository/database interactions. These proposed design signatures are not claims of implemented Java classes or approved HTTP routes; cross-module calls use public APIs/ports within the modular monolith.

All operations enforce current identity, role, workspace and object/assignment scope on the backend. Integration authentication establishes workspace scope before mapping or acceptance. Sensitive values remain protected in ordinary results and diagnostics; required writes retain history atomically. Activity diagrams describe workflows and decisions. State machines cover the SRS-defined lifecycles of sessions, purchases, Subscription periods, onboarding, configuration candidates, API keys, import batches, initial detection, recovery, cases, report generation and email delivery. Each figure identifies its object/operation scope and initial/final boundary; completion does not delete retained records. Stage labels outside the explicit case lifecycle are logical design terms, not implemented enums. Read-only views, audit attribution and unspecified review flags do not introduce invented status models. Exact external wire formats, technical limits and retry schedules remain subject to confirmed interface specifications.

3.1 Authentication and Account Access

3.1.1 Class Diagram

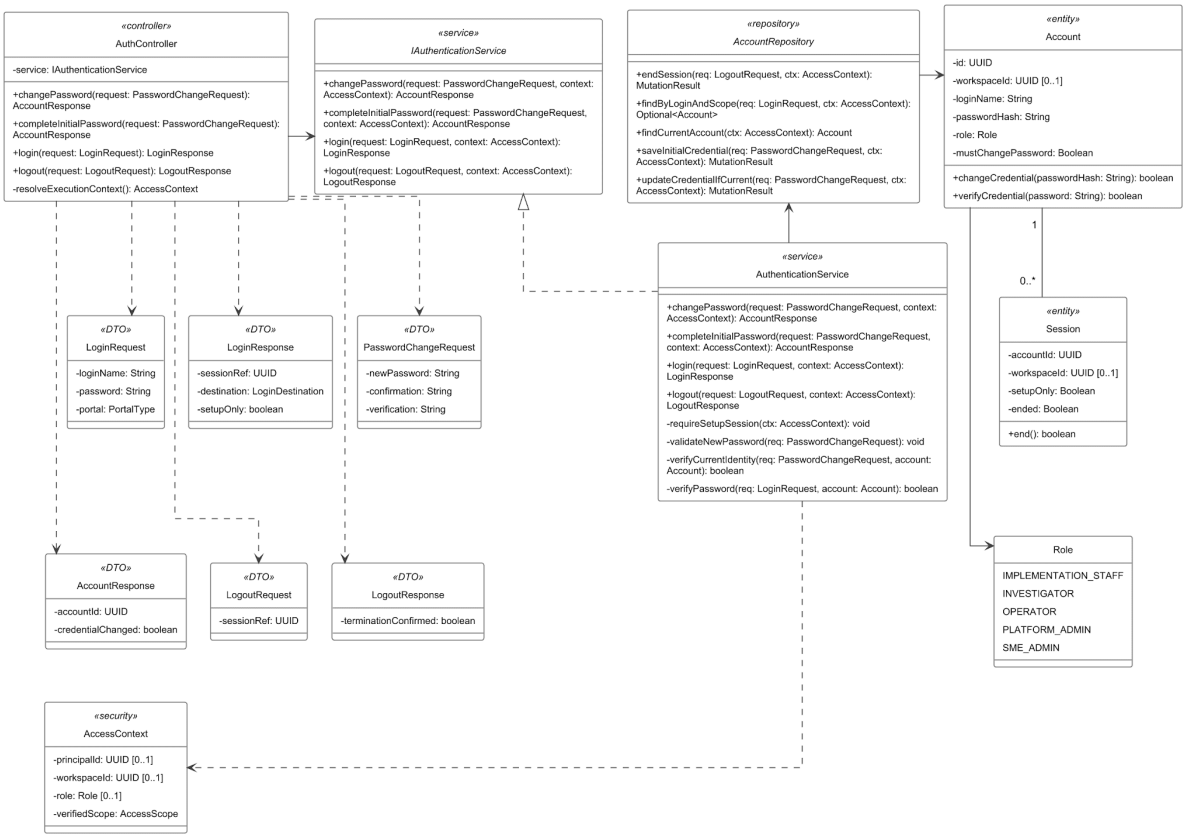


Figure 4.3.1.1.1. Authentication and Account Access Class Diagram.

3.1.2 Sequence Diagrams

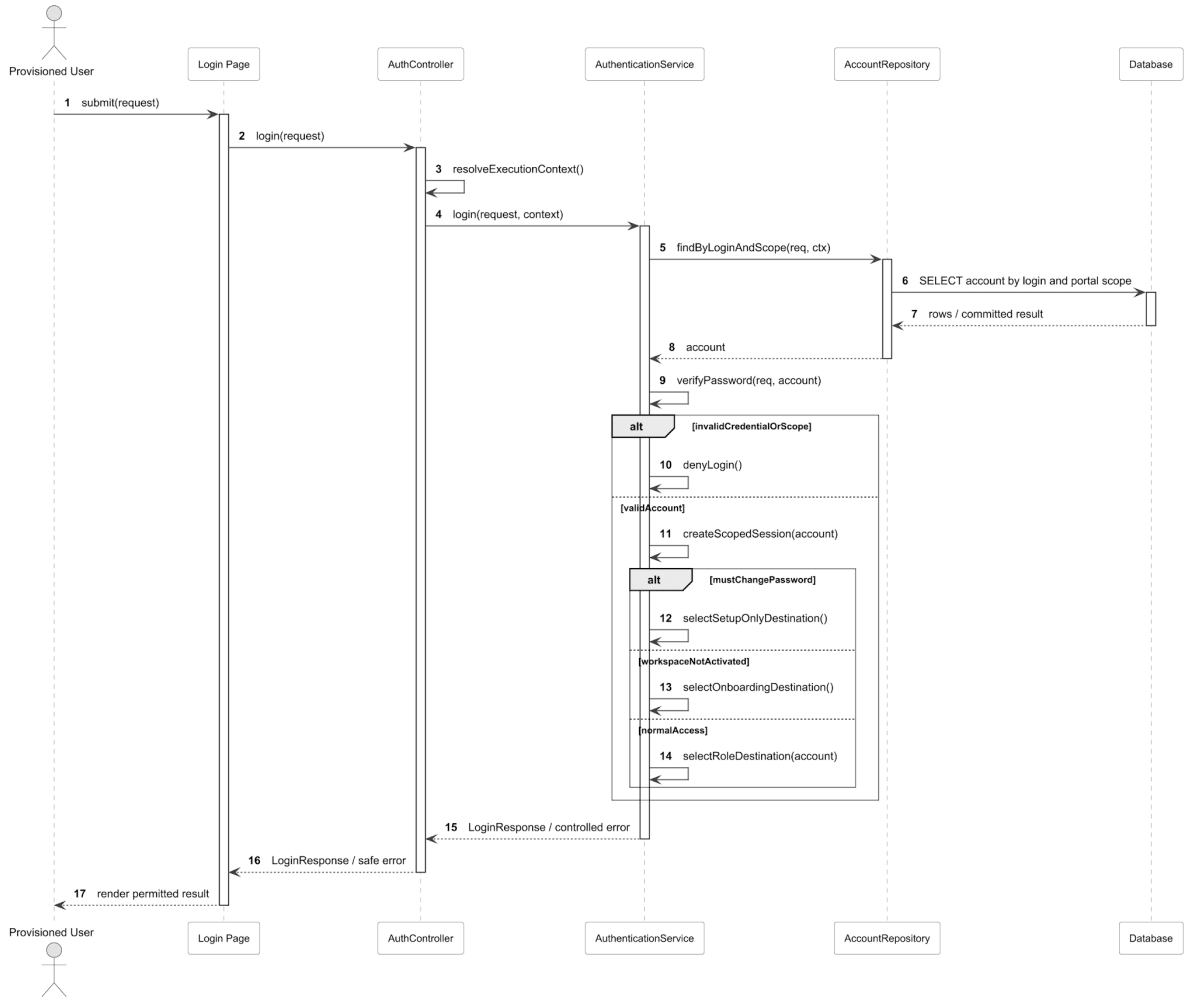


Figure 4.3.1.2.1. Log In and Select the Permitted Destination Sequence Diagram.

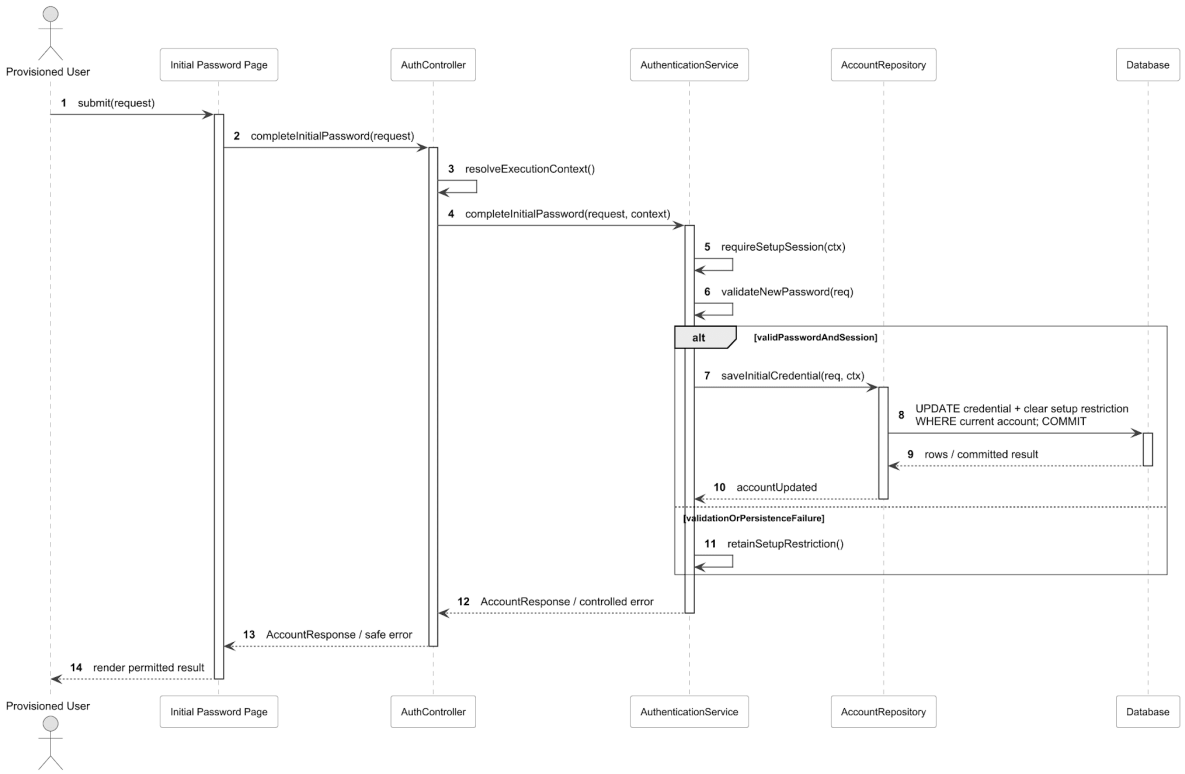


Figure 4.3.1.2.2. Complete Mandatory Initial Password Change Sequence Diagram.

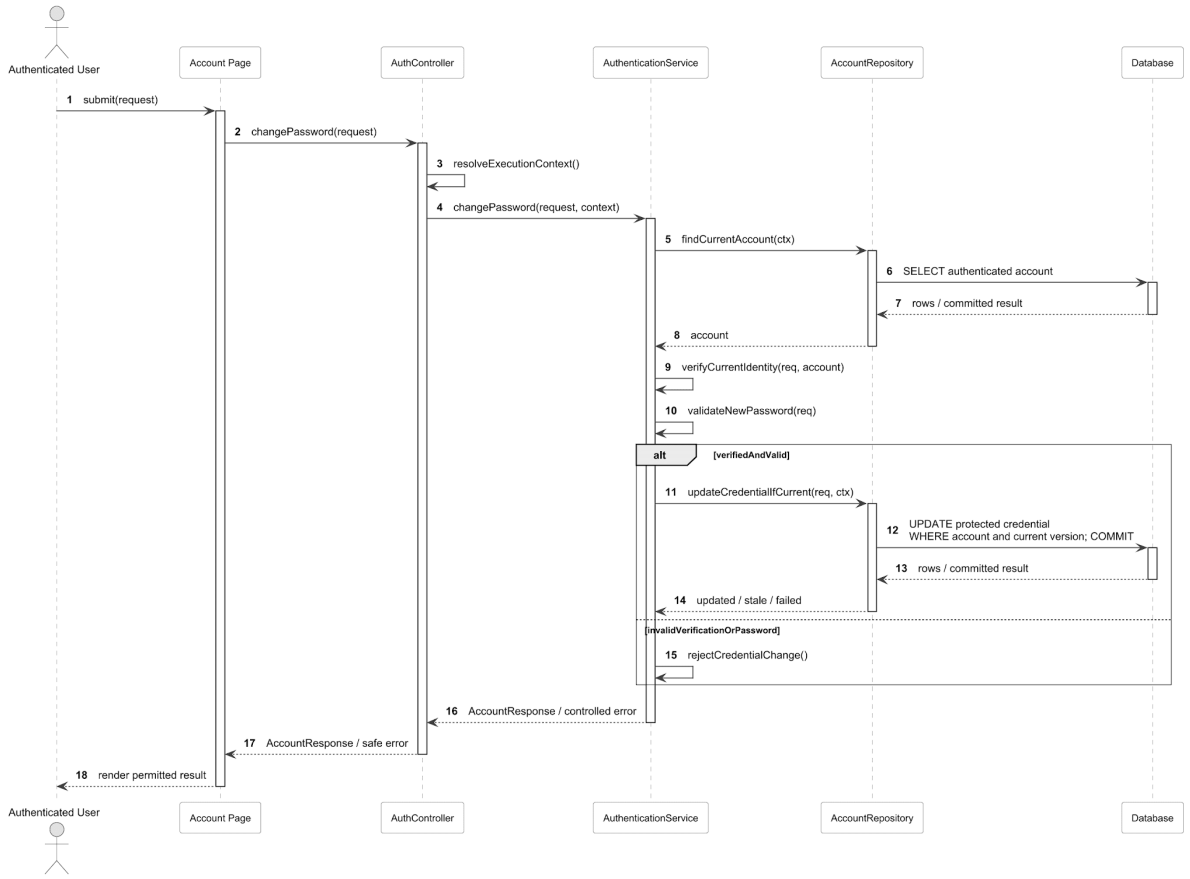


Figure 4.3.1.2.3. Change Current Password Sequence Diagram.

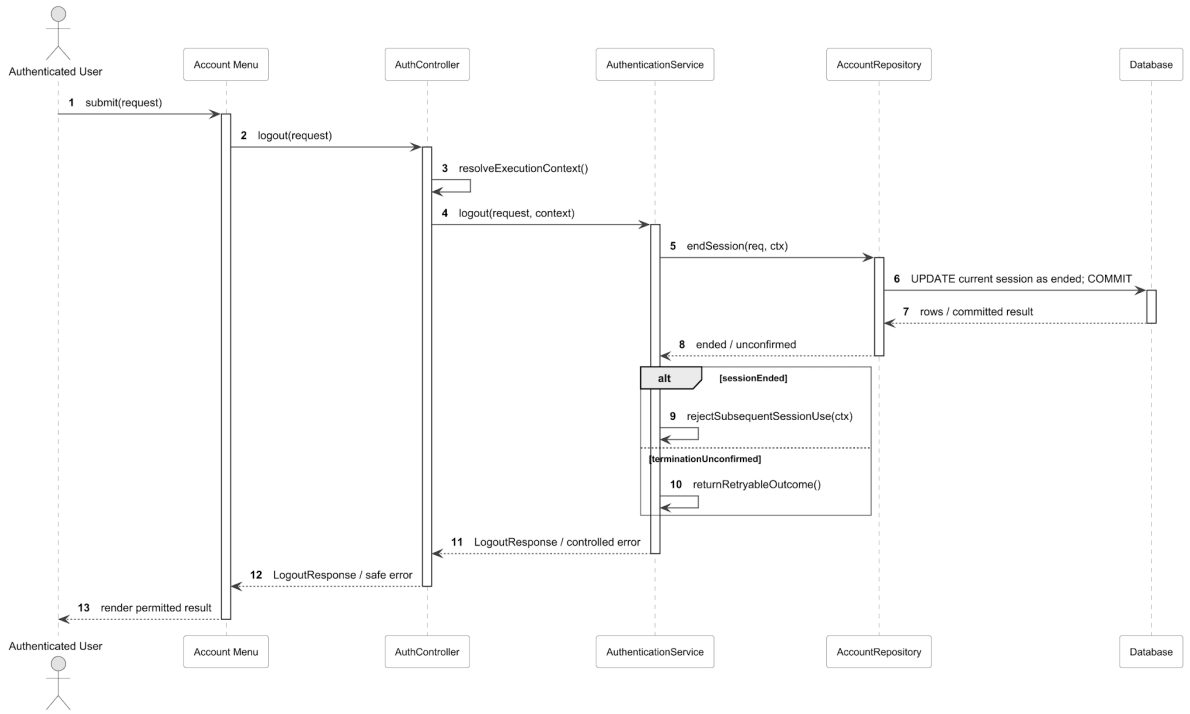


Figure 4.3.1.2.4. Log Out and Reject the Ended Session Sequence Diagram.

3.1.3 Activity Diagrams

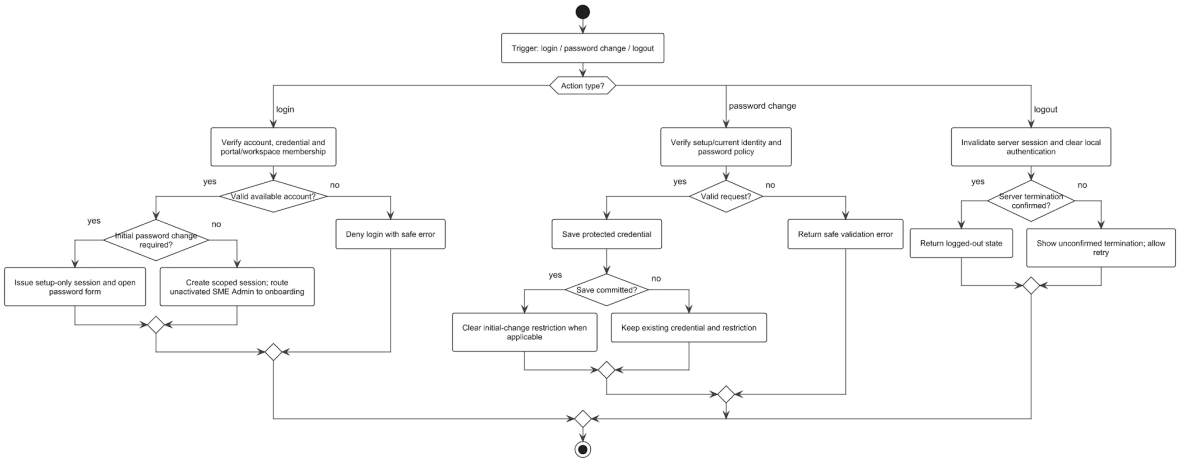


Figure 4.3.1.3.1. Authenticated Session Lifecycle Activity Diagram.

3.1.4 State Machine Diagrams

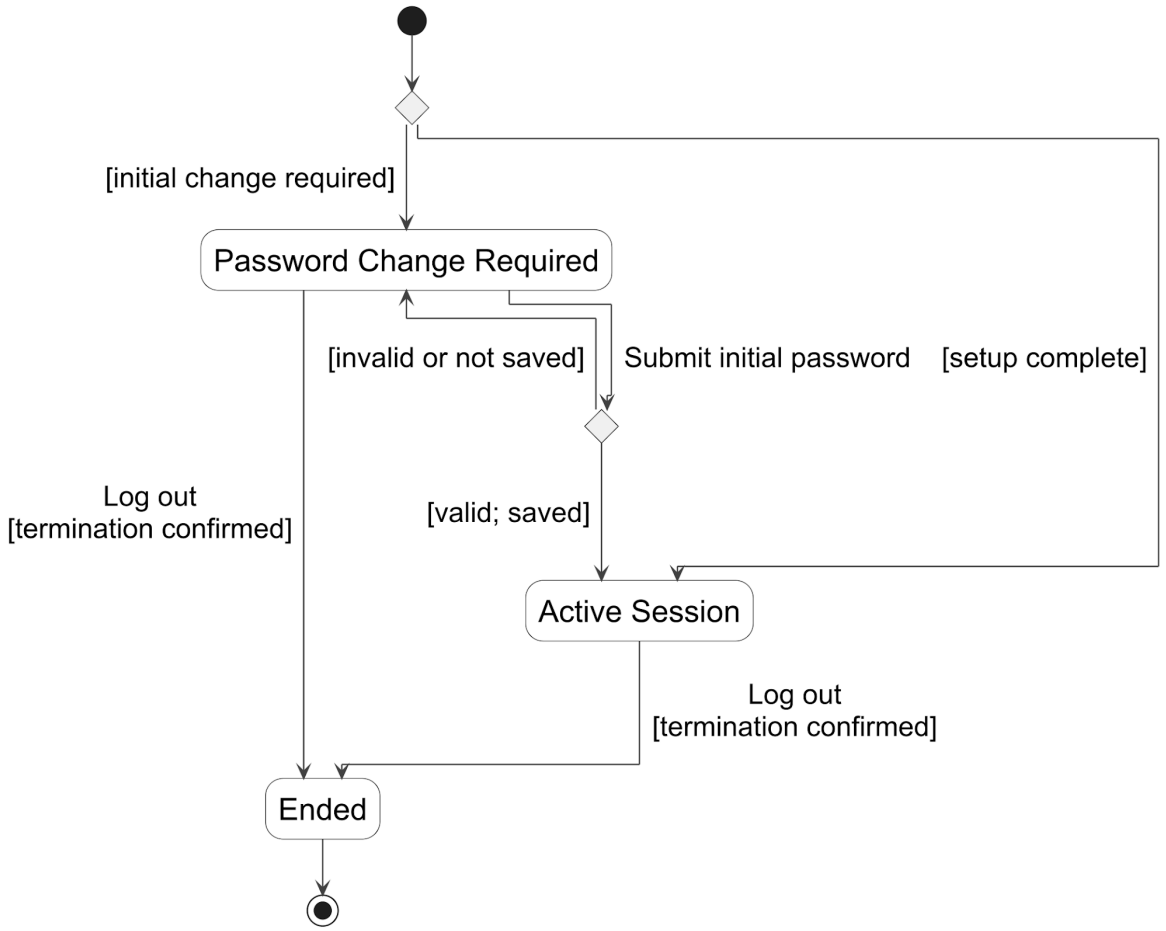


Figure 4.3.1.4.1. Authenticated Session State Machine Diagram.

3.2 Customer and Workspace Management

3.2.1 Class Diagram

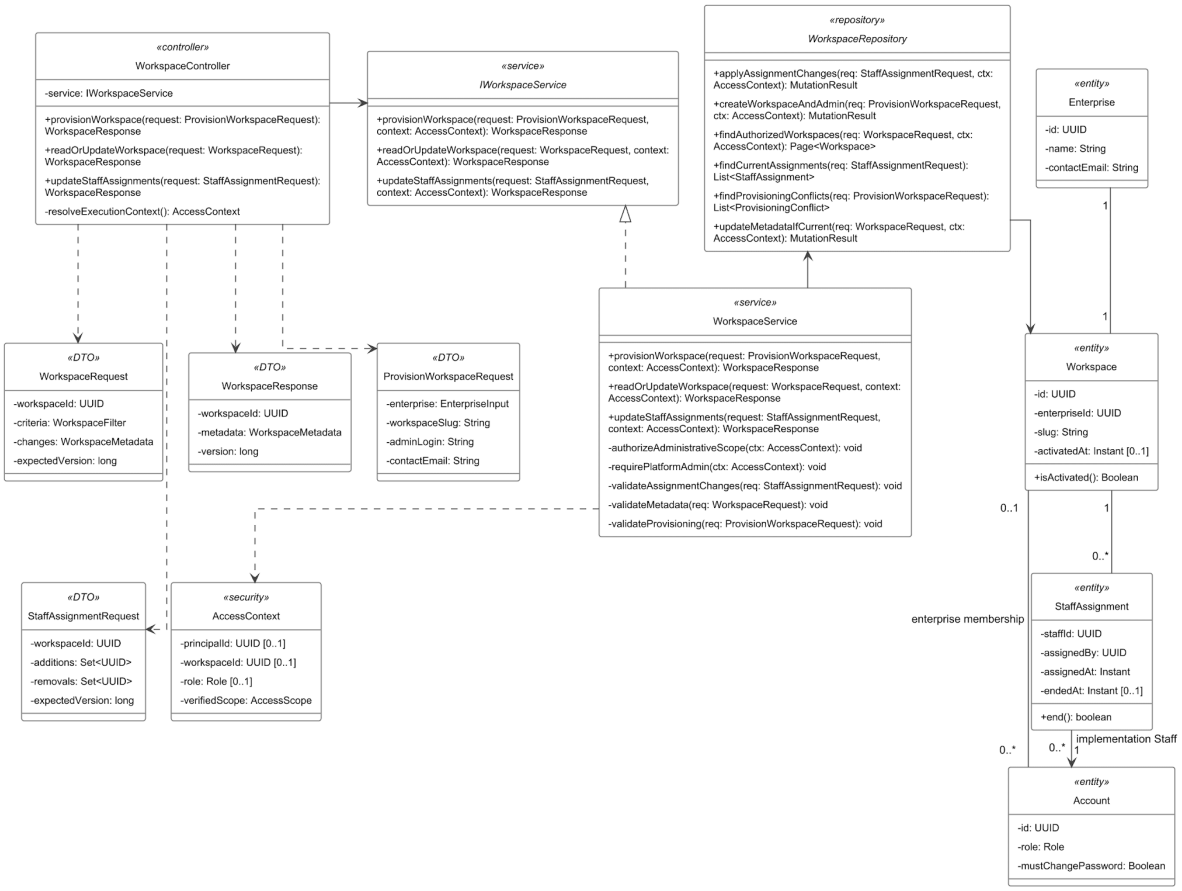


Figure 4.3.2.1.1. Customer and Workspace Management Class Diagram.

3.2.2 Sequence Diagrams

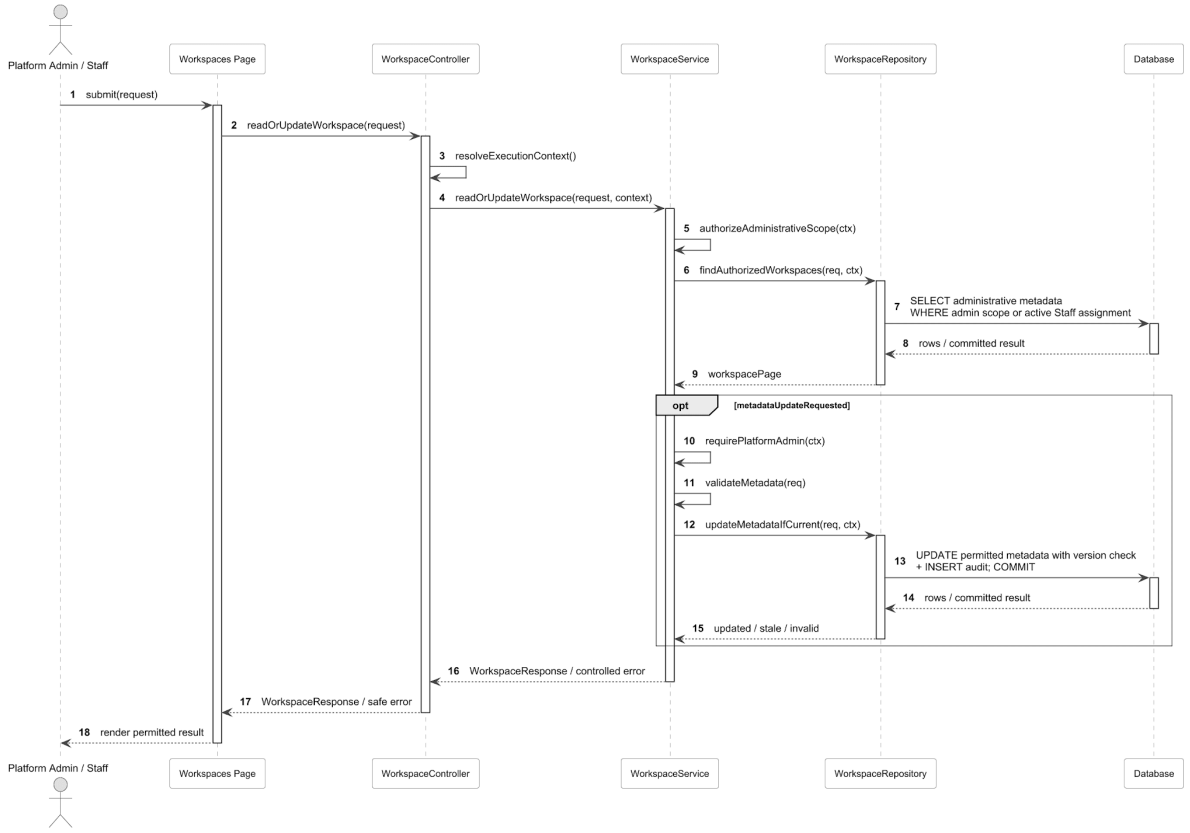


Figure 4.3.2.2.1. List, Inspect and Update Administrative Workspace Information Sequence Diagram.

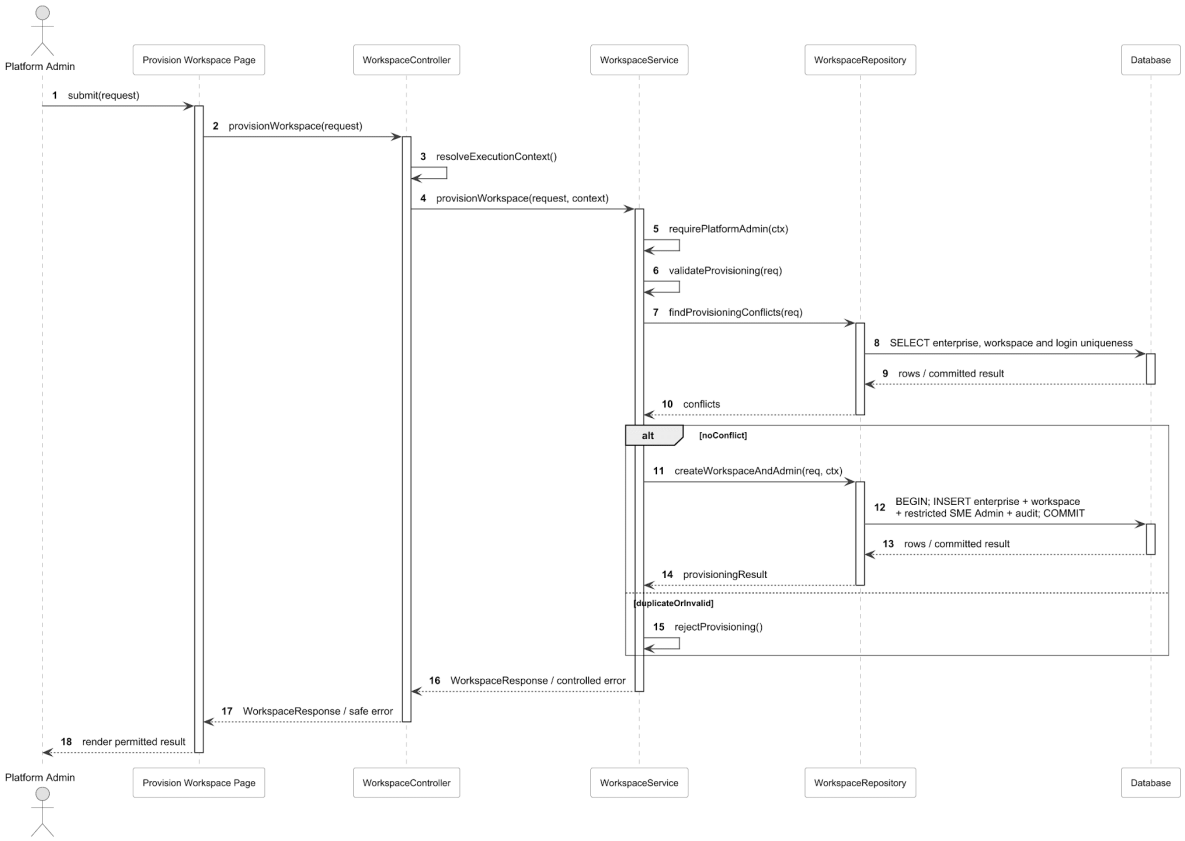


Figure 4.3.2.2.2. Provision Enterprise, Workspace and Initial SME Admin Sequence Diagram.

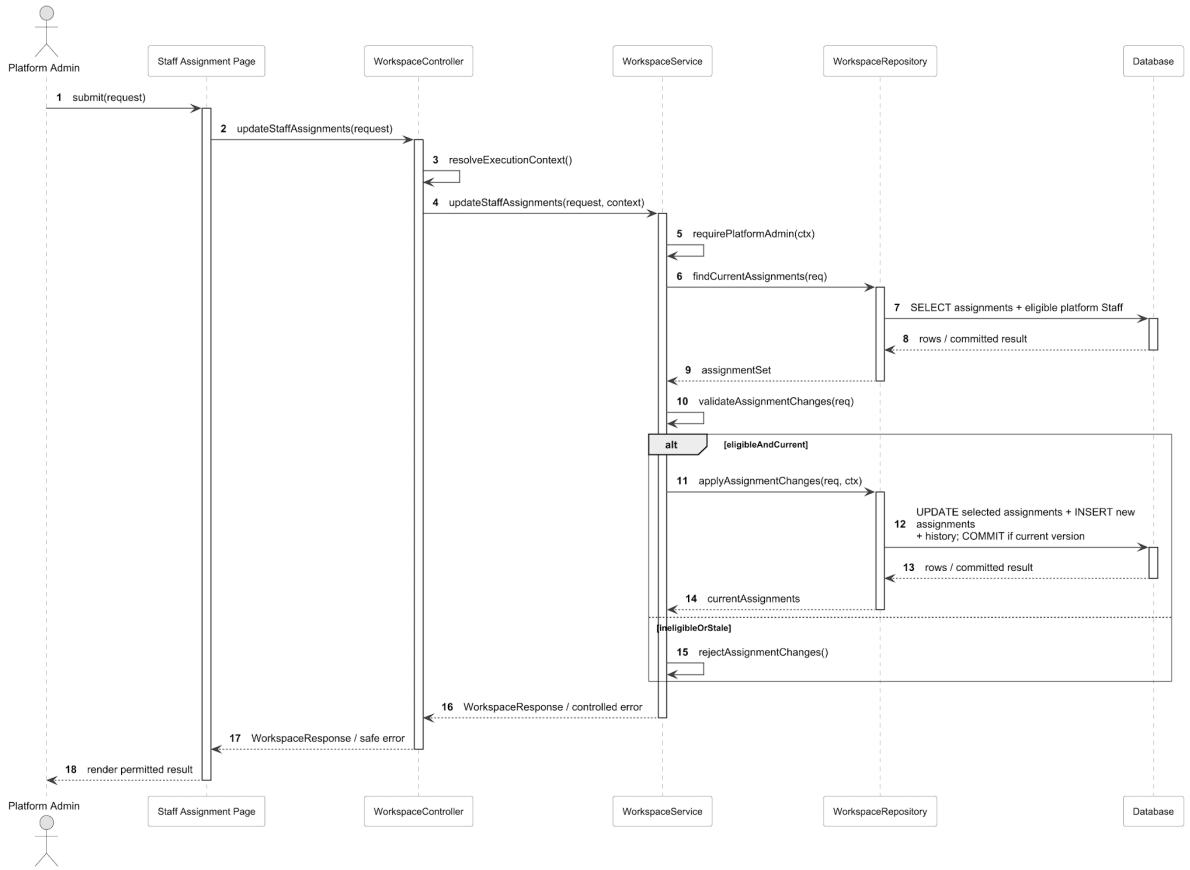


Figure 4.3.2.2.3. Add or Remove Concurrent Implementation Staff Assignments Sequence Diagram.

3.3 Plan and Subscription Management

3.3.1 Class Diagram

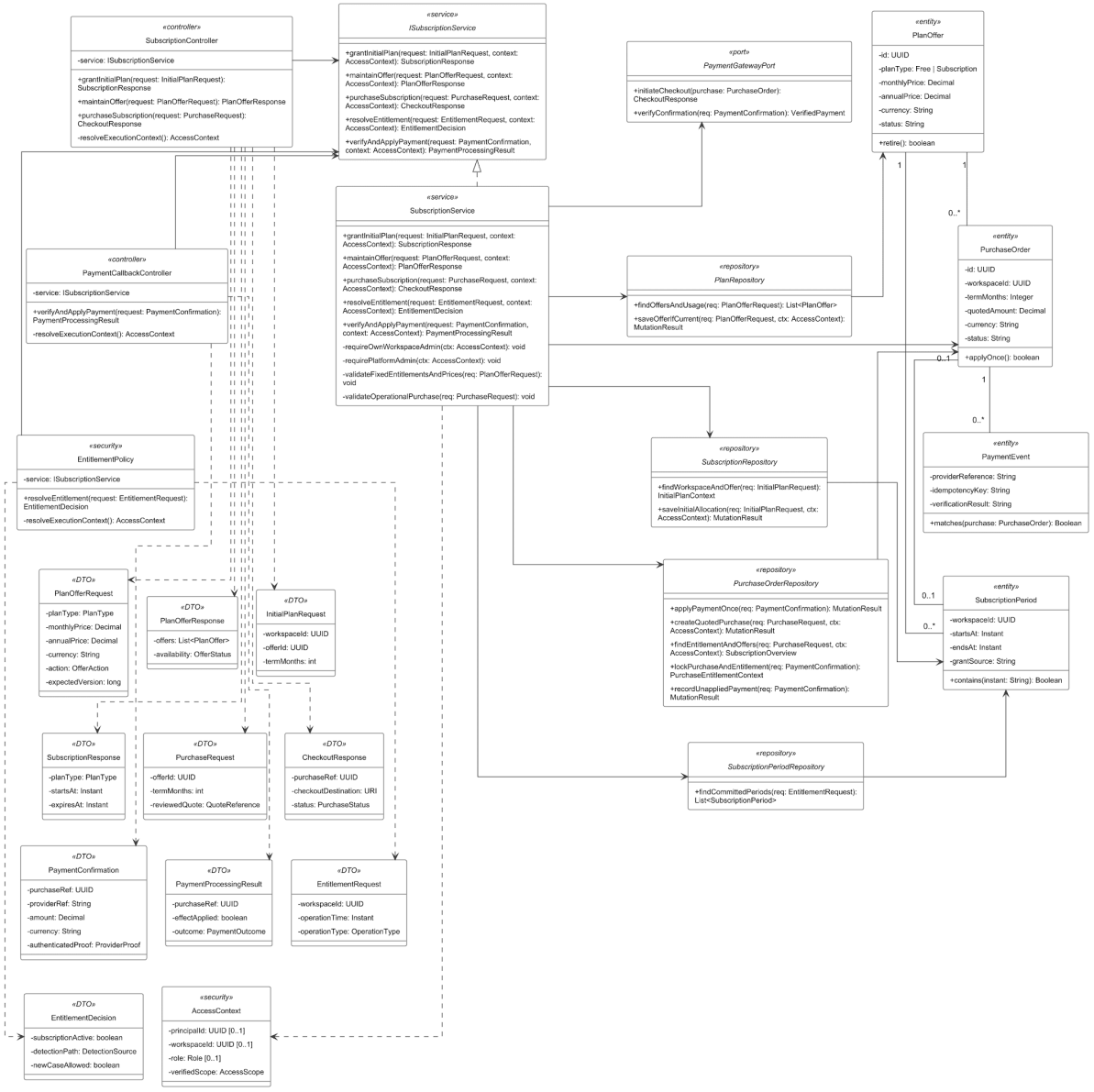


Figure 4.3.3.1.1. Plan and Subscription Management Class Diagram.

3.3.2 Sequence Diagrams

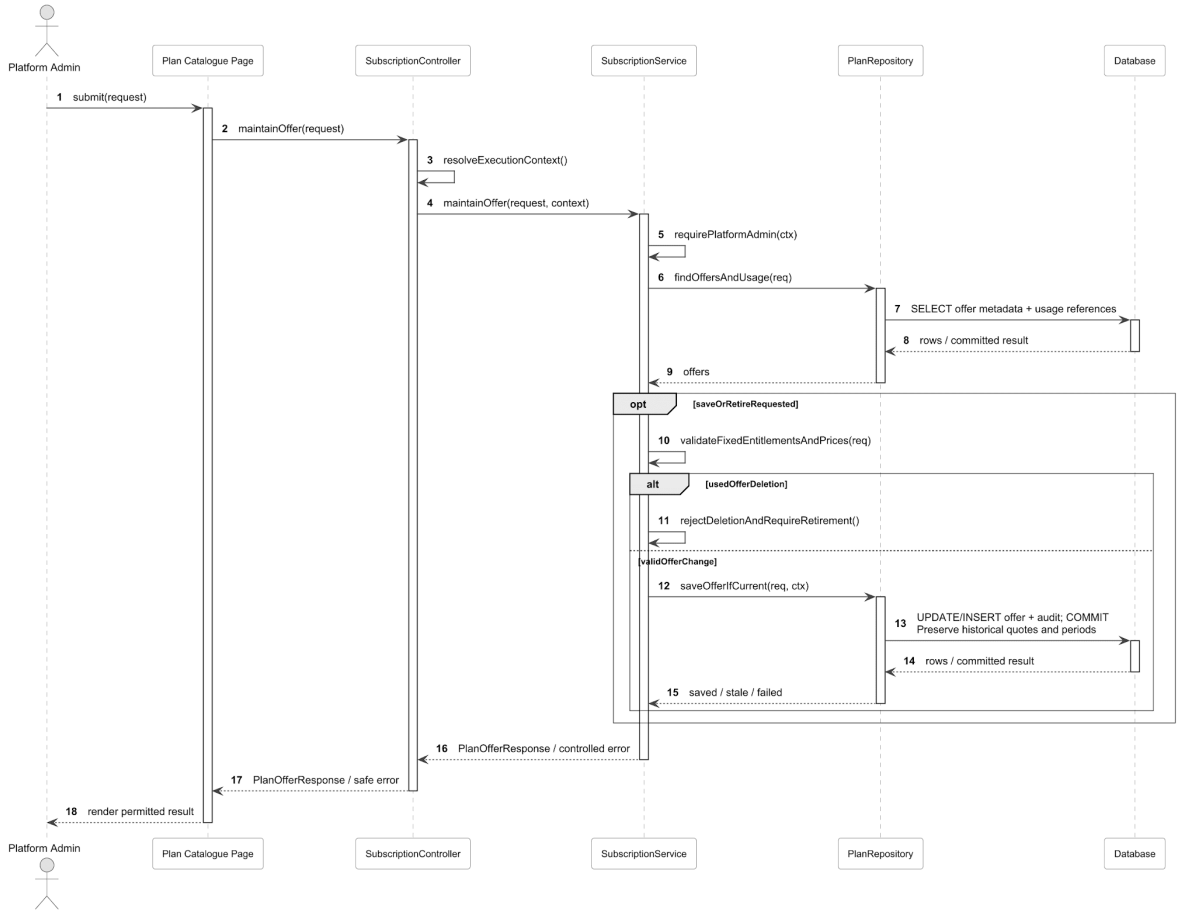


Figure 4.3.3.2.1. Review and Maintain Plan Offers Sequence Diagram.

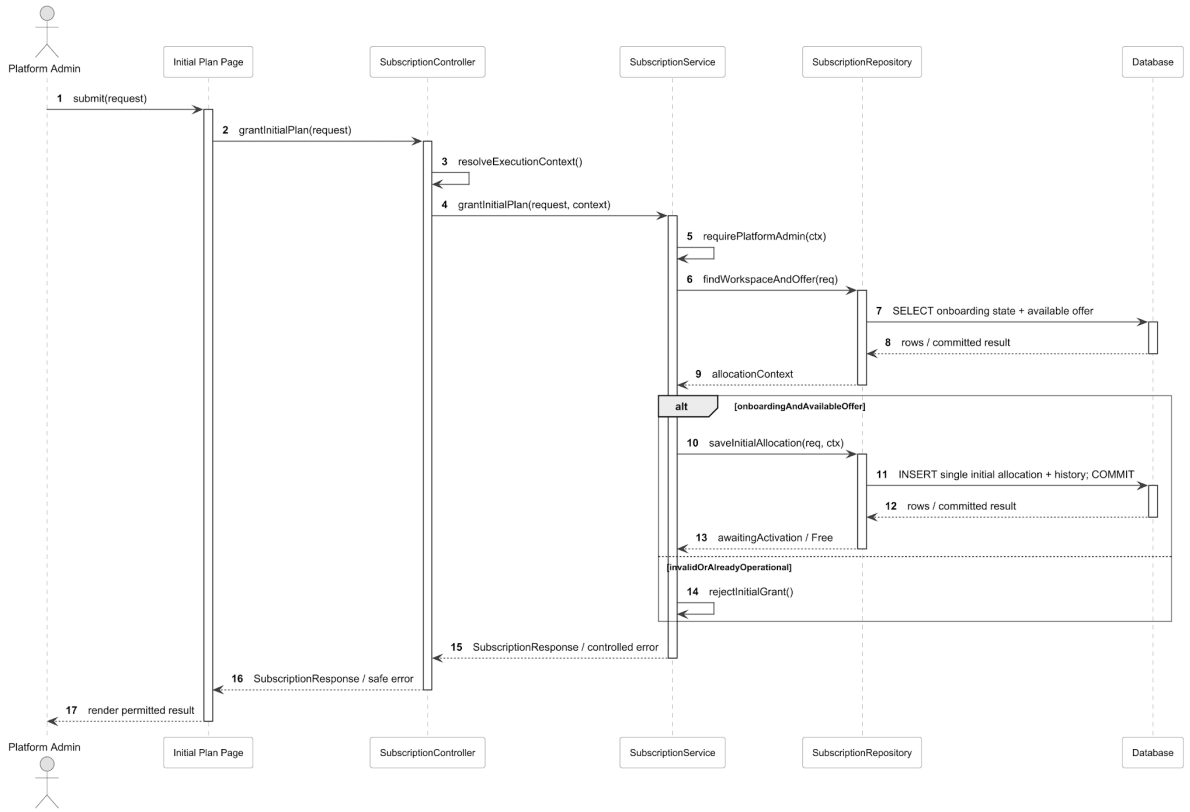


Figure 4.3.3.2.2. Grant the Initial Workspace Plan Sequence Diagram.

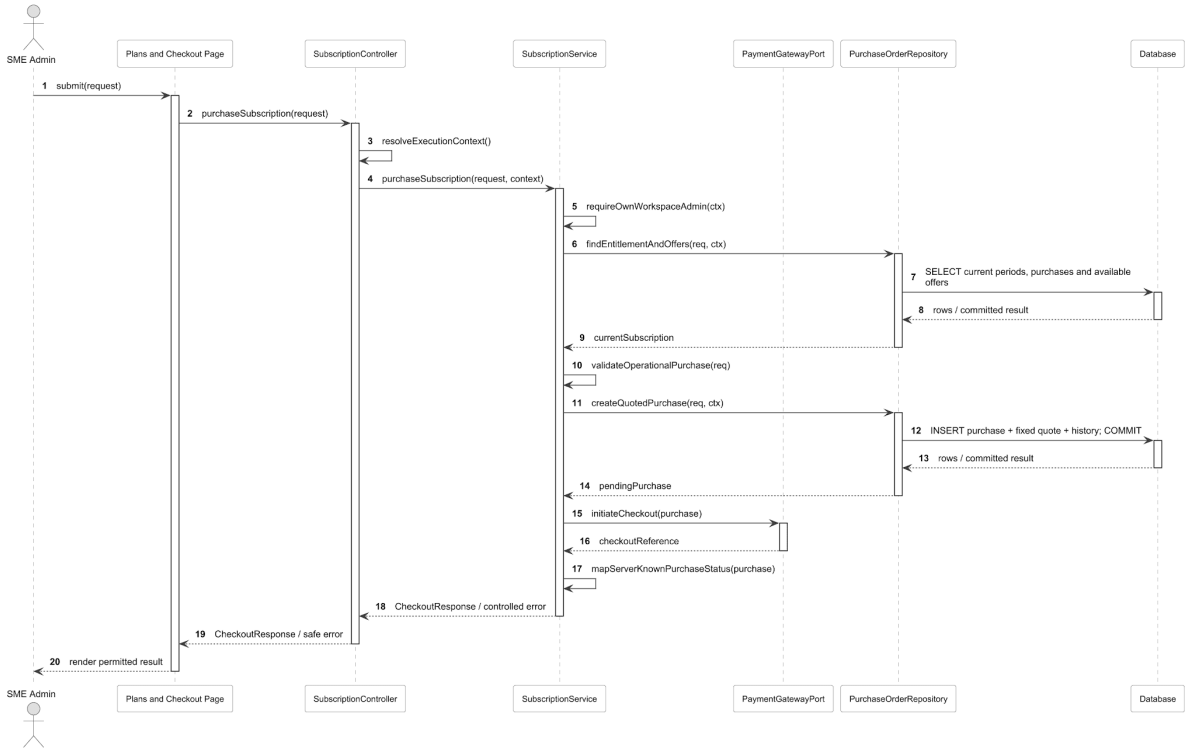


Figure 4.3.3.2.3. View Subscription and Initiate Purchase or Renewal Sequence Diagram.

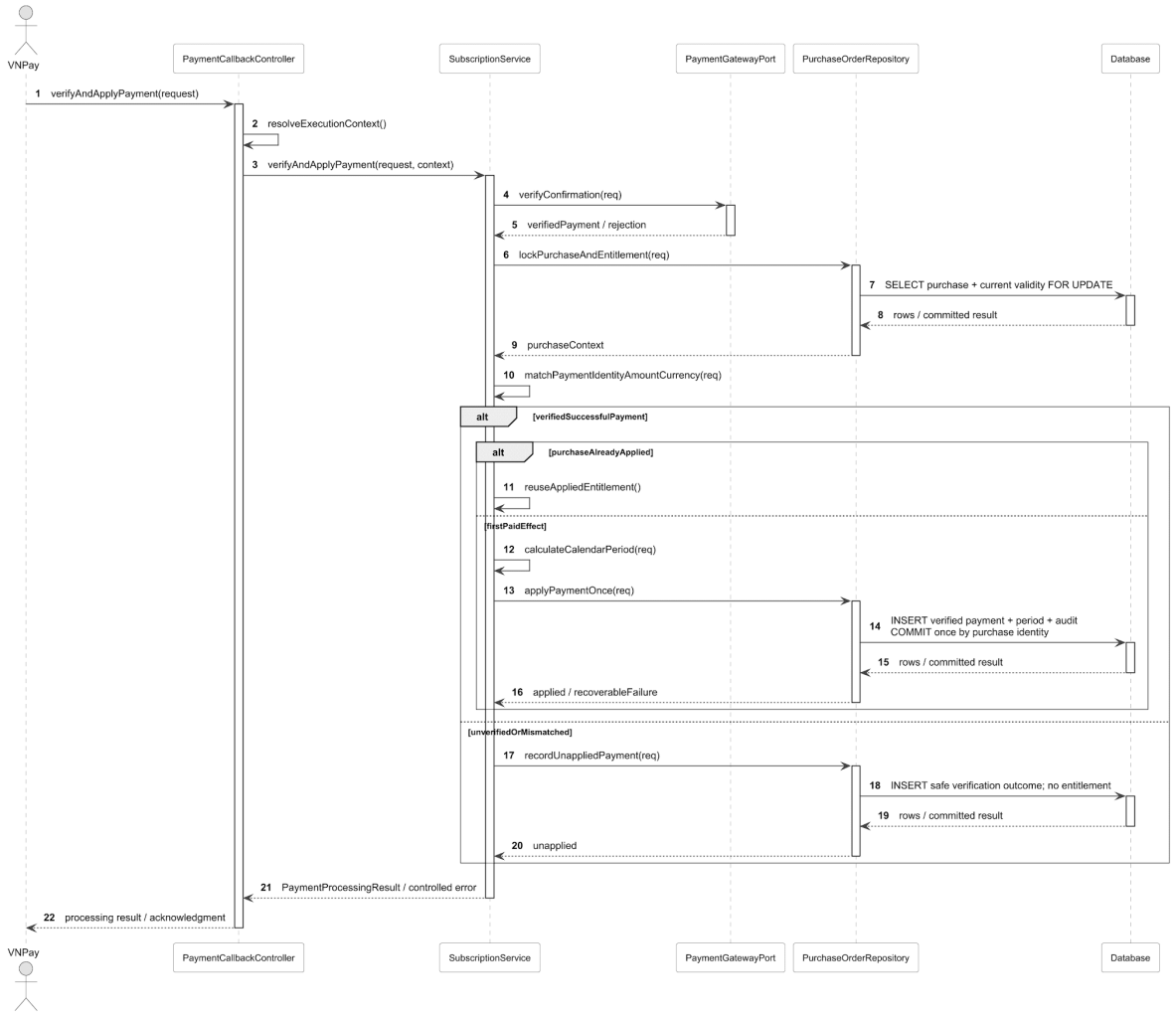


Figure 4.3.3.2.4. Verify Payment and Apply One Calendar Entitlement Effect Sequence Diagram.

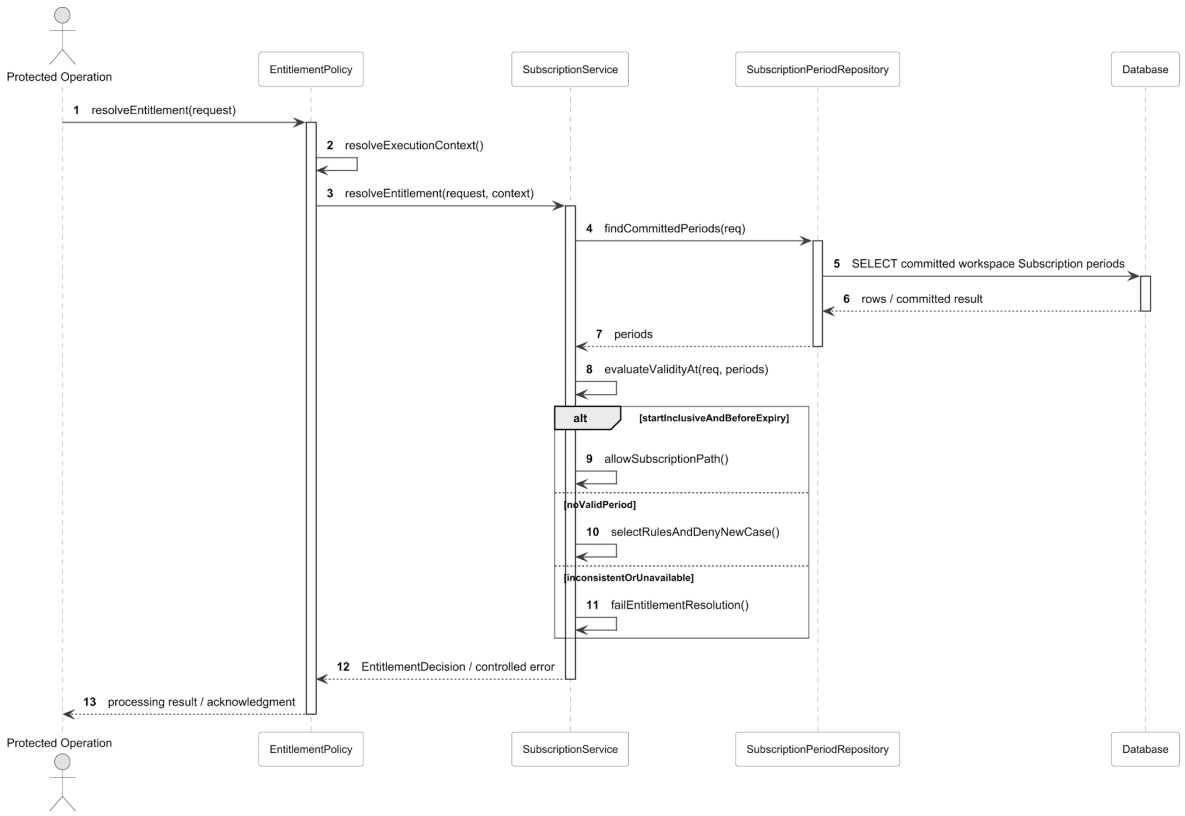


Figure 4.3.3.2.5. Enforce Expiry at New Operations and Preserve Existing Work Sequence Diagram.

3.3.3 Activity Diagrams

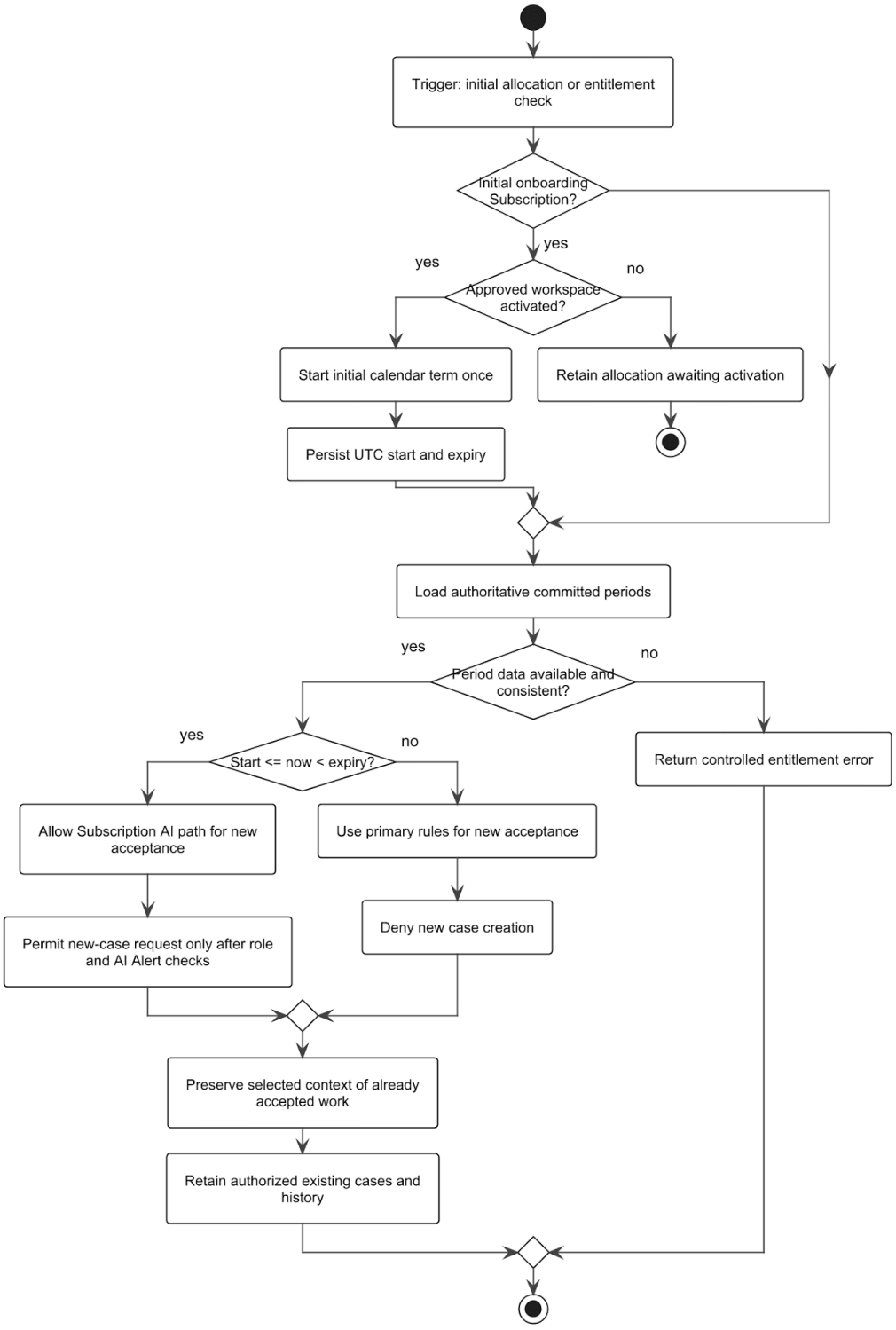


Figure 4.3.3.3.1. Subscription Entitlement Lifecycle Activity Diagram.

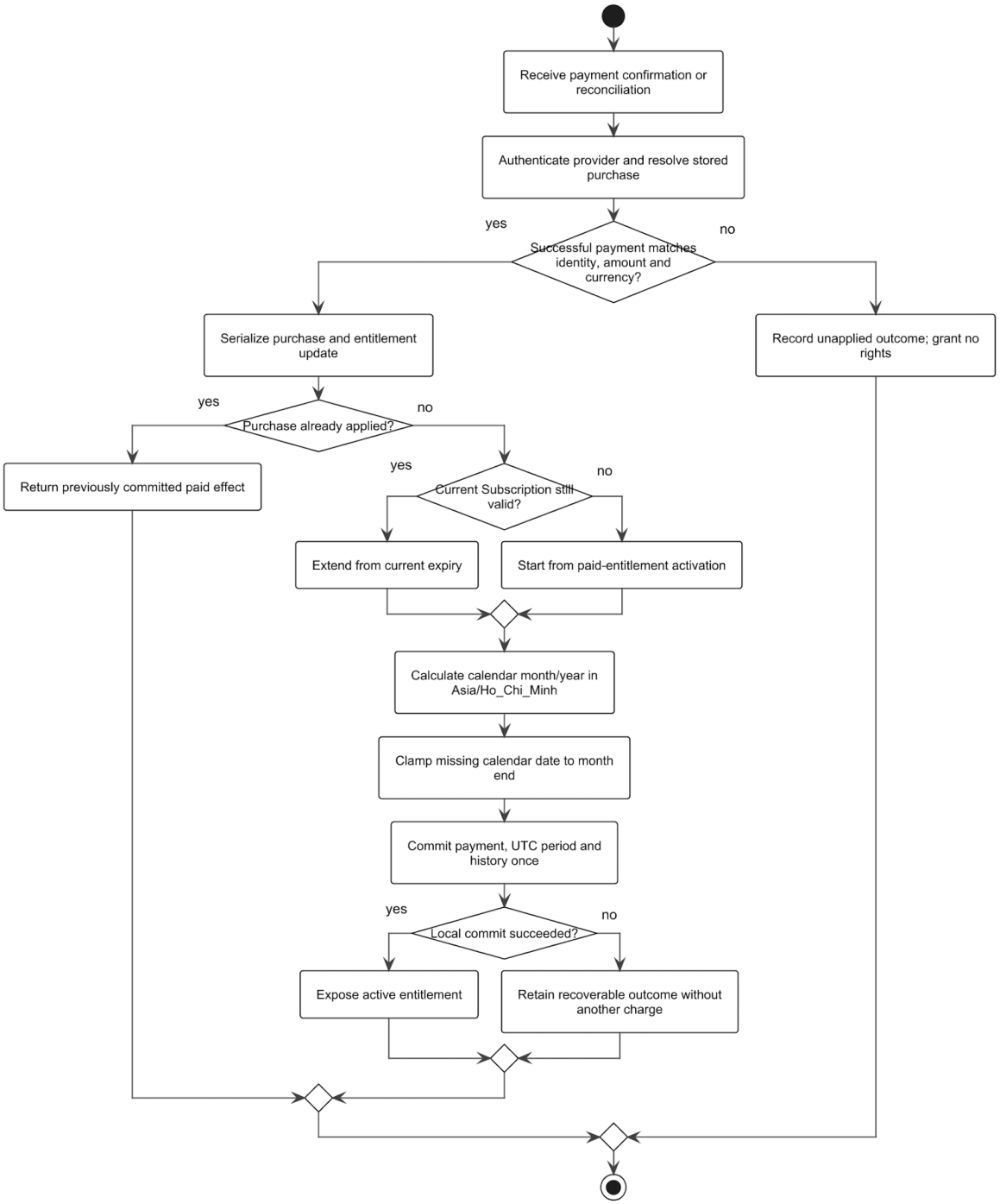


Figure 4.3.3.3.2. Purchase Verification and Local Entitlement Application Activity Diagram.

3.3.4 State Machine Diagrams

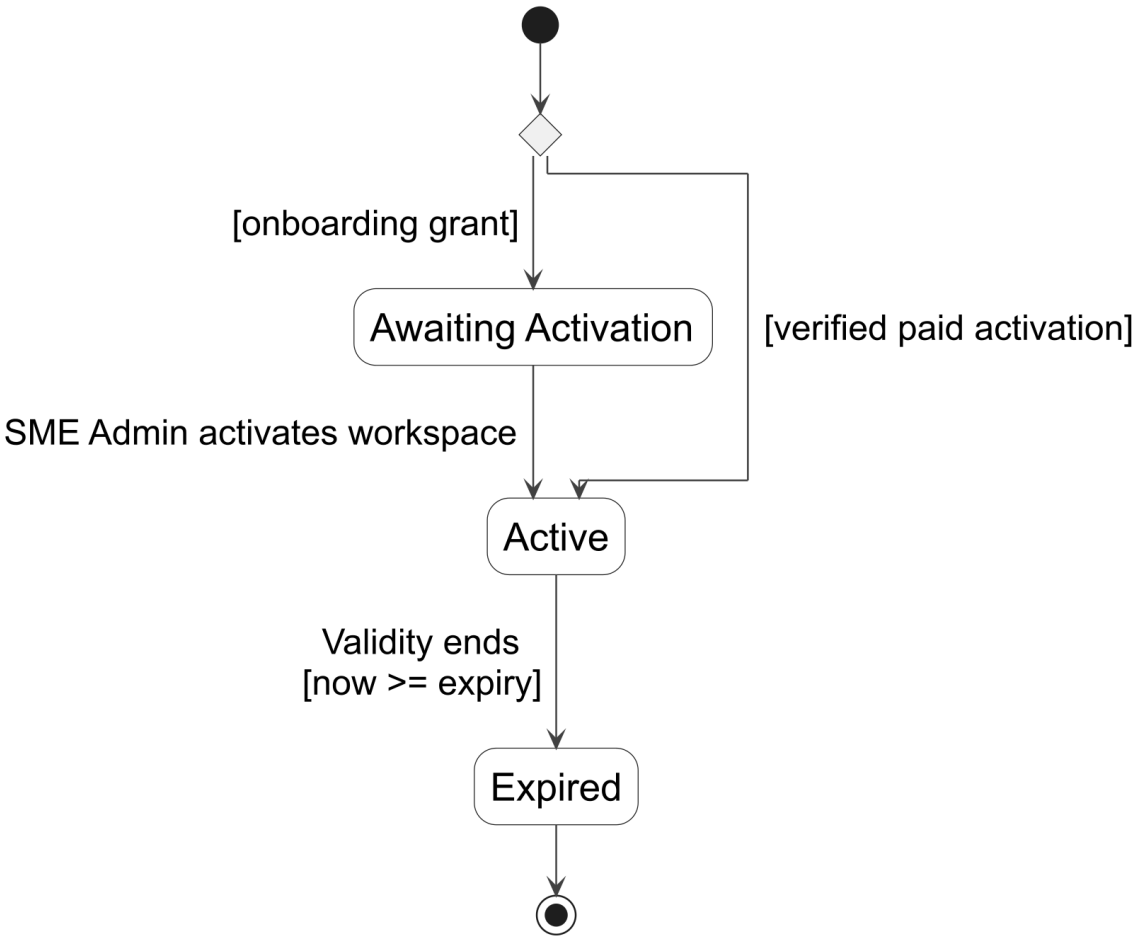


Figure 4.3.3.4.1. Subscription Validity Period State Machine Diagram.

Scope: one continuous Subscription validity period, not the workspace's lifetime or a payment attempt. An initial onboarding grant waits for approved workspace activation; a paid period is created only after verified payment and local entitlement application succeed once. Free has no Subscription period. Active means start <= authoritative time < expiry. A verified renewal before expiry extends the same active period; after expiry it starts a new period, not a transition out of Expired. Calculate calendar months/years in Asia/Ho_Chi_Minh and store instants in UTC. Duplicate payments and repeated workspace activation never extend a period twice. Expiry uses the latest committed validity, keeps accepted transaction context and existing cases/history, and denies new cases. AI failure does not expire Subscription. The final node ends this period; it does not delete history or terminate the workspace.

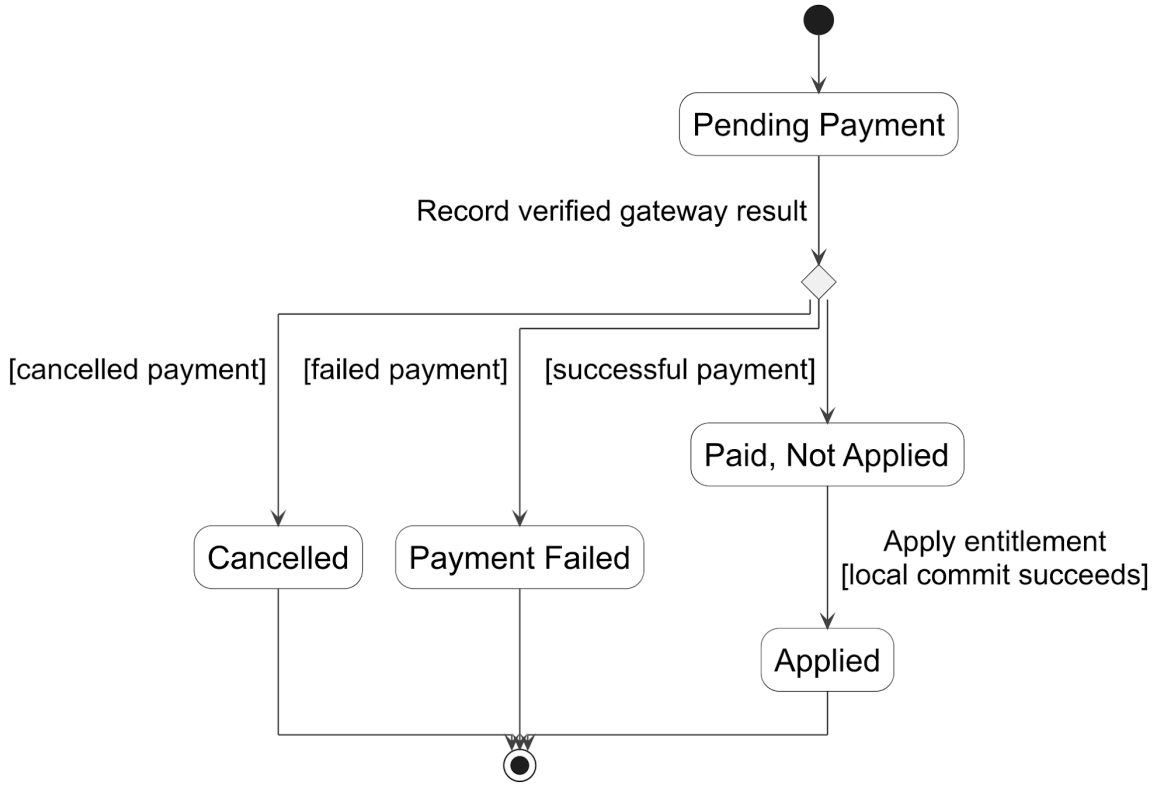


Figure 4.3.3.4.2. Subscription Purchase State Machine Diagram.

SRS 3.4.5-3.4.6: one stored purchase quote for an available offer, owned by SME Admin of an operational workspace. Verify provider identity, purchase/workspace, amount, currency and final status on the server. Pending/unverified/mismatched callbacks do not grant rights or enter a final-result branch; browser redirects are not payment proof. Successful payment with a local application failure remains Paid, Not Applied for reconciliation without another charge. Duplicate/concurrent confirmations have one entitlement effect. Applied ends this purchase's handling; the Subscription validity period is modeled separately. Failed/cancelled checkout grants no rights; safe retry must retain its purchase/payment identity and cannot duplicate an applied effect.

3.4 Workspace Onboarding and Activation

3.4.1 Class Diagram

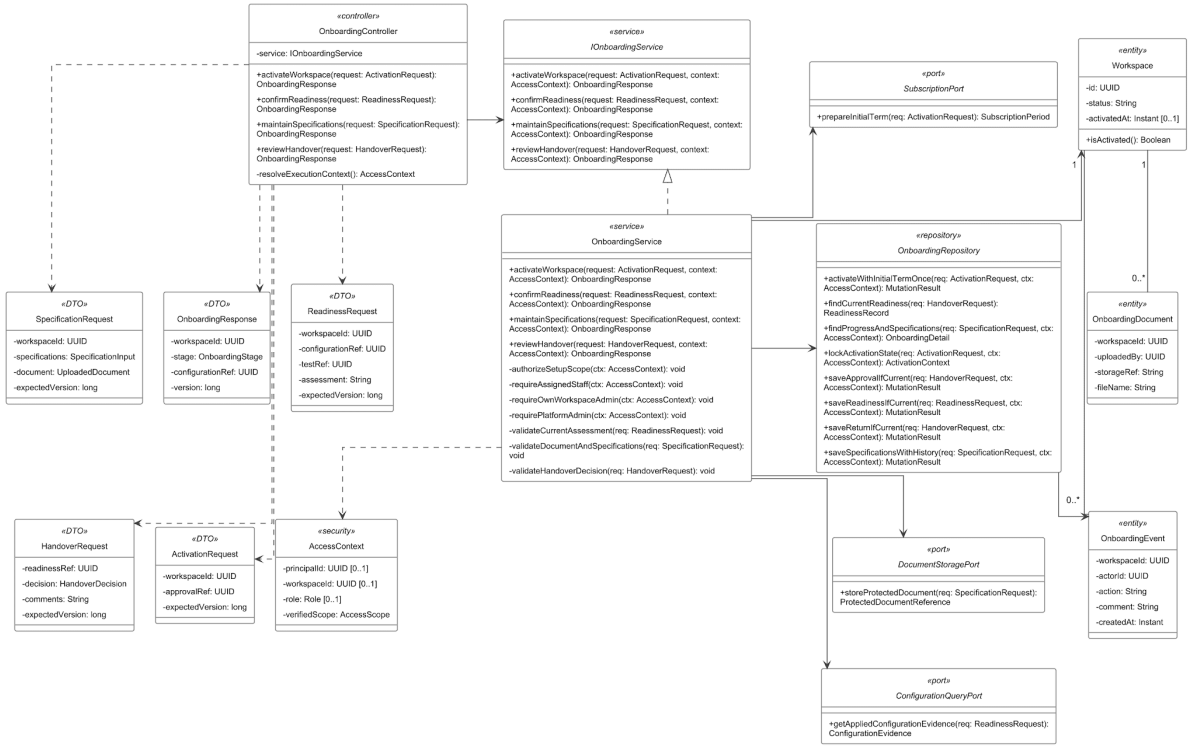


Figure 4.3.4.1.1. Workspace Onboarding and Activation Class Diagram.

3.4.2 Sequence Diagrams

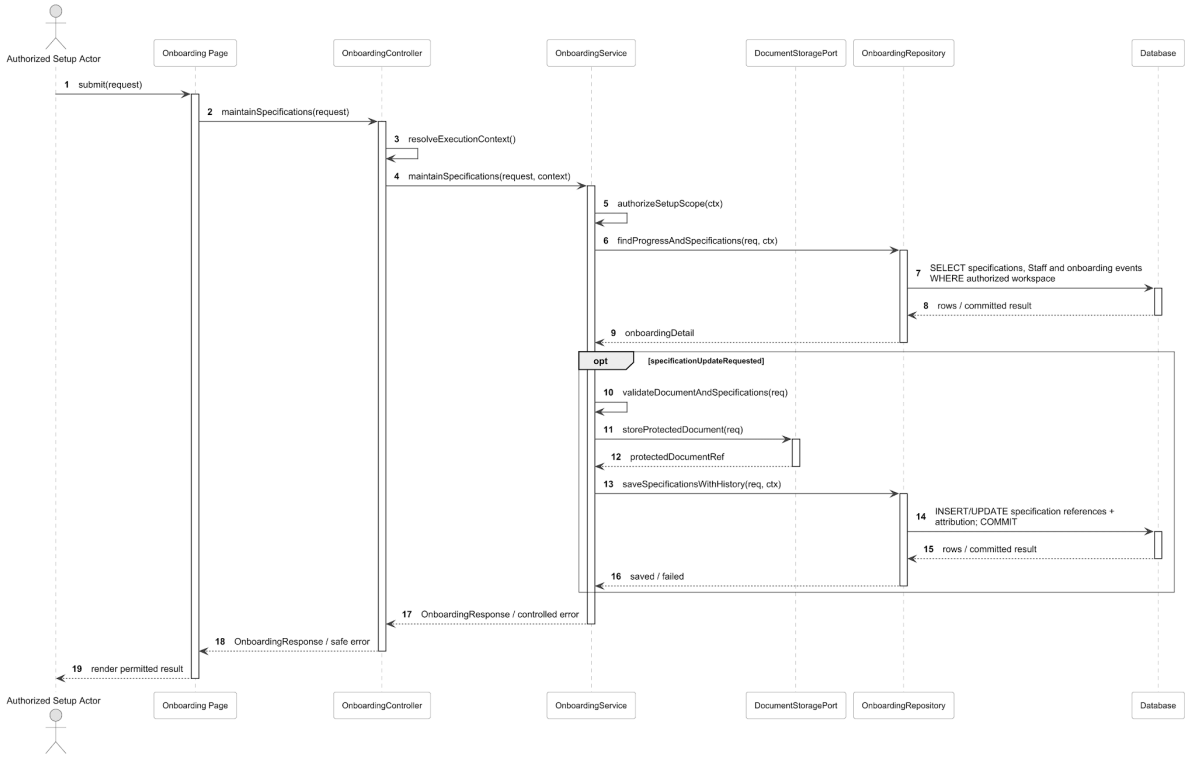


Figure 4.3.4.2.1. Maintain Specifications and Track Onboarding Progress Sequence Diagram.

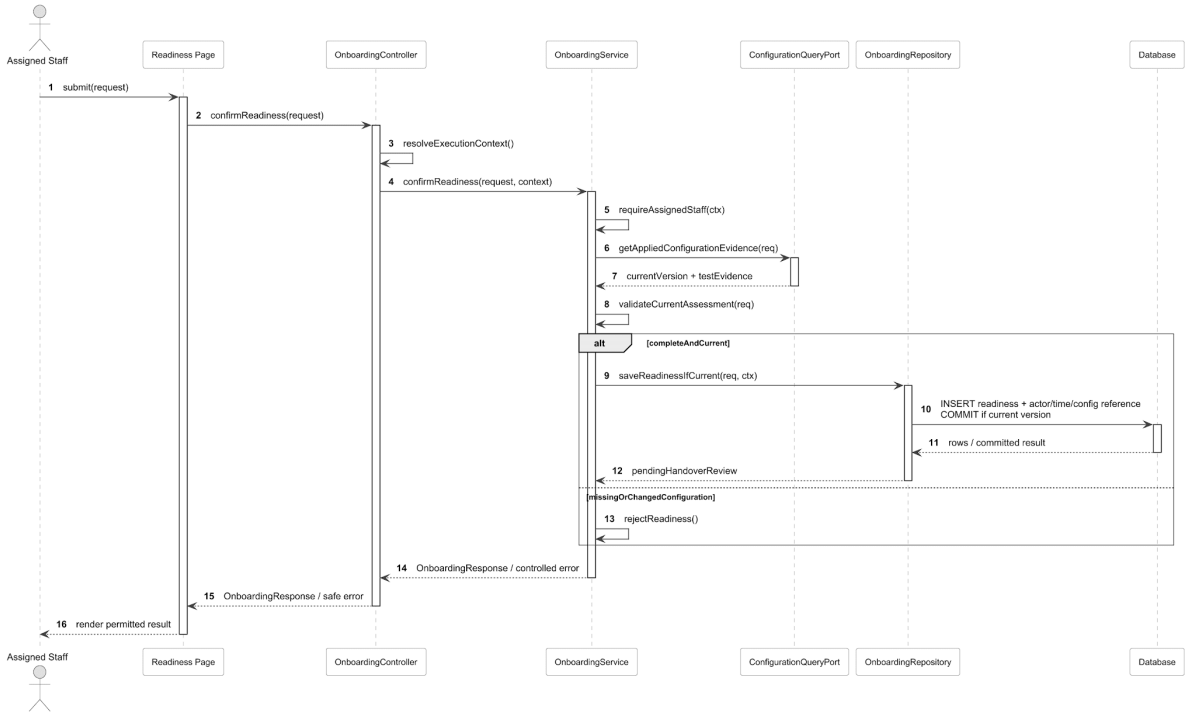


Figure 4.3.4.2.2. Confirm Implementation Readiness Sequence Diagram.

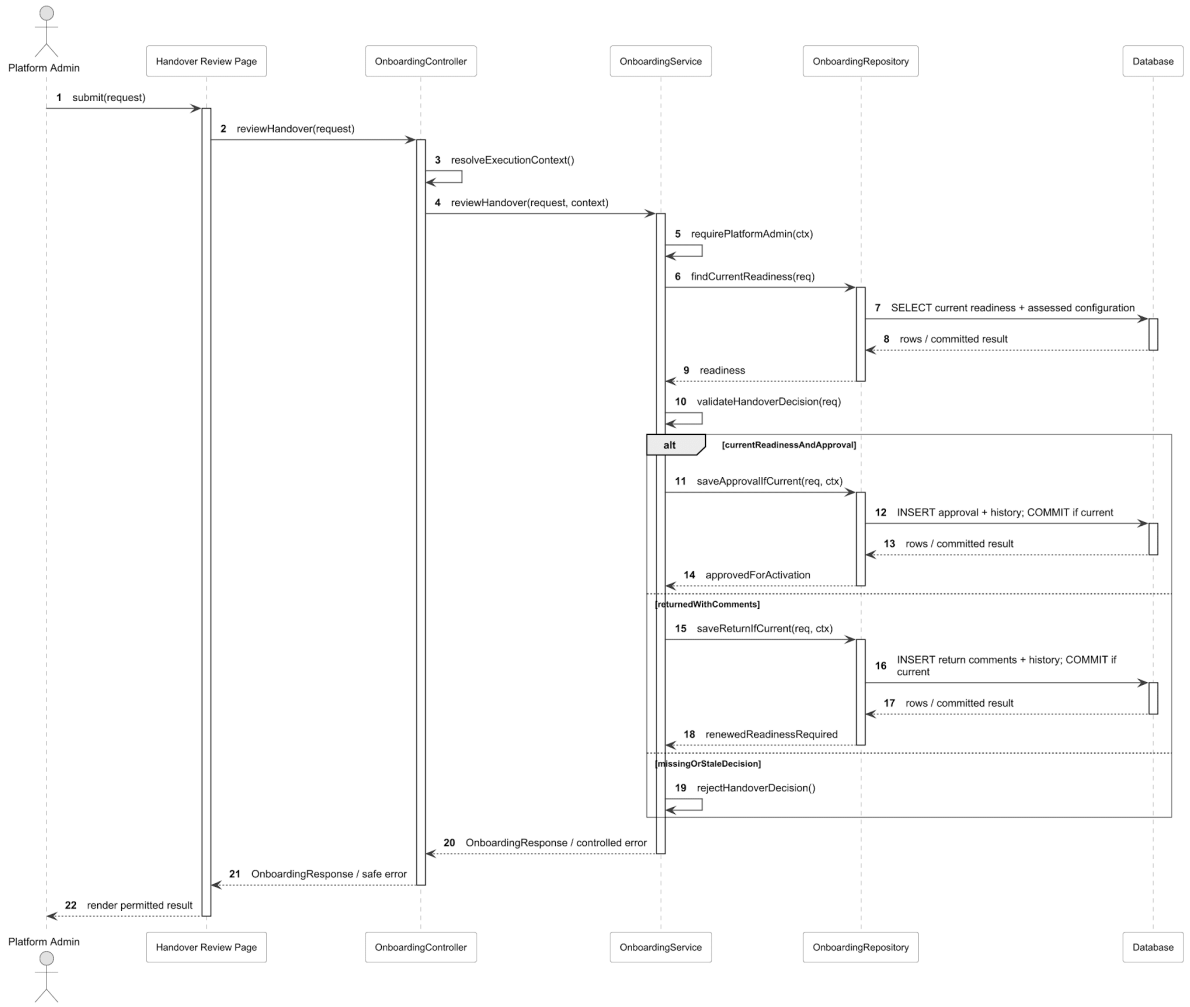


Figure 4.3.4.2.3. Approve or Return Workspace Handover Sequence Diagram.

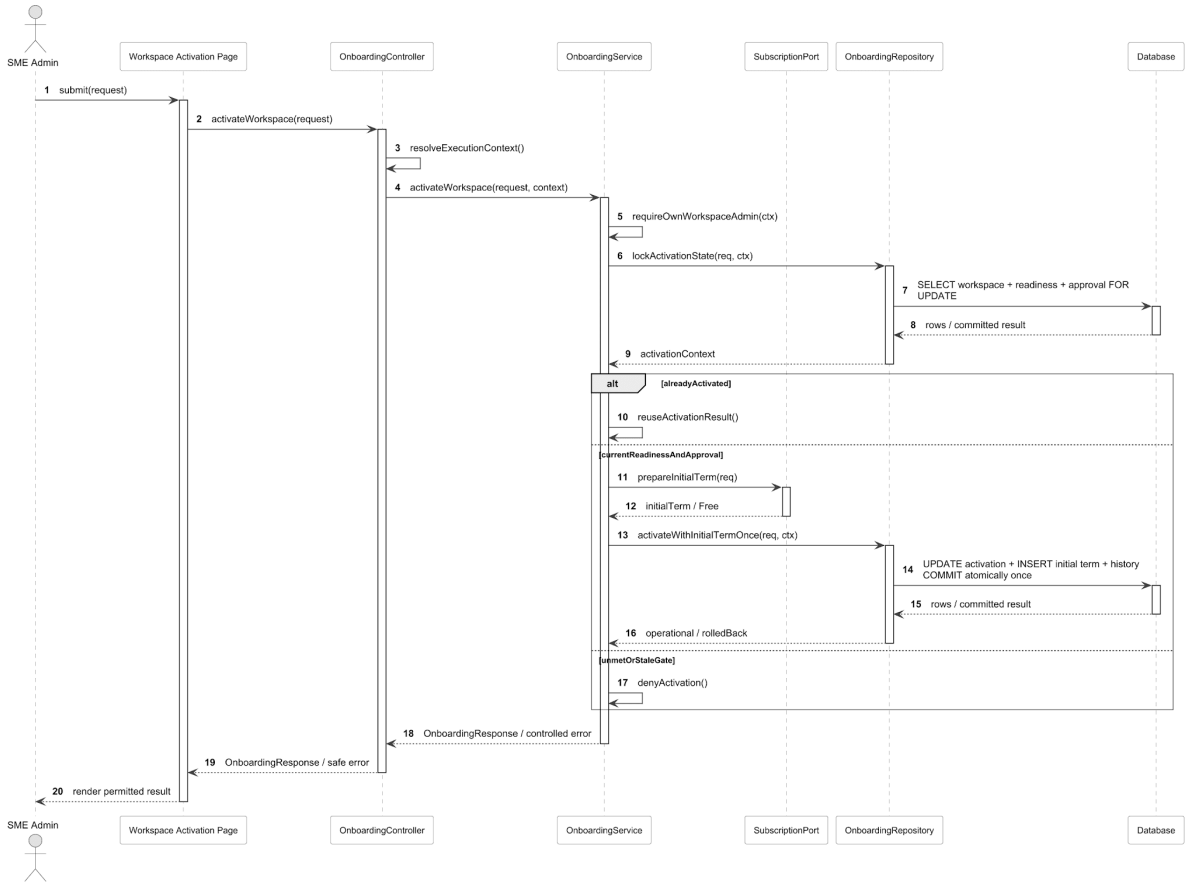


Figure 4.3.4.2.4. Activate Approved Workspace and Start the Initial Term Once Sequence Diagram.

3.4.3 Activity Diagrams

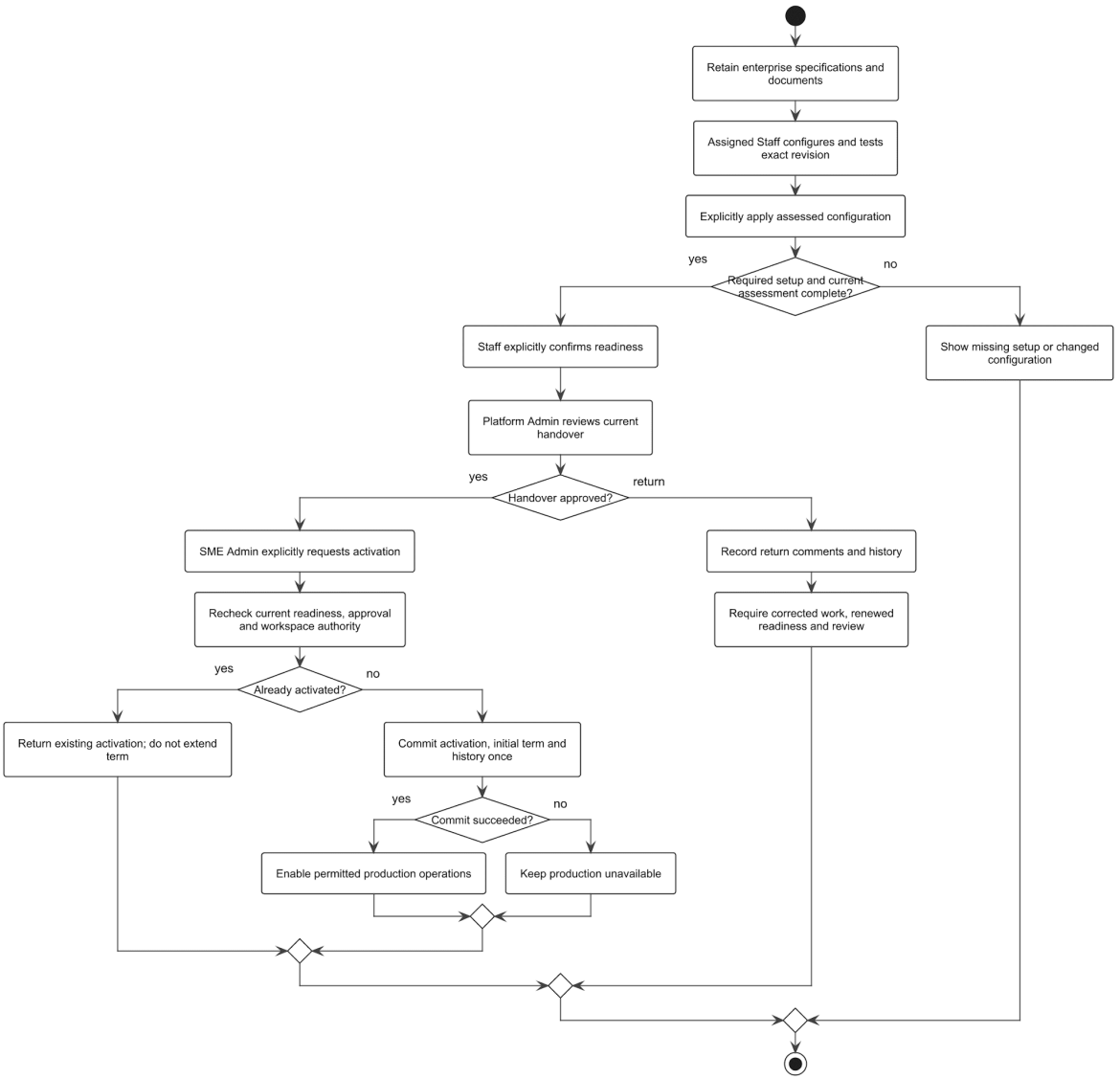


Figure 4.3.4.3.1. Workspace Readiness, Handover and Activation Lifecycle Activity Diagram.

3.4.4 State Machine Diagrams

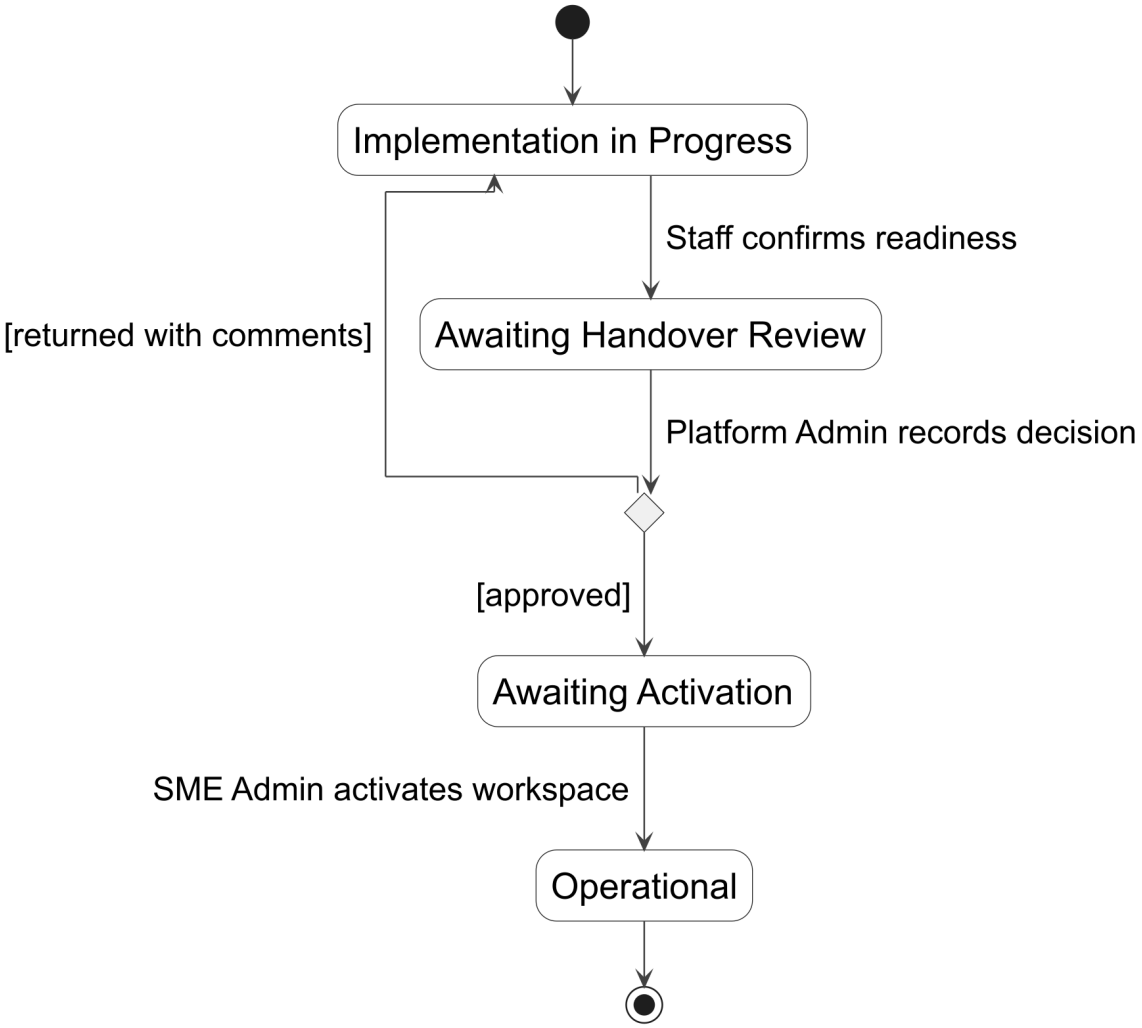


Figure 4.3.4.4.1. Workspace Onboarding State Machine Diagram.

Scope: onboarding of one newly provisioned workspace, ending when it becomes Operational. Platform Admin creates the customer, isolated workspace and initial SME Admin account (SRS 3.3.2); provisioning does not enable production or start an initial Subscription. Specifications, protected onboarding files, Staff assignment, field/mapping/rule configuration and sample testing belong to implementation. Assigned Staff explicitly assesses the current configuration before confirming readiness; test success alone is insufficient. Platform Admin approves current Staff readiness or returns work with required comments. Returned work needs renewed readiness and review. SME Admin activates only with current readiness and approval; activation and the initial entitlement term commit once with history. Missing setup, stale configuration/approval, unauthorized requests or persistence failure leave the last committed stage unchanged. Logical stage labels are design terms, not implemented enums. The final node ends onboarding only; the workspace remains available for ongoing operation and Staff support.

3.5 Enterprise Member and Permission Management

3.5.1 Class Diagram

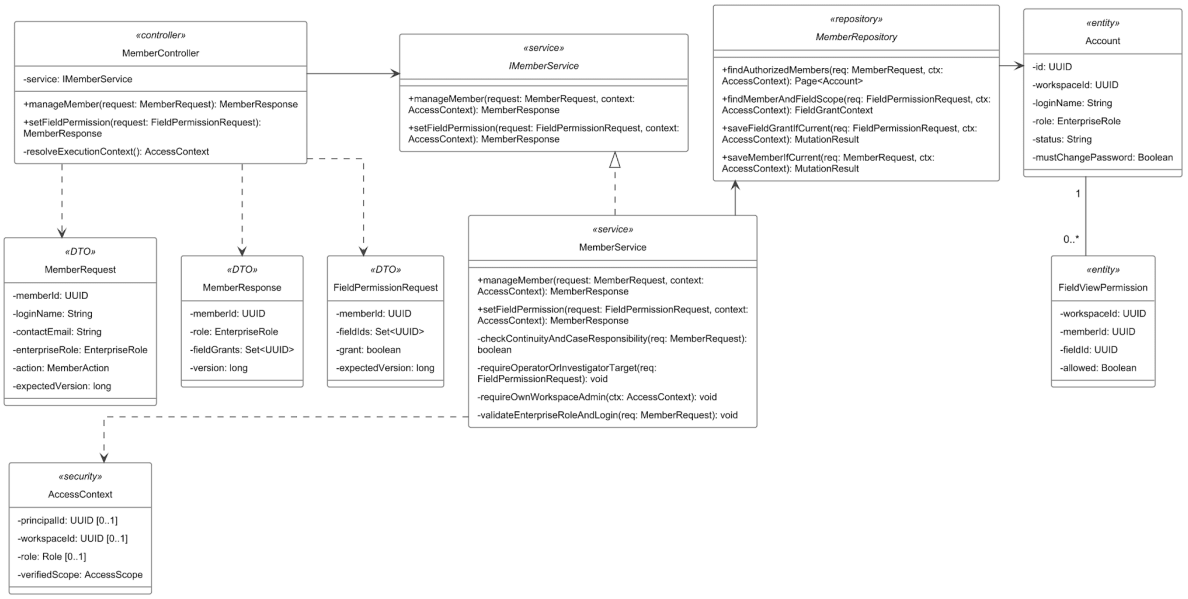


Figure 4.3.5.1.1. Enterprise Member and Permission Management Class Diagram.

3.5.2 Sequence Diagrams

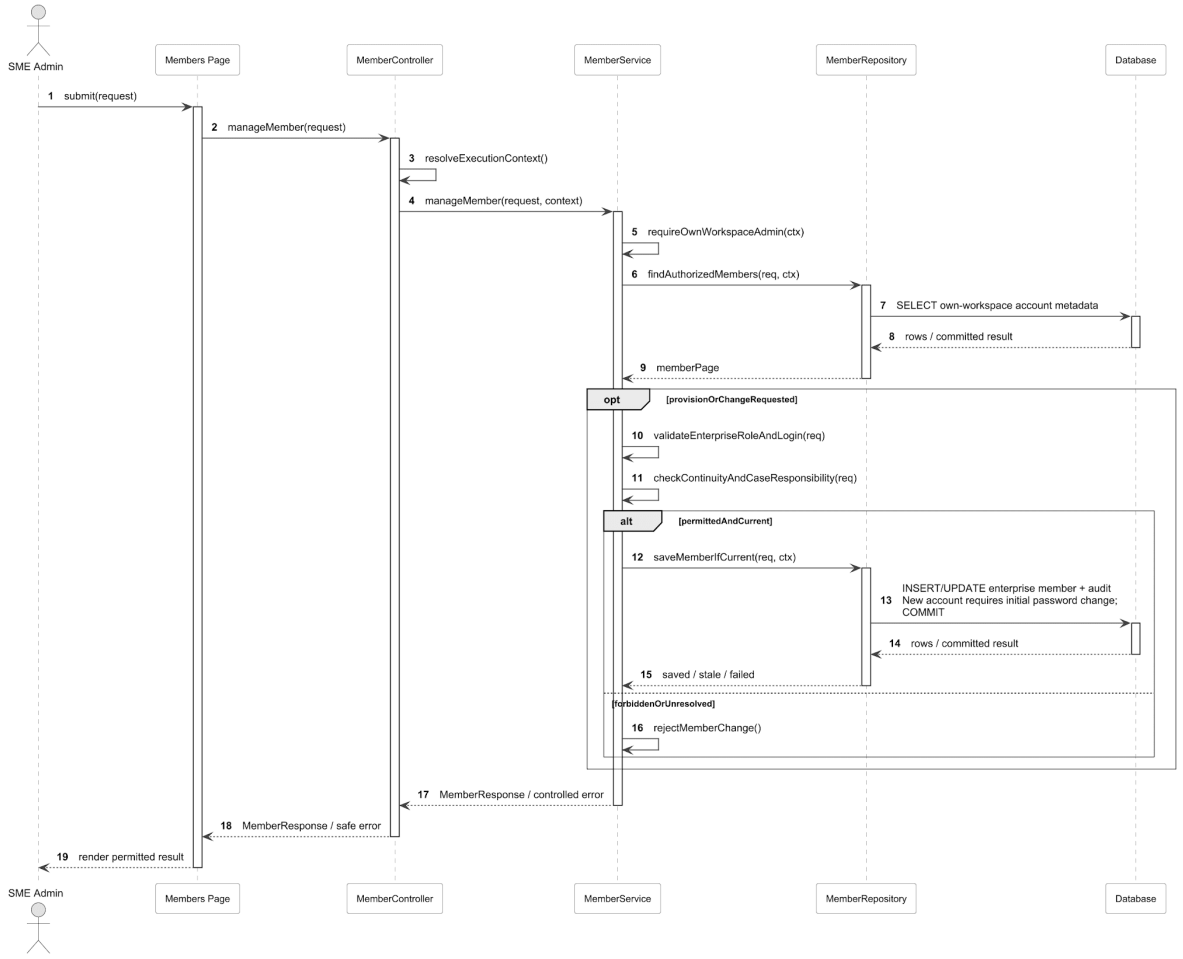


Figure 4.3.5.2.1. List, Provision and Update Enterprise Members Sequence Diagram.

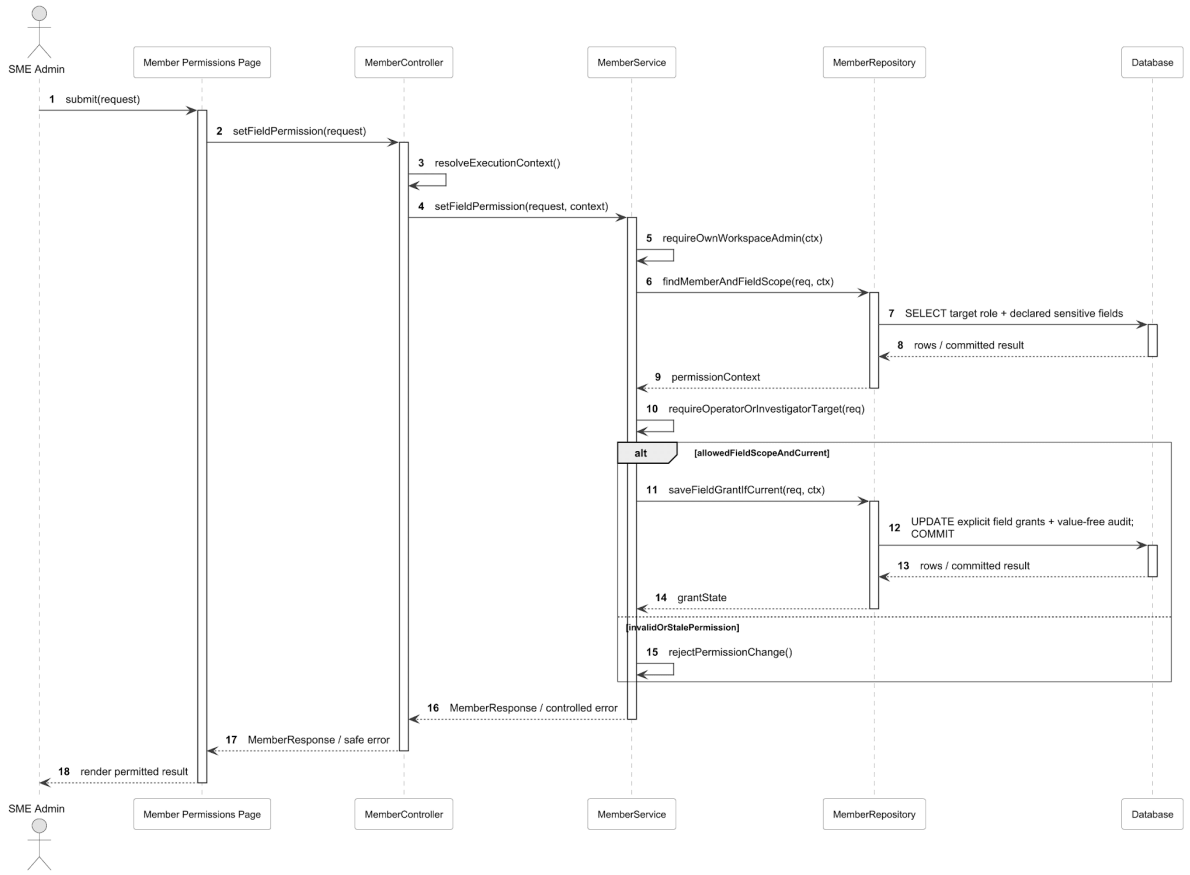


Figure 4.3.5.2.2. Grant or Revoke Explicit Sensitive-Field Viewing Permission Sequence Diagram.

3.6 Detection Configuration Management

3.6.1 Class Diagram

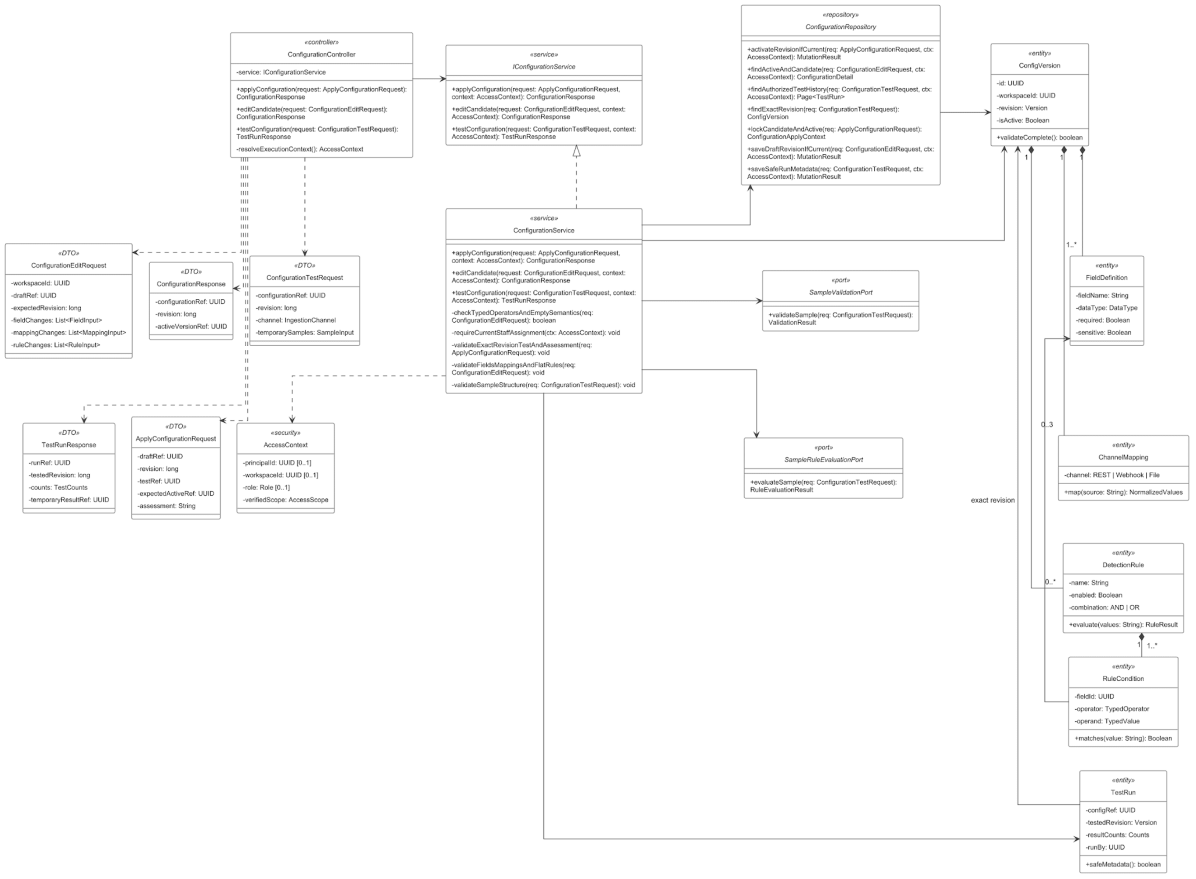


Figure 4.3.6.1.1. Detection Configuration Management Class Diagram.

3.6.2 Sequence Diagrams

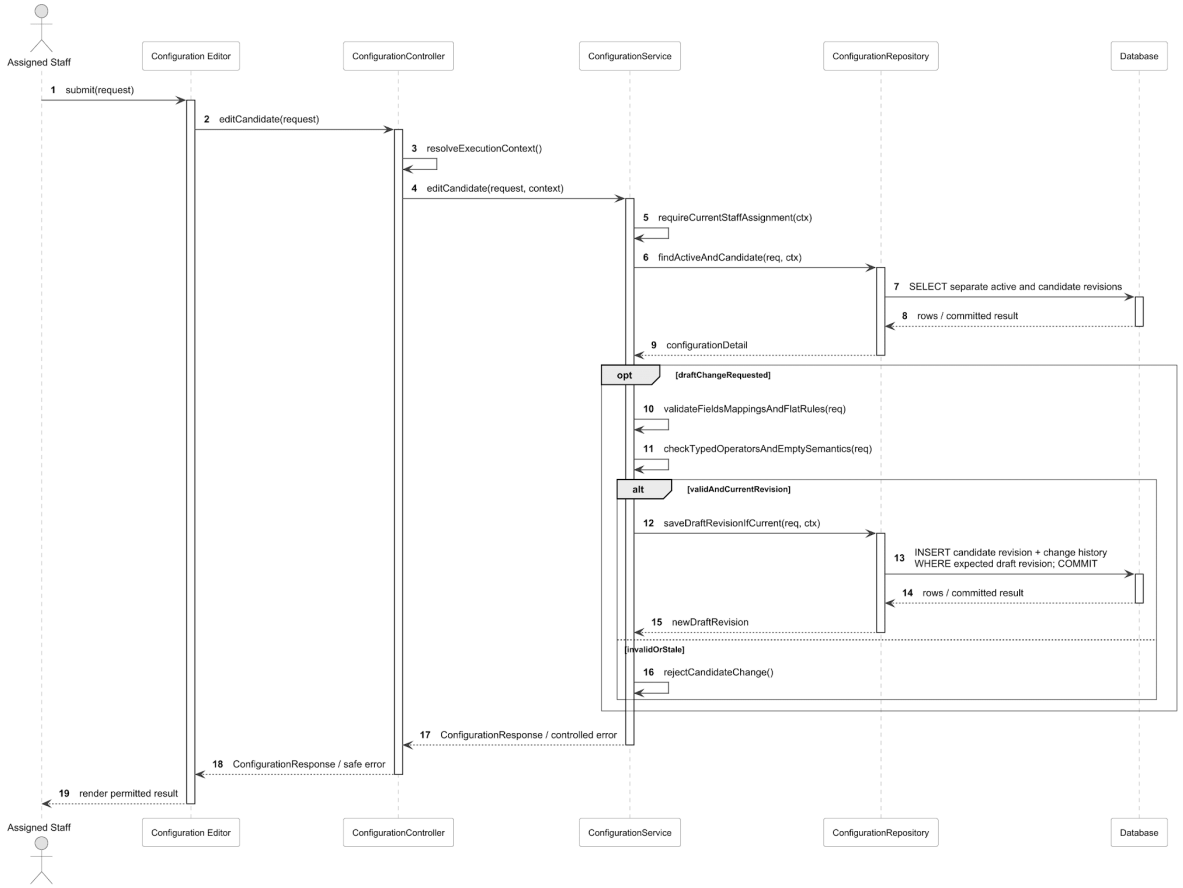


Figure 4.3.6.2.1. View and Edit Candidate Fields, Mappings and Rules Sequence Diagram.

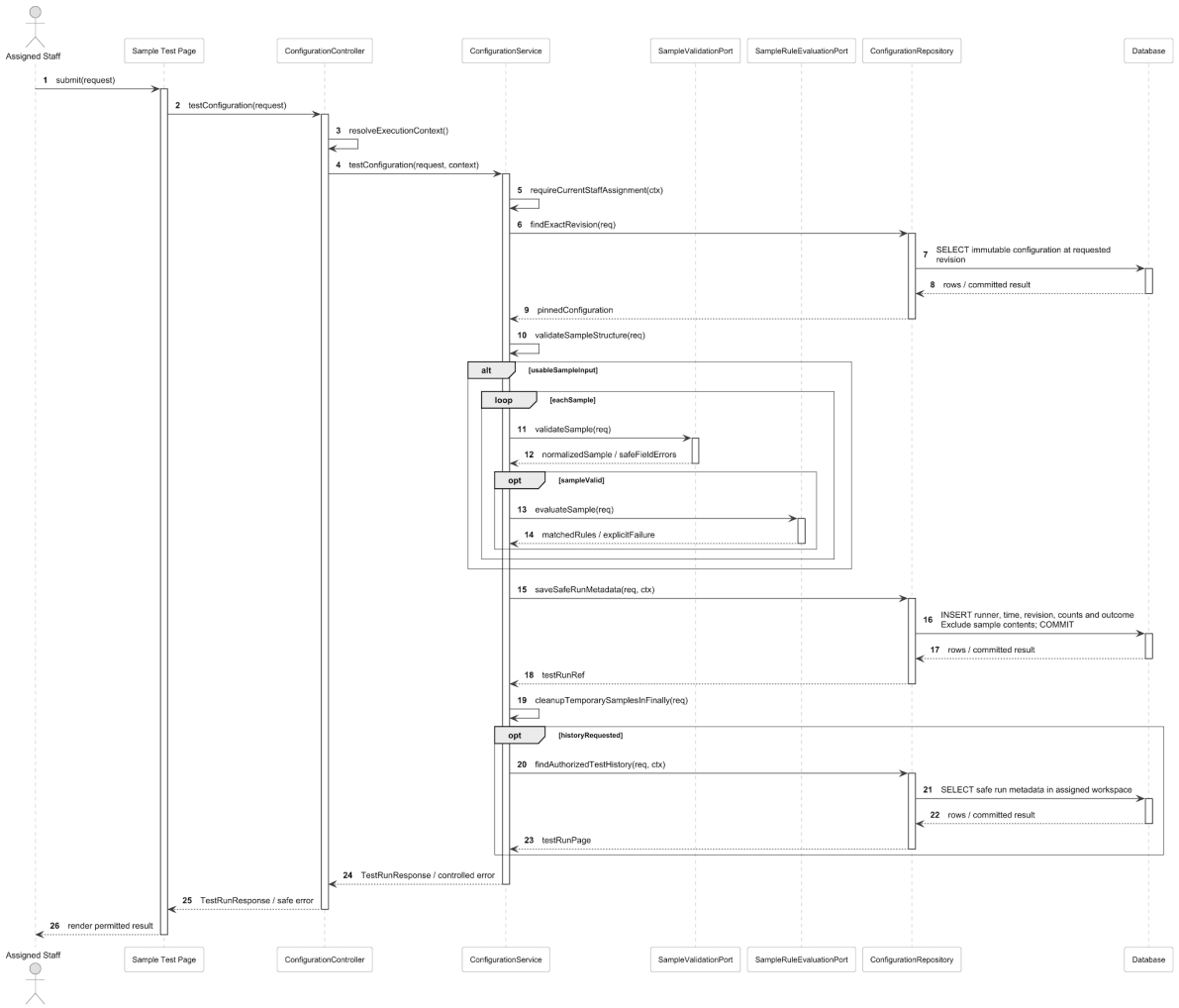


Figure 4.3.6.2.2. Test an Exact Revision and Review Safe Run History Sequence Diagram.

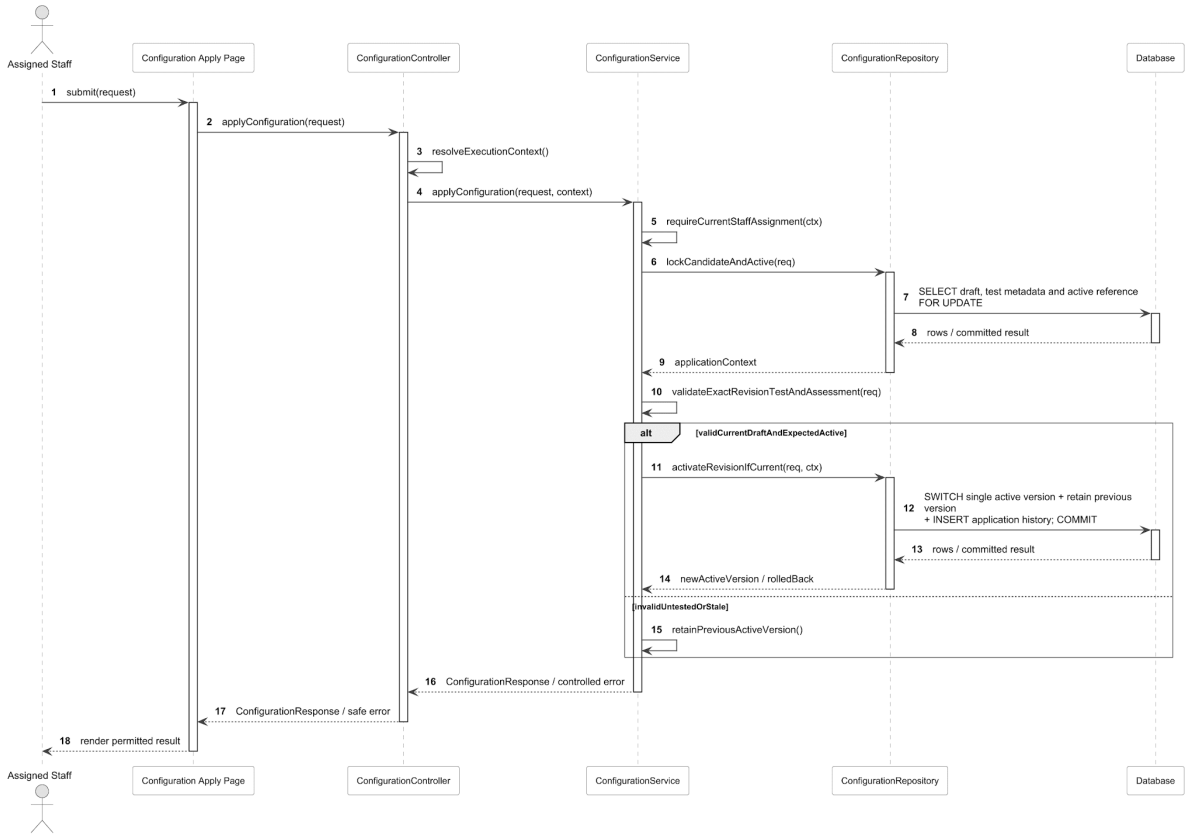


Figure 4.3.6.2.3. Apply the Tested Candidate Atomically Sequence Diagram.

3.6.3 Activity Diagrams

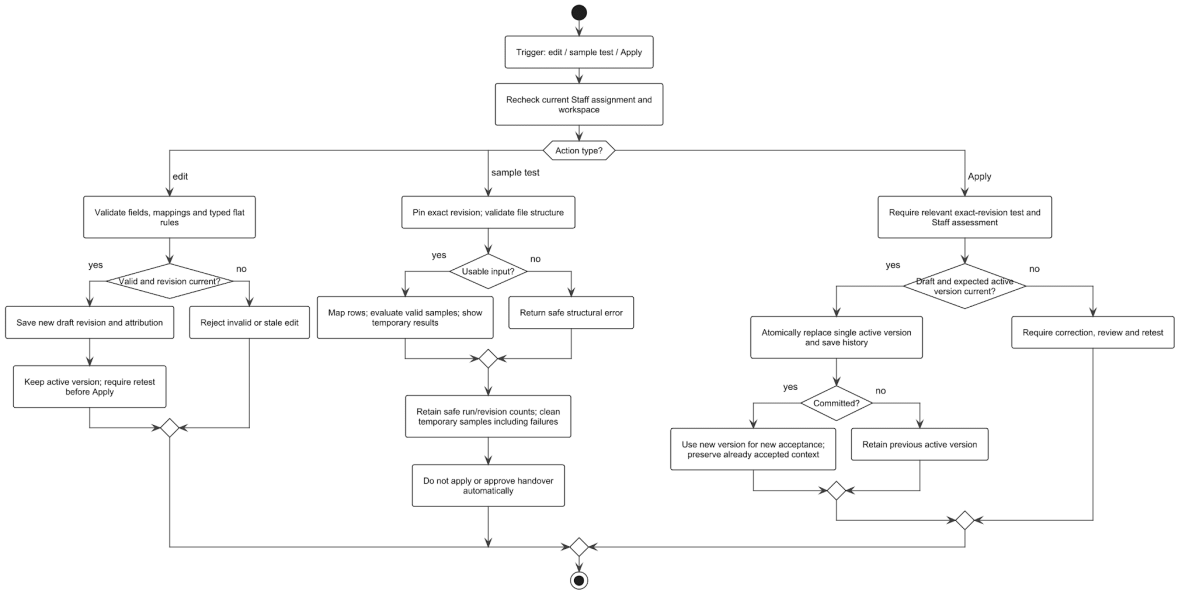


Figure 4.3.6.3.1. Candidate Configuration Testing and Application Lifecycle Activity Diagram.

3.6.4 State Machine Diagrams

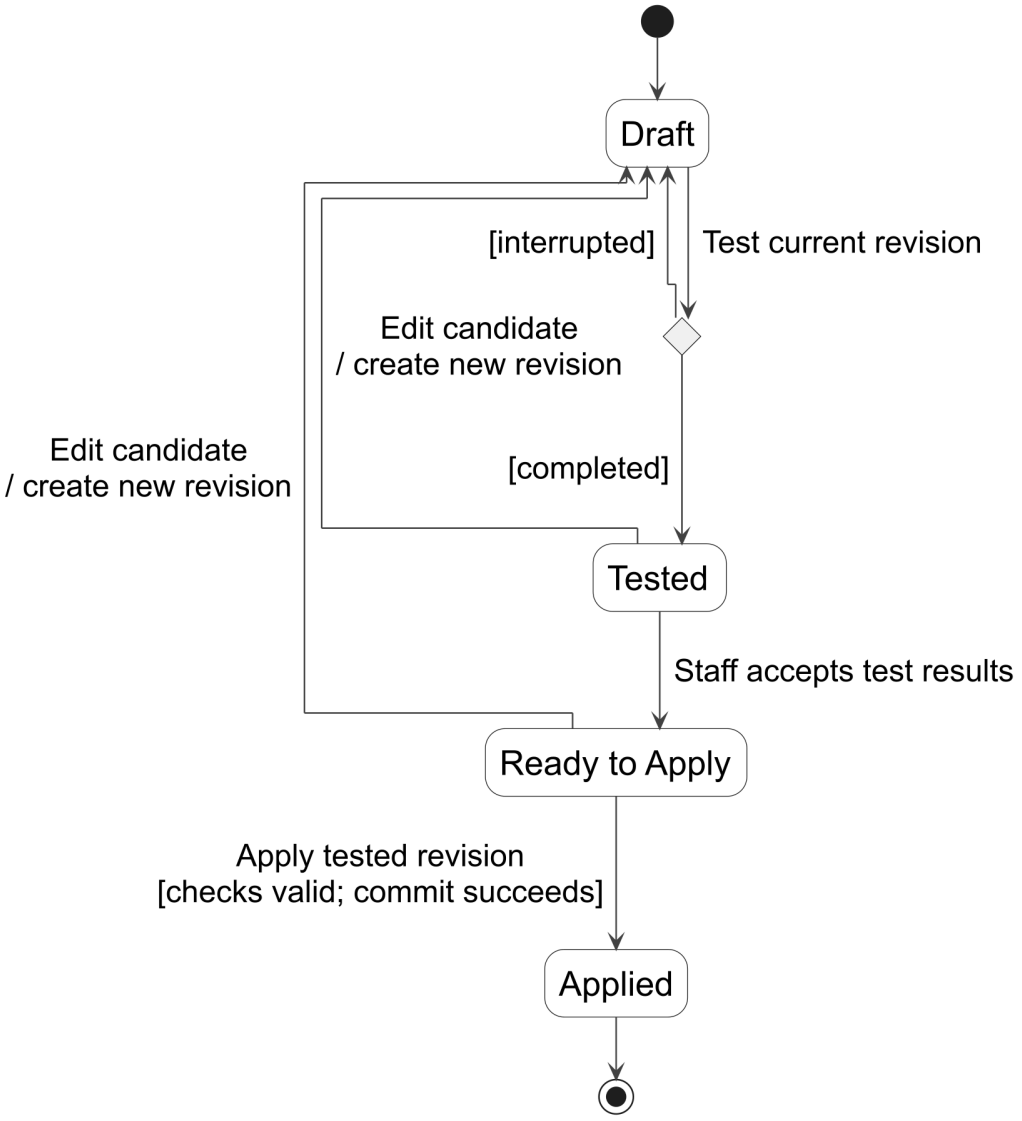


Figure 4.3.6.4.1. Configuration Candidate State Machine Diagram.

3.7 Integration and Transaction Ingestion

3.7.1 Class Diagram

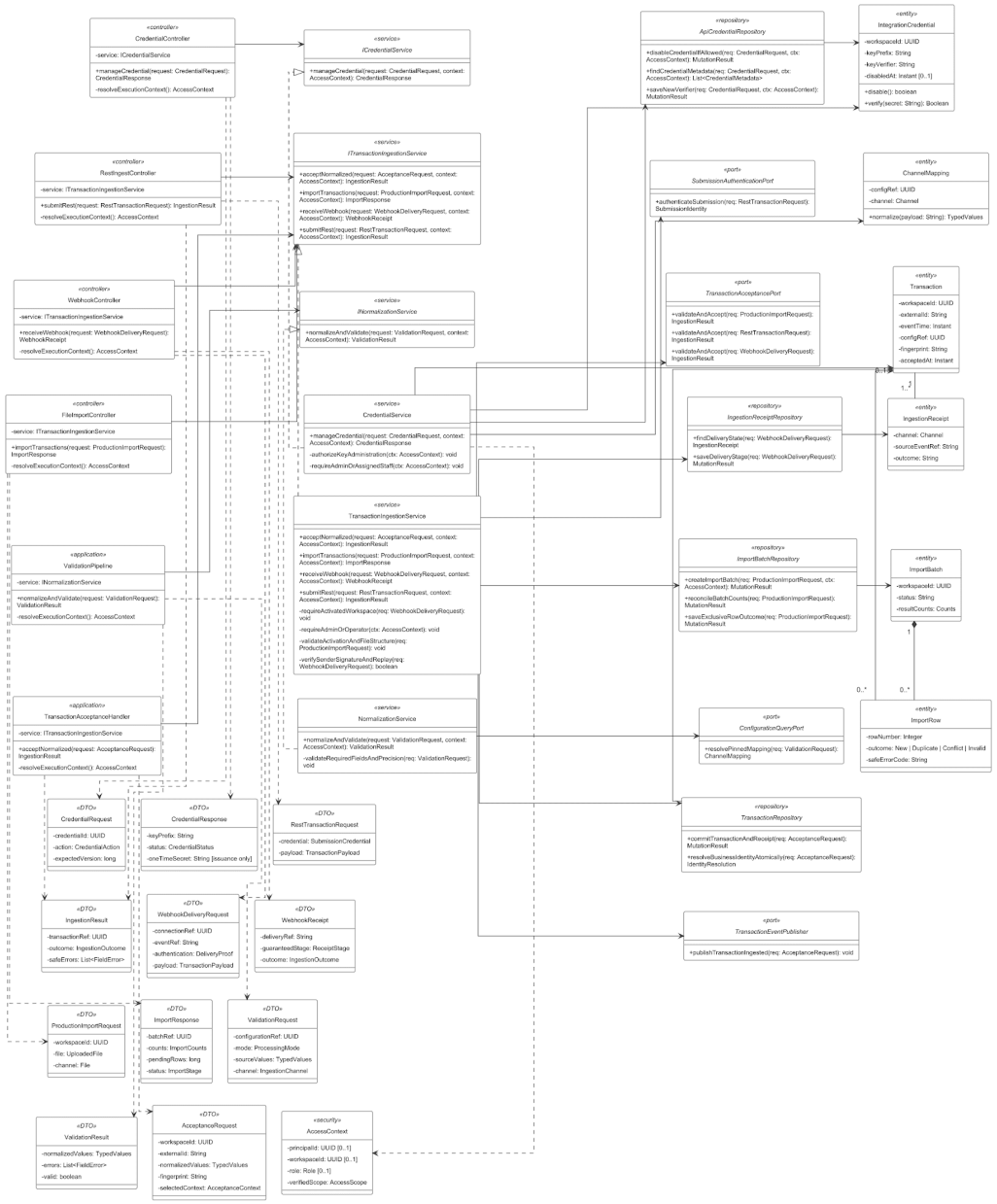


Figure 4.3.7.1.1. Integration and Transaction Ingestion Class Diagram.

3.7.2 Sequence Diagrams

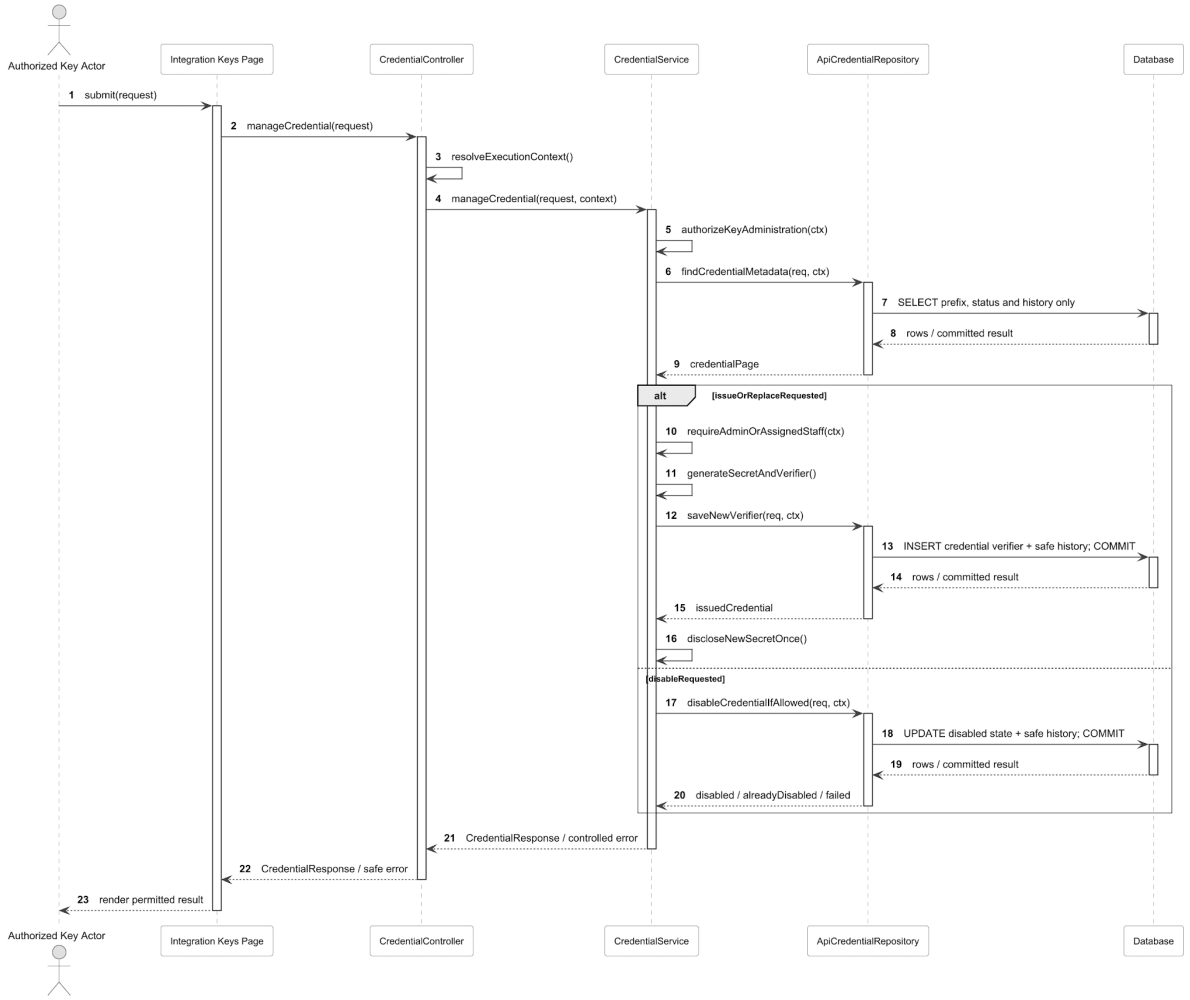


Figure 4.3.7.2.1. Inspect, Issue, Replace and Disable Workspace API Keys Sequence Diagram.

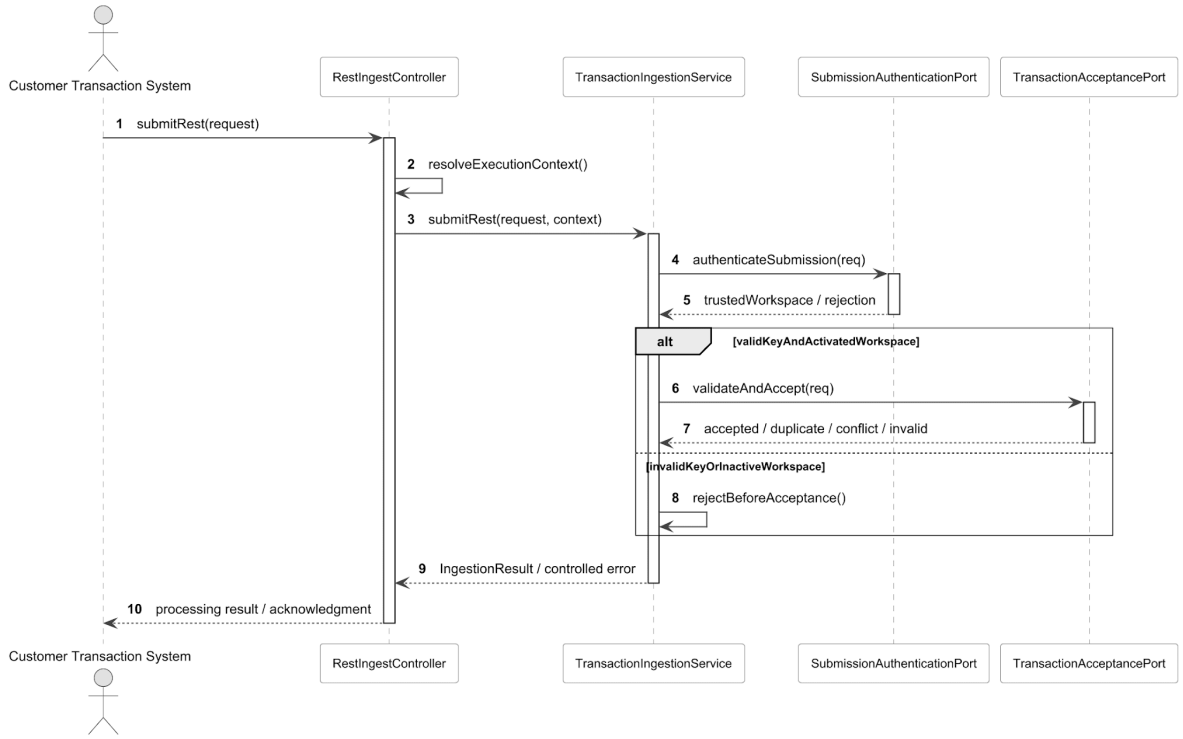


Figure 4.3.7.2.2. Submit Production Transactions through REST API Sequence Diagram.

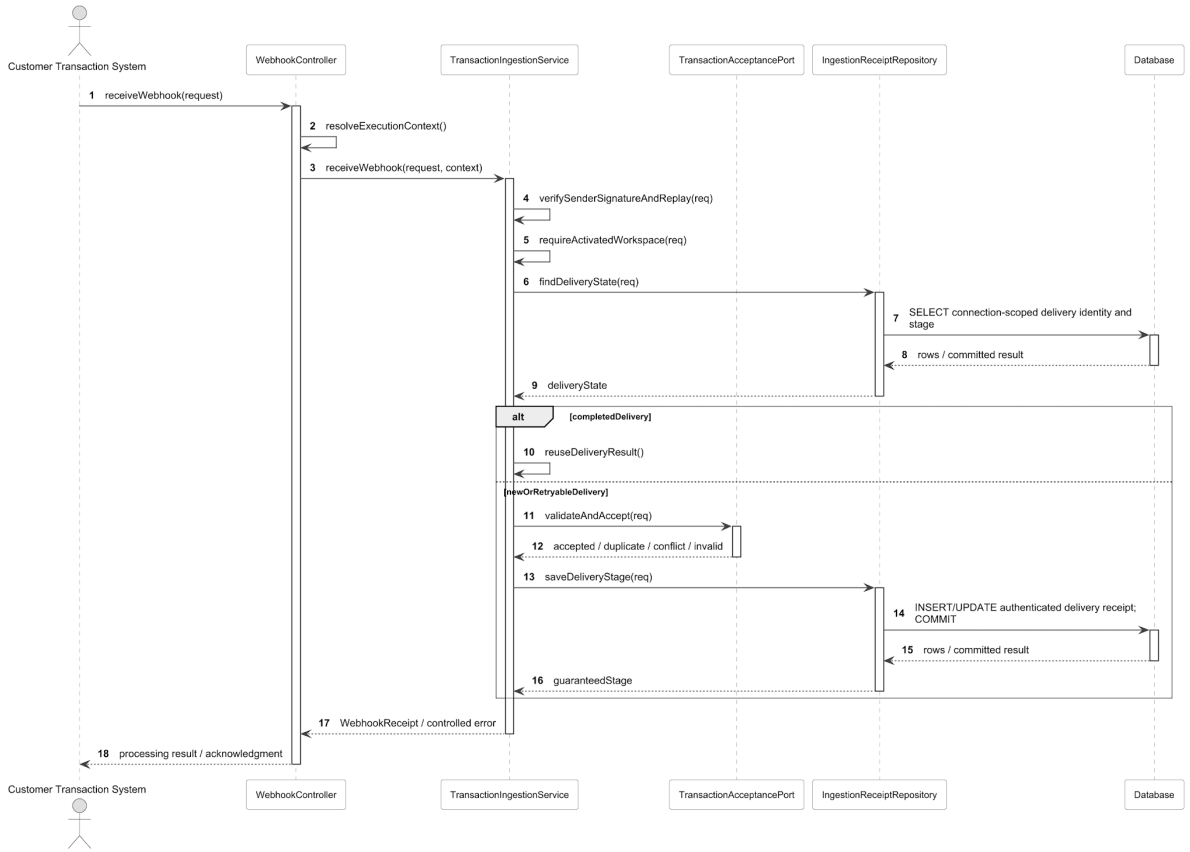


Figure 4.3.7.2.3. Authenticate Incoming Webhook and Handle Delivery Retries Sequence Diagram.

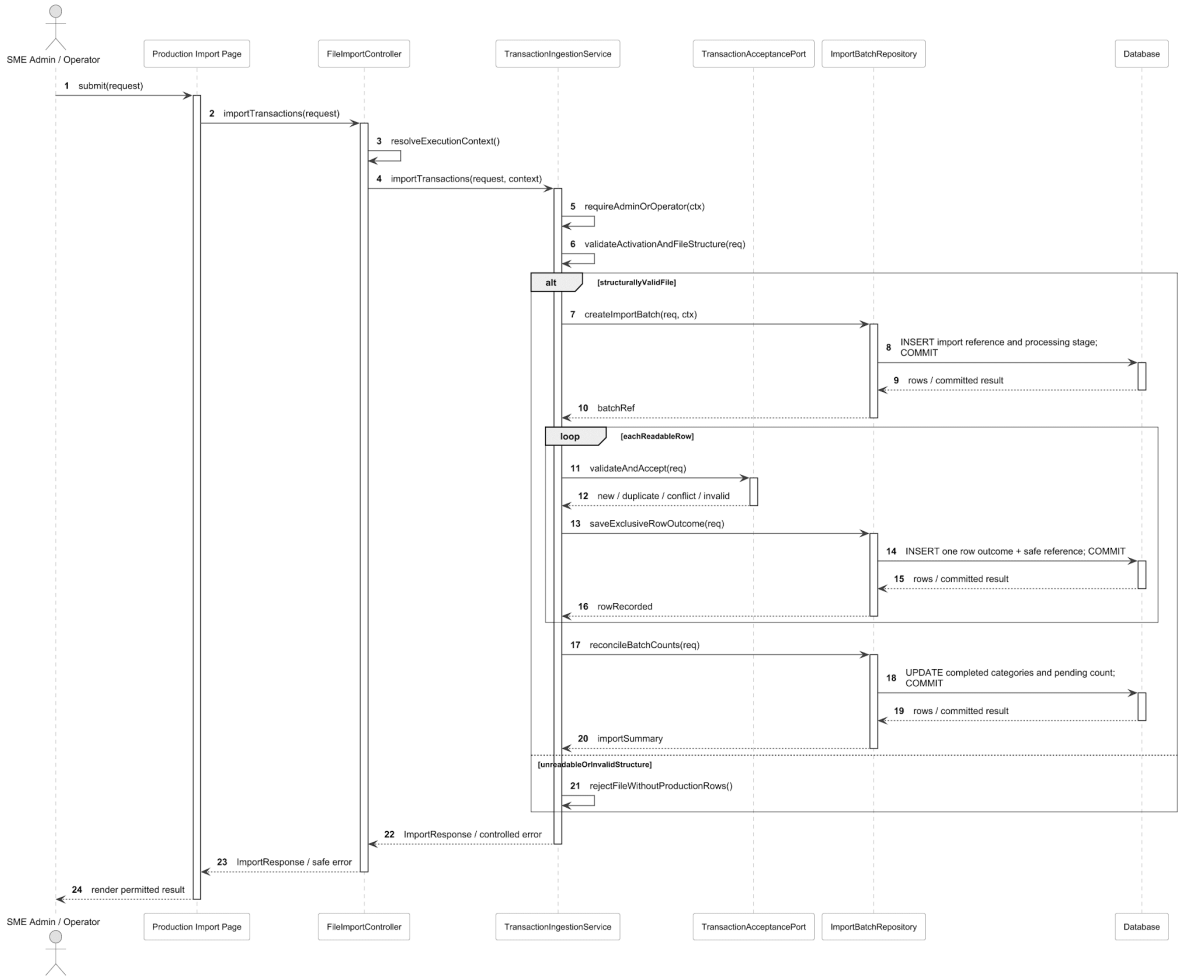


Figure 4.3.7.2.4. Import Production CSV/Excel with Reconciled Row Outcomes Sequence Diagram.

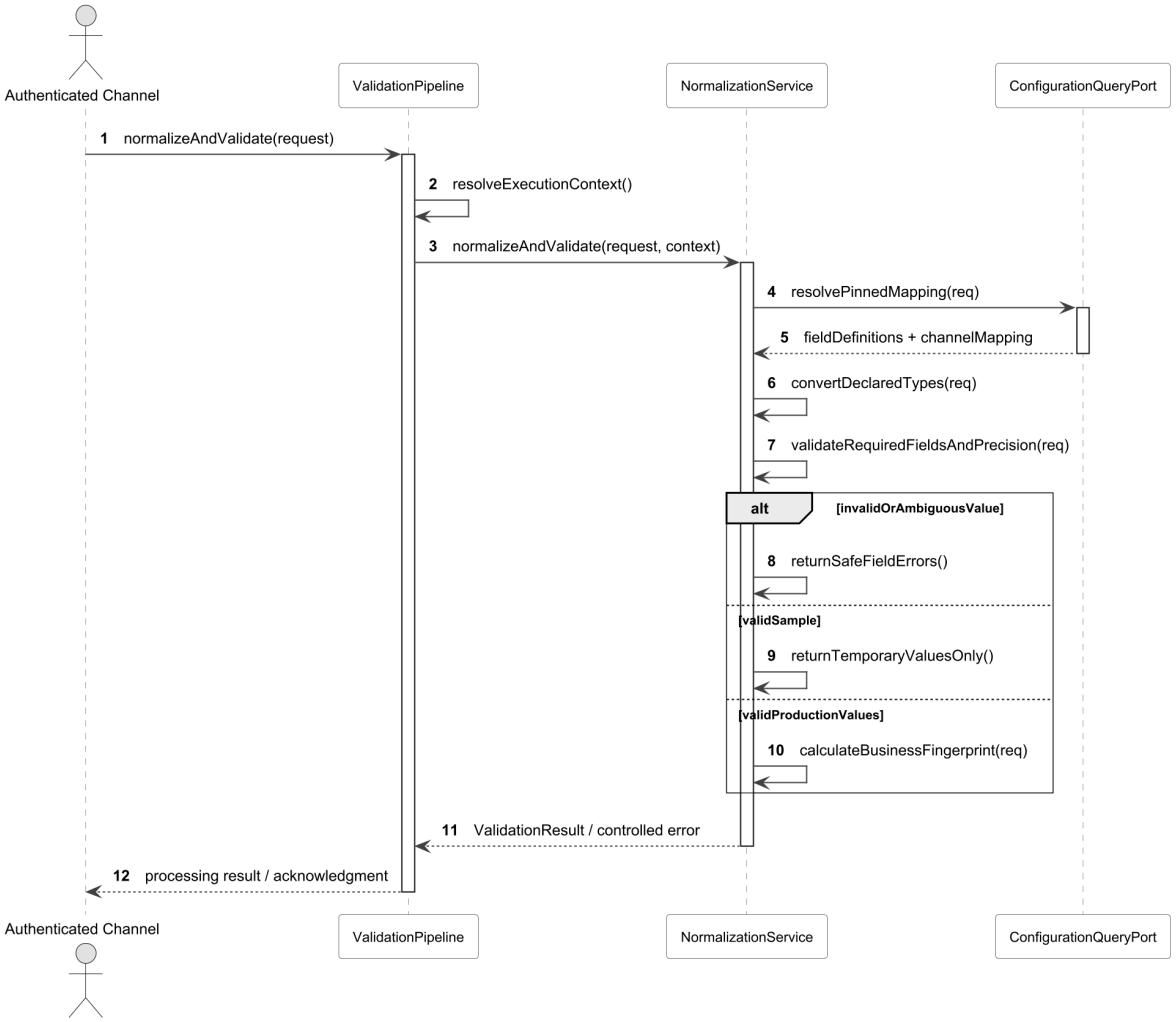


Figure 4.3.7.2.5. Normalize and Validate Declared Channel Values Sequence Diagram.

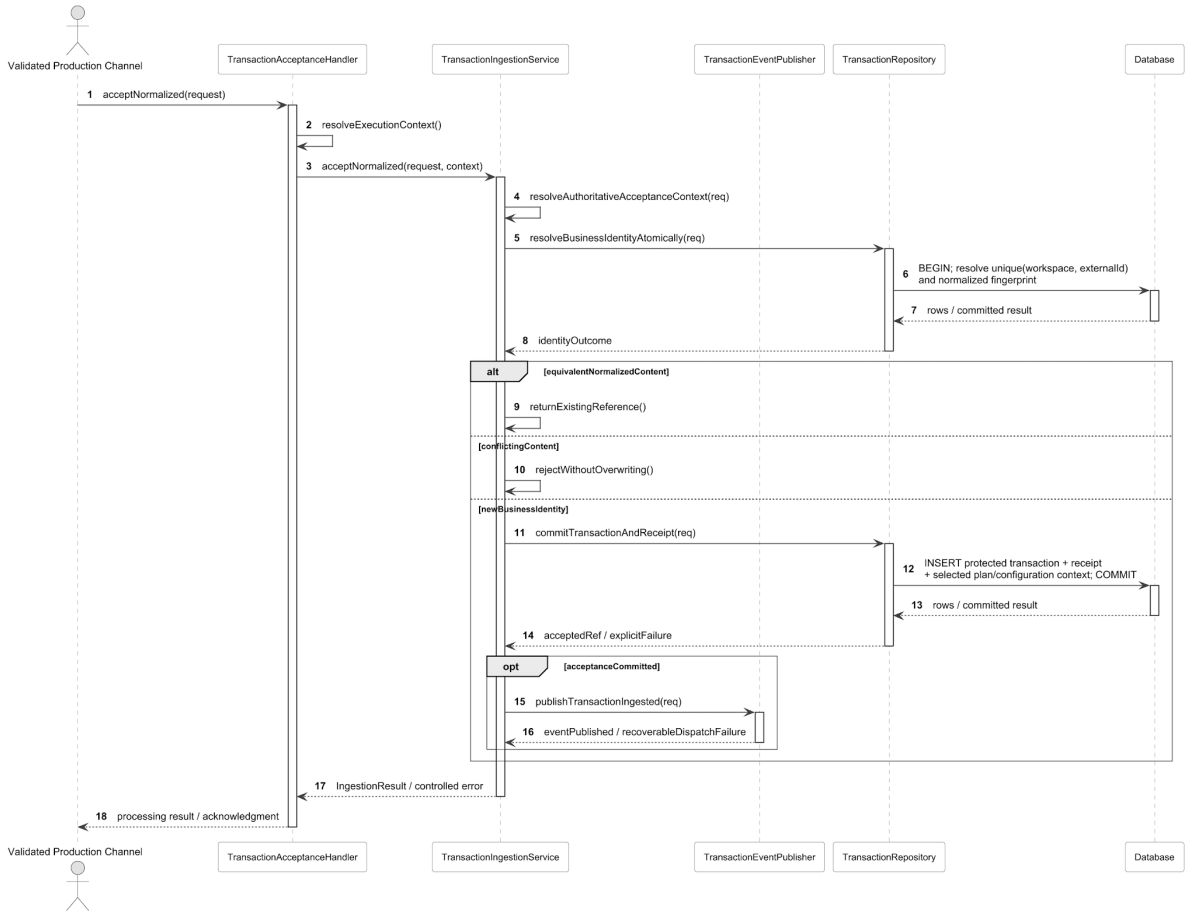


Figure 4.3.7.2.6. Atomically Resolve Cross-Channel Business Identity and Accept Once Sequence Diagram.

3.7.3 Activity Diagrams

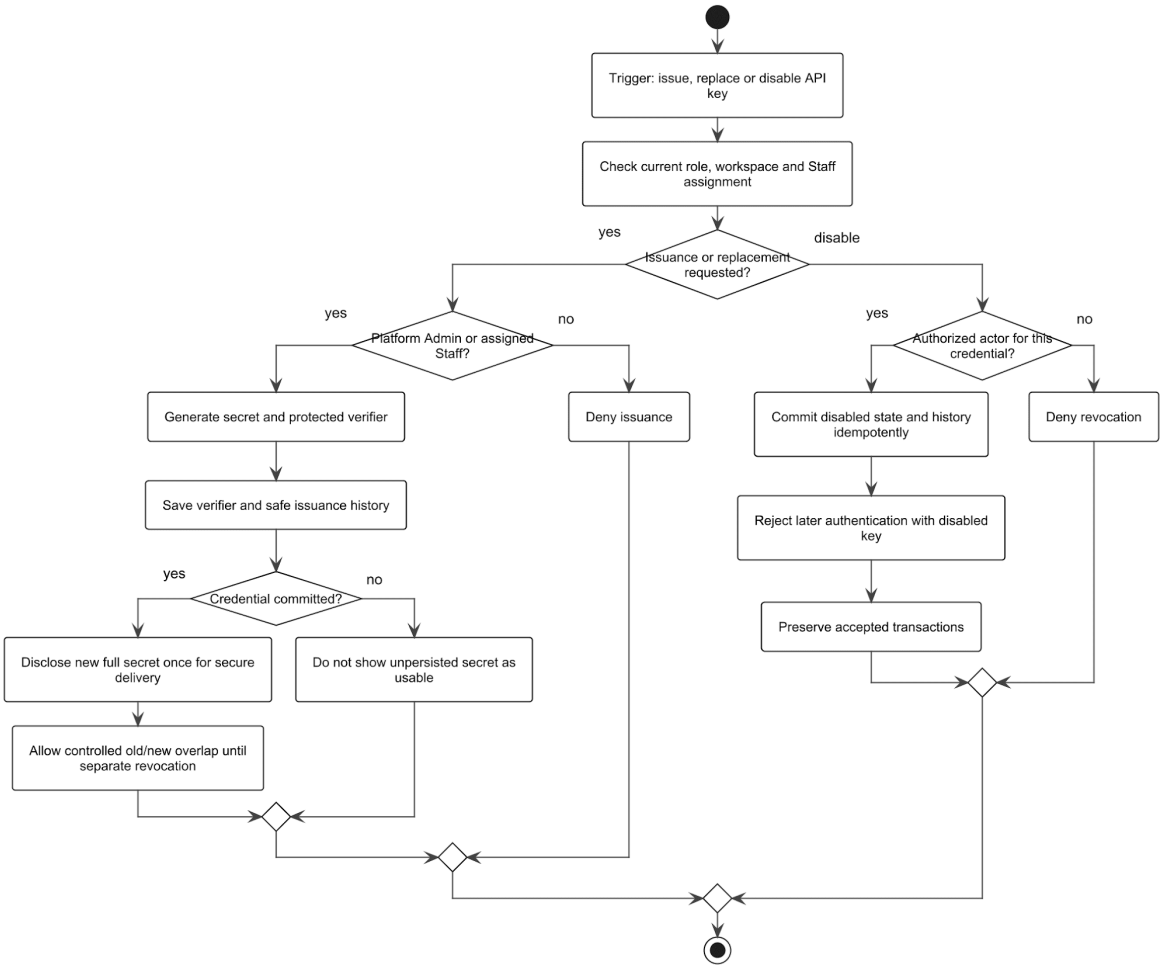


Figure 4.3.7.3.1. Workspace API Credential Lifecycle Activity Diagram.

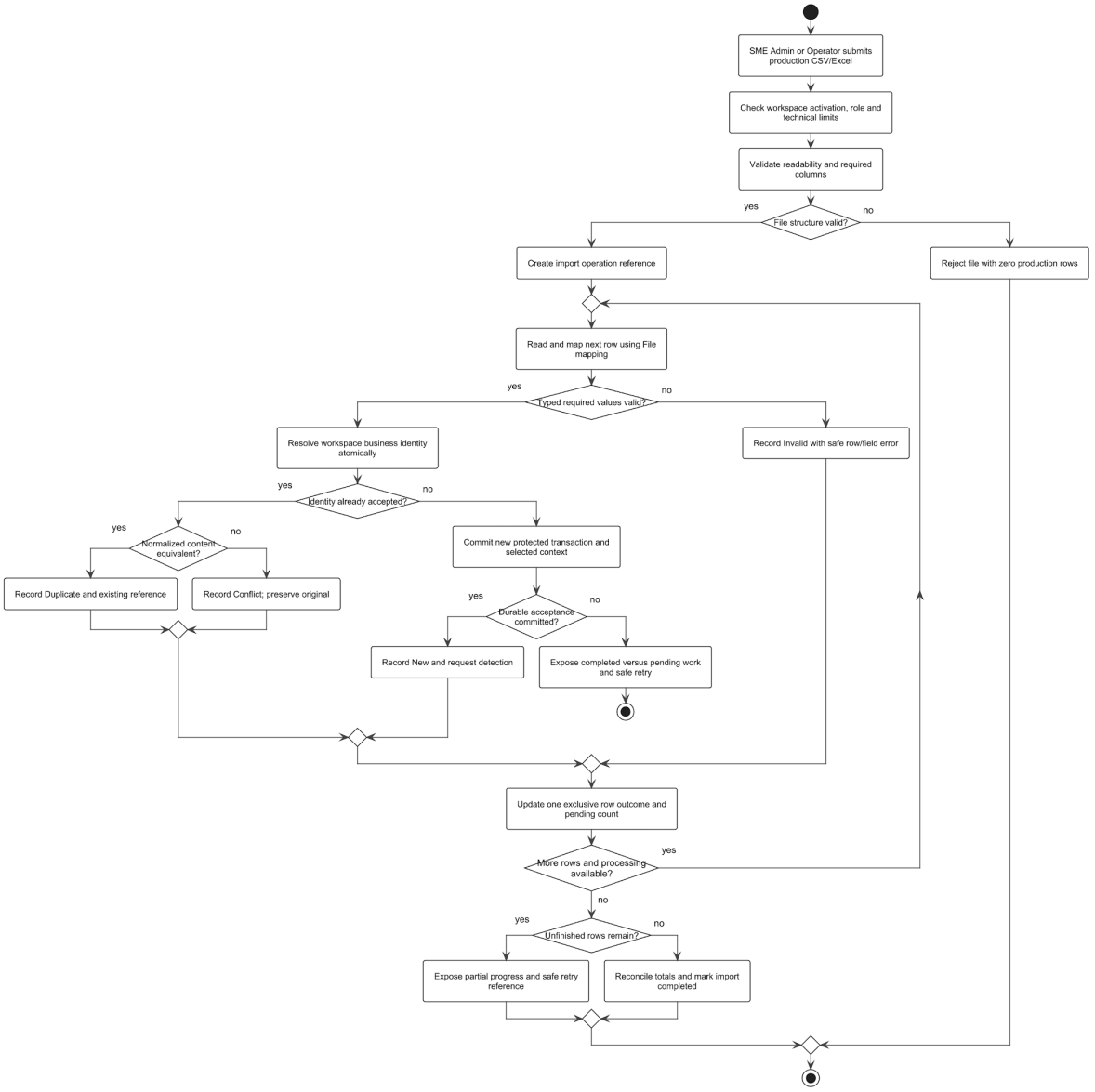


Figure 4.3.7.3.2. Production Import Processing Lifecycle Activity Diagram.

3.7.4 State Machine Diagrams

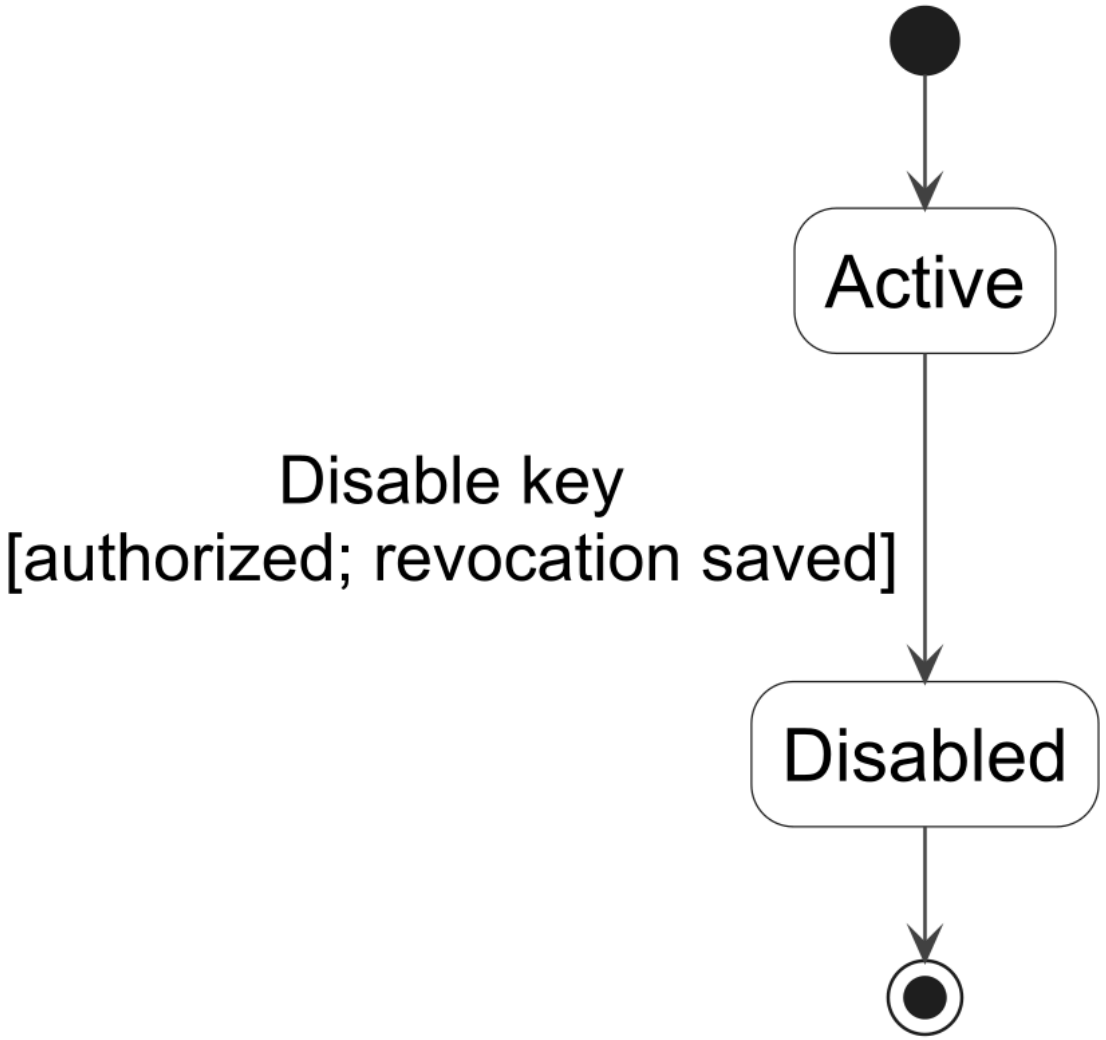


Figure 4.3.7.4.1. Workspace API Credential State Machine Diagram.

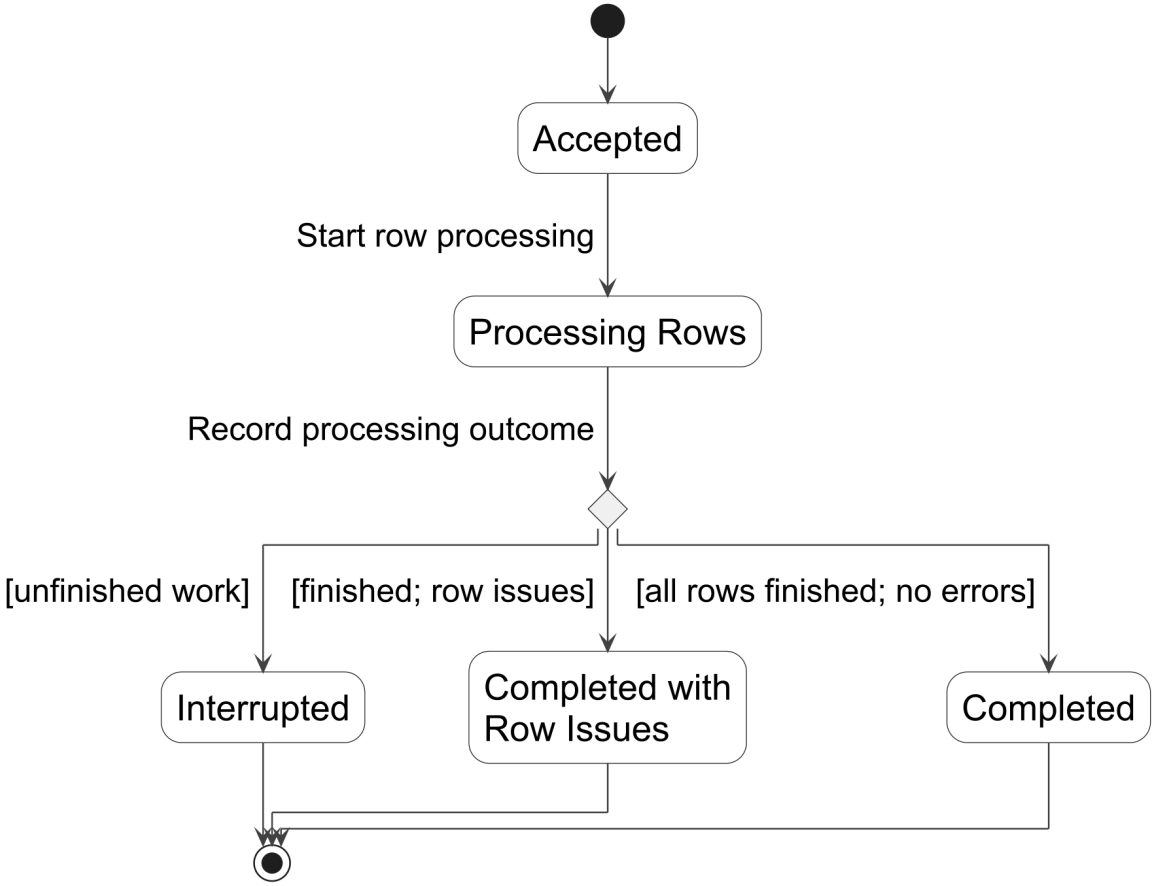


Figure 4.3.7.4.2. Production Import Batch State Machine Diagram.

SRS 3.8.6-3.8.7: one import run starts after SME Admin/Operator authorization, workspace activation and supported-file/required-column validation. Structurally rejected files never reach row processing or create production rows. CSV and Excel share the approved file mapping. Classify each row as accepted, duplicate, conflicting or invalid; totals reconcile without double counting. Process independent valid rows and send newly accepted transactions into detection. Completed means all rows were handled without invalid/conflicting rows; duplicates remain distinct from newly accepted rows. Row processing completion is not a claim that asynchronous detection has finished. Interrupted runs expose finished versus unfinished work. A controlled retry uses retained progress and workspace transaction identity rather than accepting completed rows twice. Final nodes end this processing run's outcome, not the retained batch reference, row summaries or accepted transactions. Safe result access never reveals unauthorized plaintext.

3.8 Rule-Based Detection

3.8.1 Class Diagram

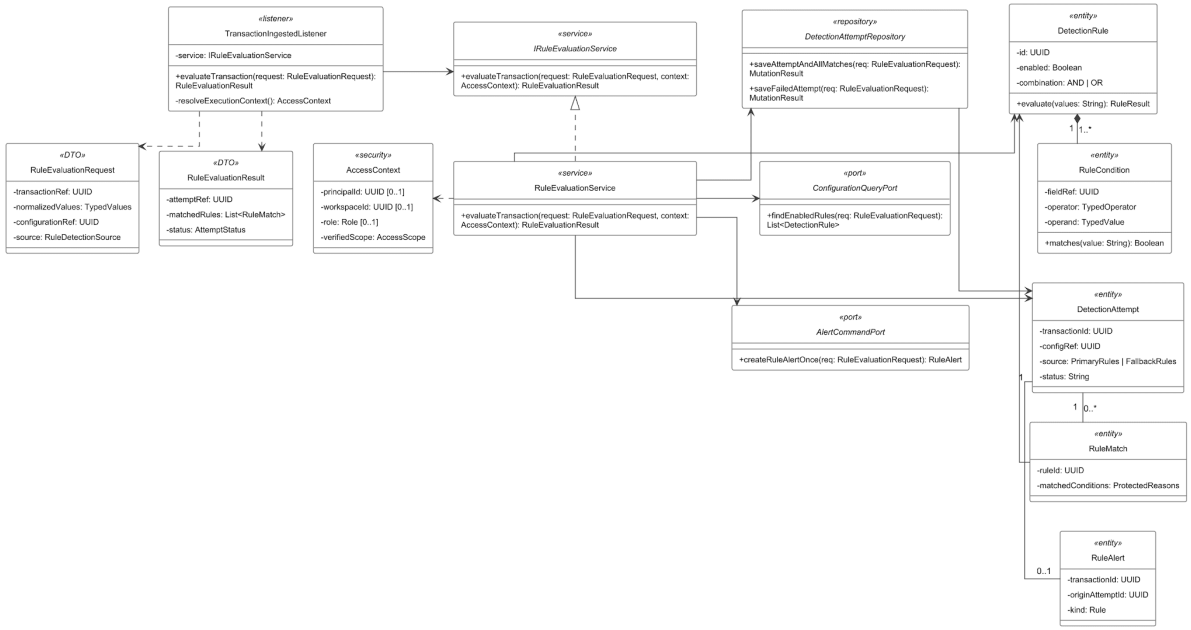


Figure 4.3.8.1.1. Rule-Based Detection Class Diagram.

3.8.2 Sequence Diagrams

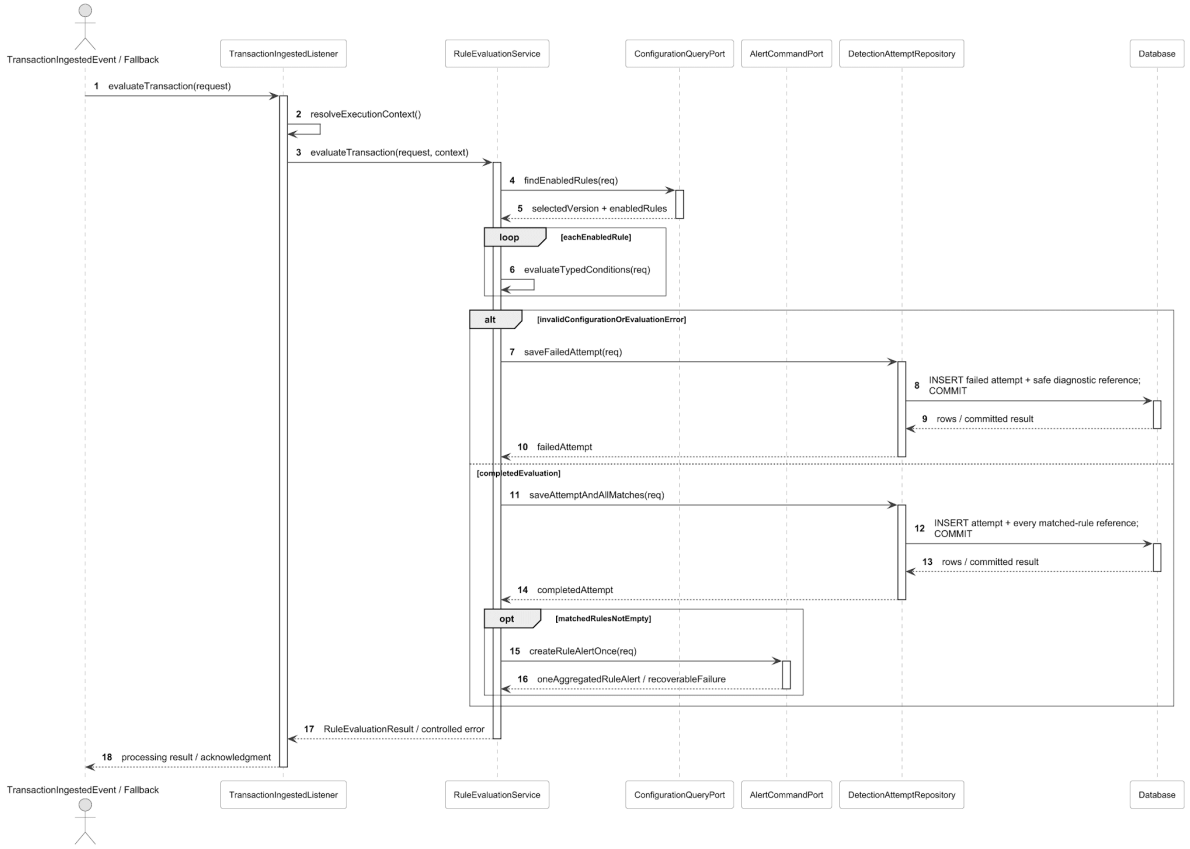


Figure 4.3.8.2.1. Evaluate All Enabled Rules and Create One Aggregated Alert Sequence Diagram.

3.9 AI Detection and Rule-Based Fallback

3.9.1 Class Diagram

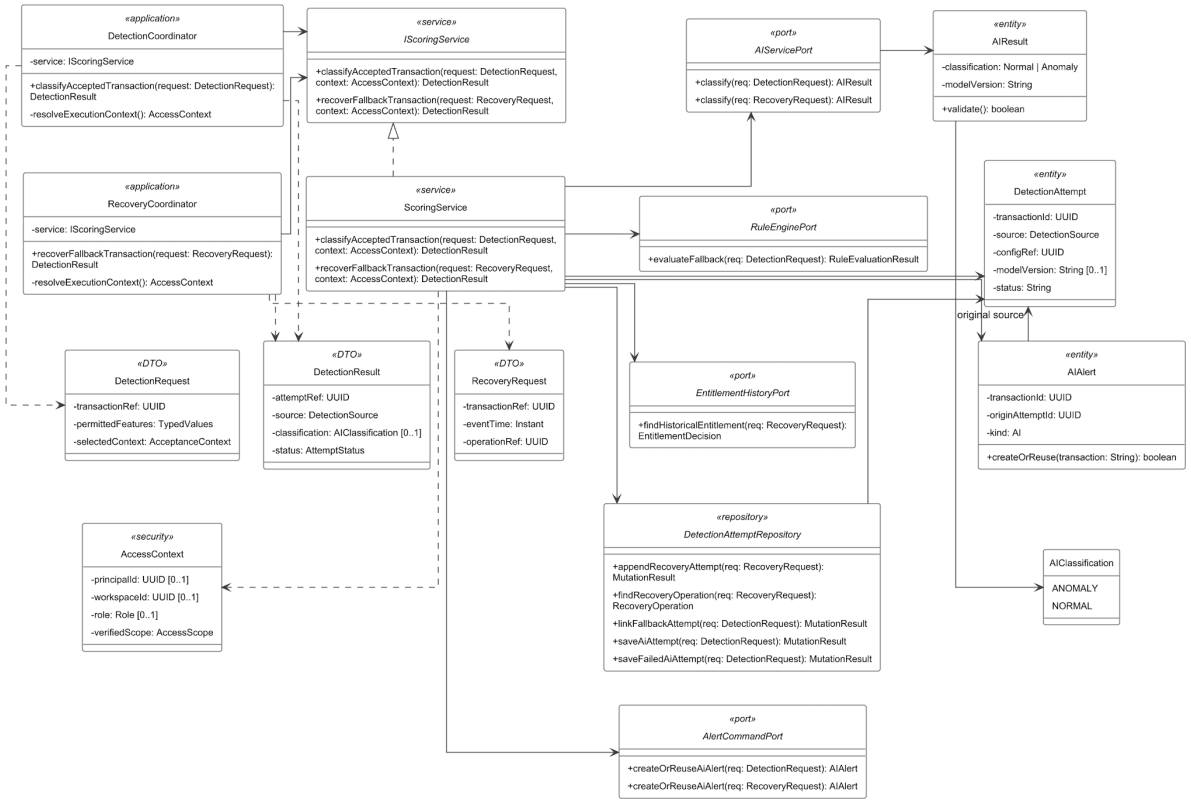


Figure 4.3.9.1.1. AI Detection and Rule-Based Fallback Class Diagram.

3.9.2 Sequence Diagrams

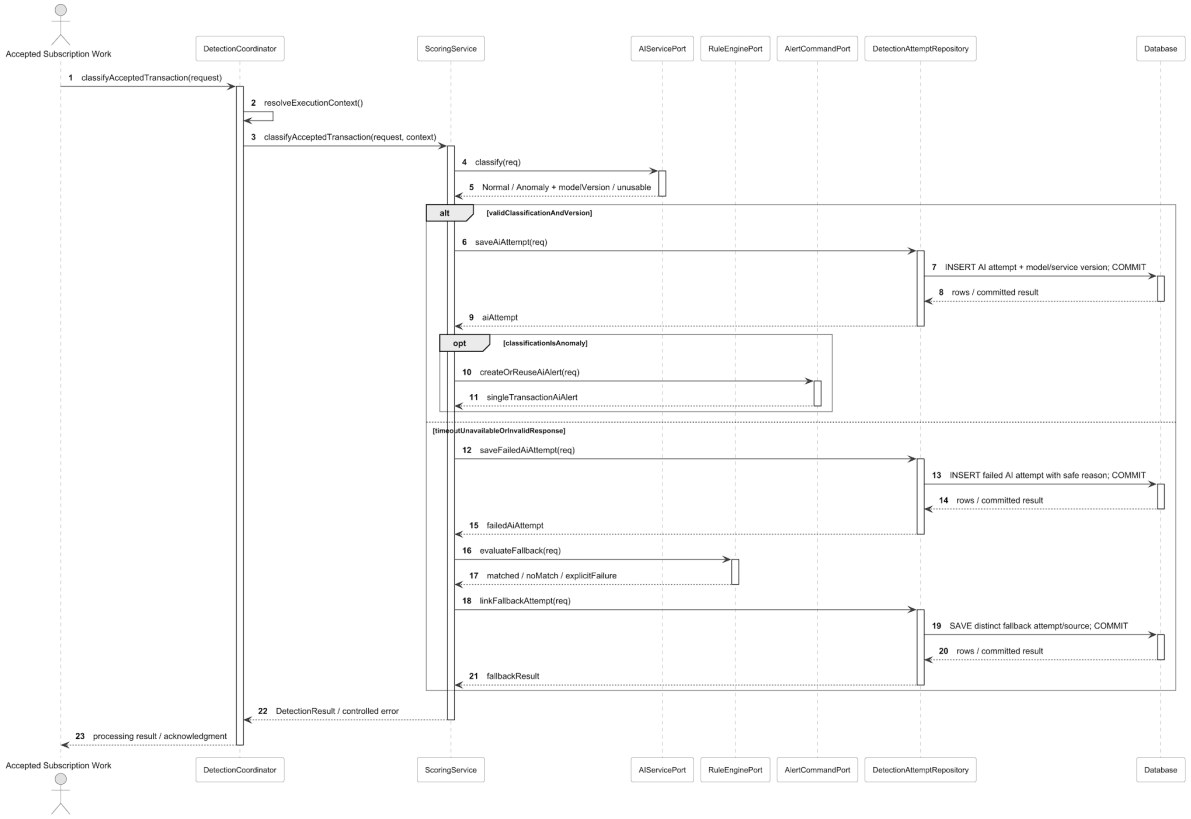


Figure 4.3.9.2.1. Classify Subscription Transaction and Fall Back on Unusable AI Sequence Diagram.

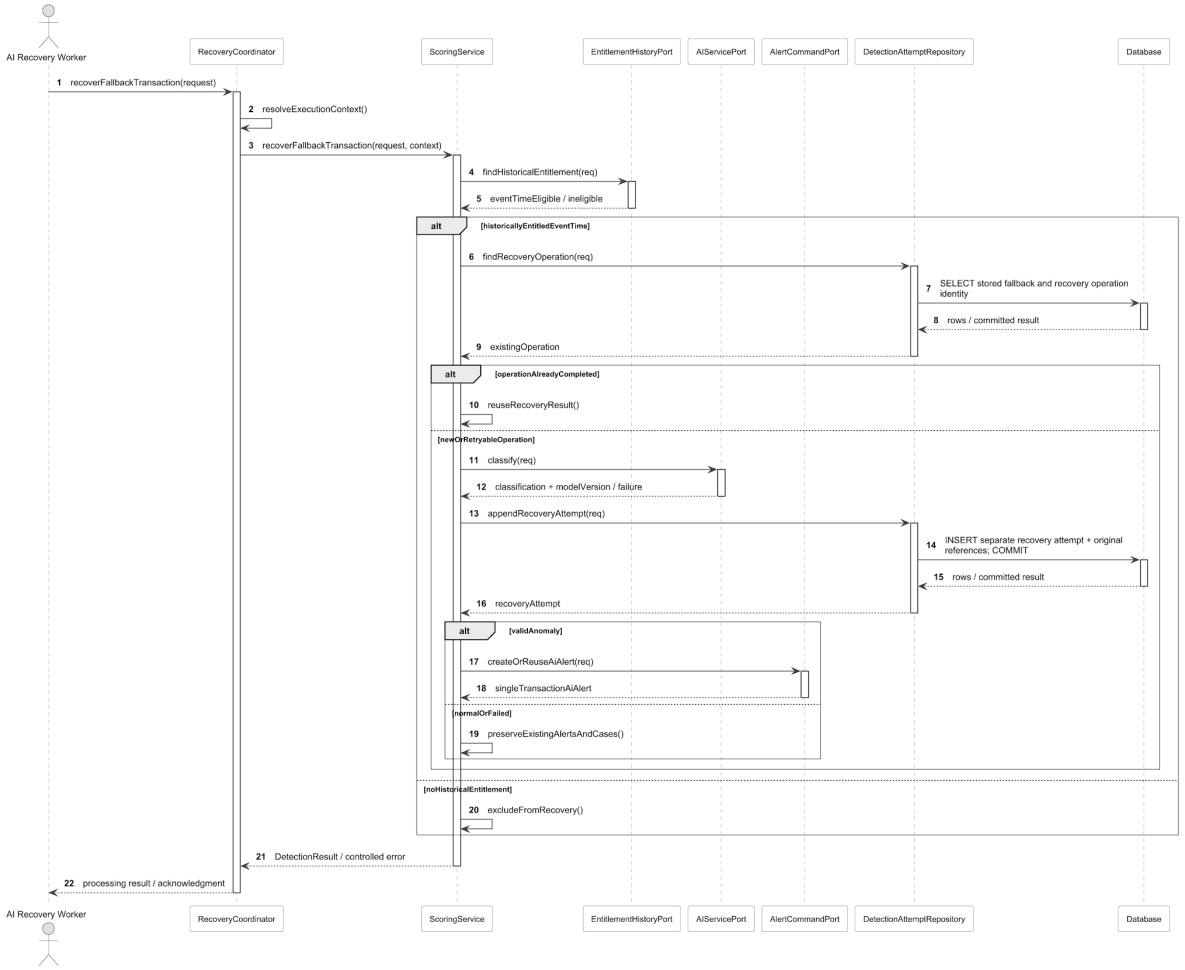


Figure 4.3.9.2.2. Recover Historically Eligible Fallback Work and Reconcile AI Alerts Sequence Diagram.

3.9.3 Activity Diagrams

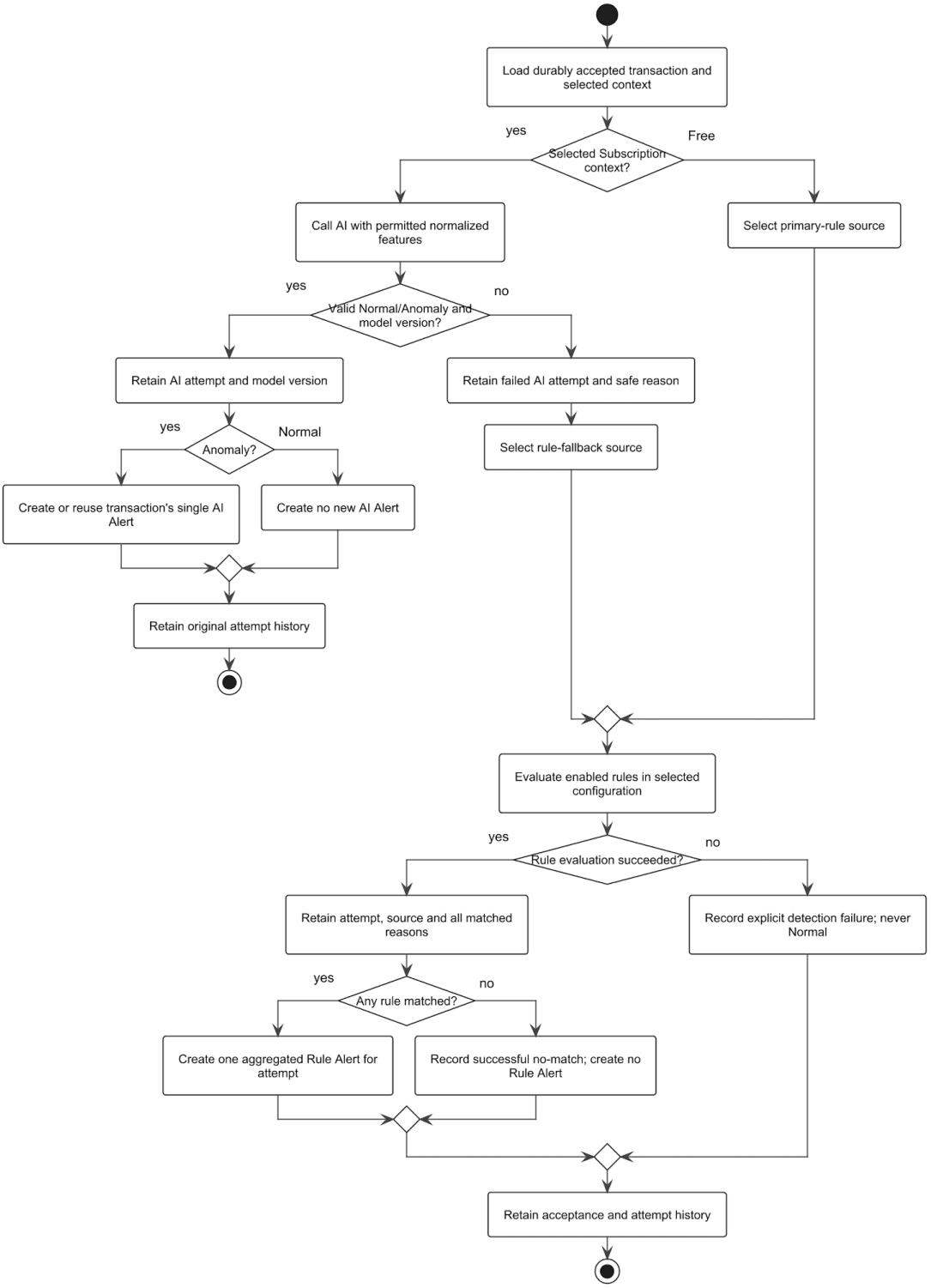


Figure 4.3.9.3.1. Transaction Detection and Rule-Based Fallback Processing Activity Diagram.

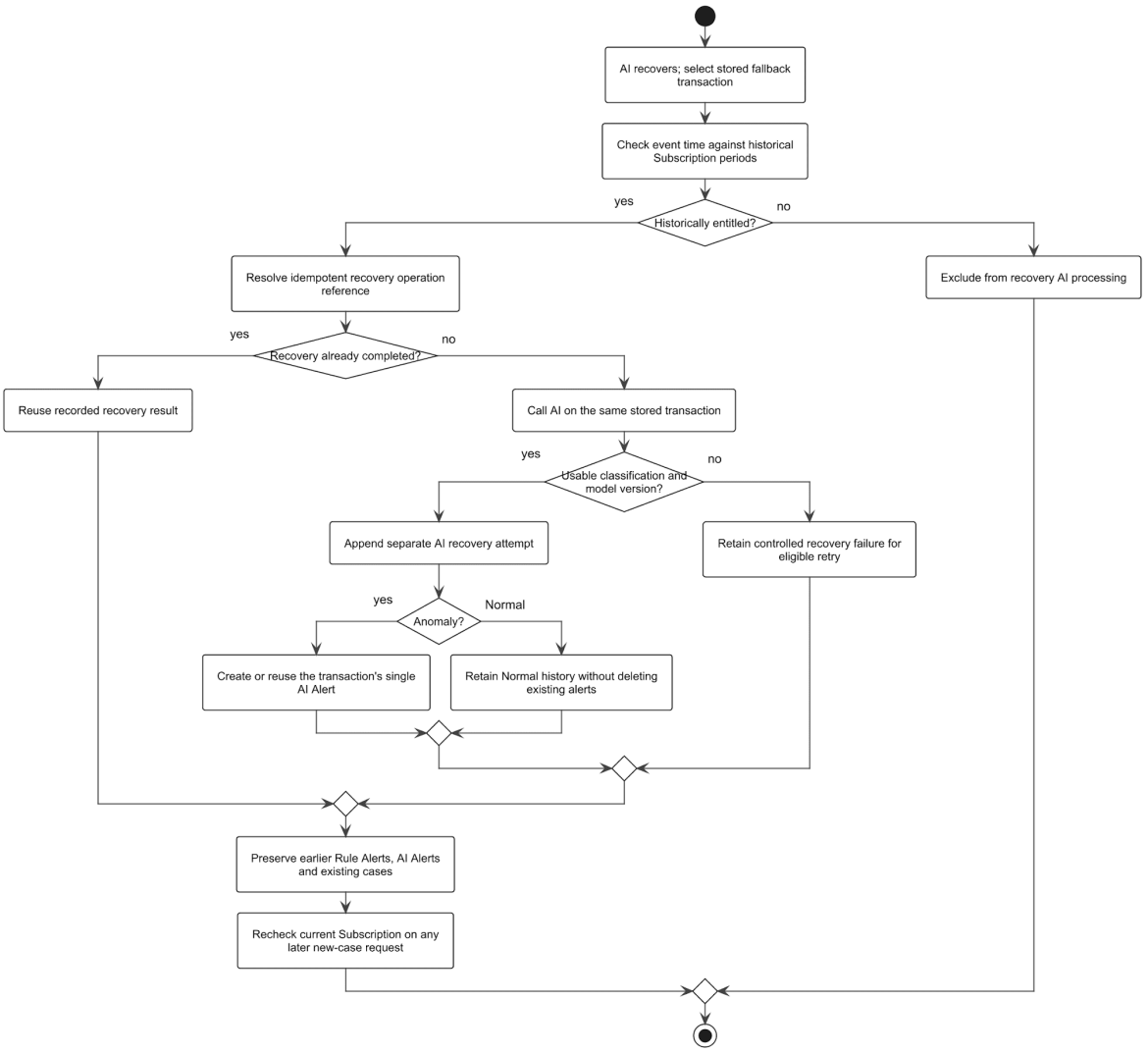


Figure 4.3.9.3.2. Historically Entitled AI Recovery Processing Activity Diagram.

3.9.4 State Machine Diagrams

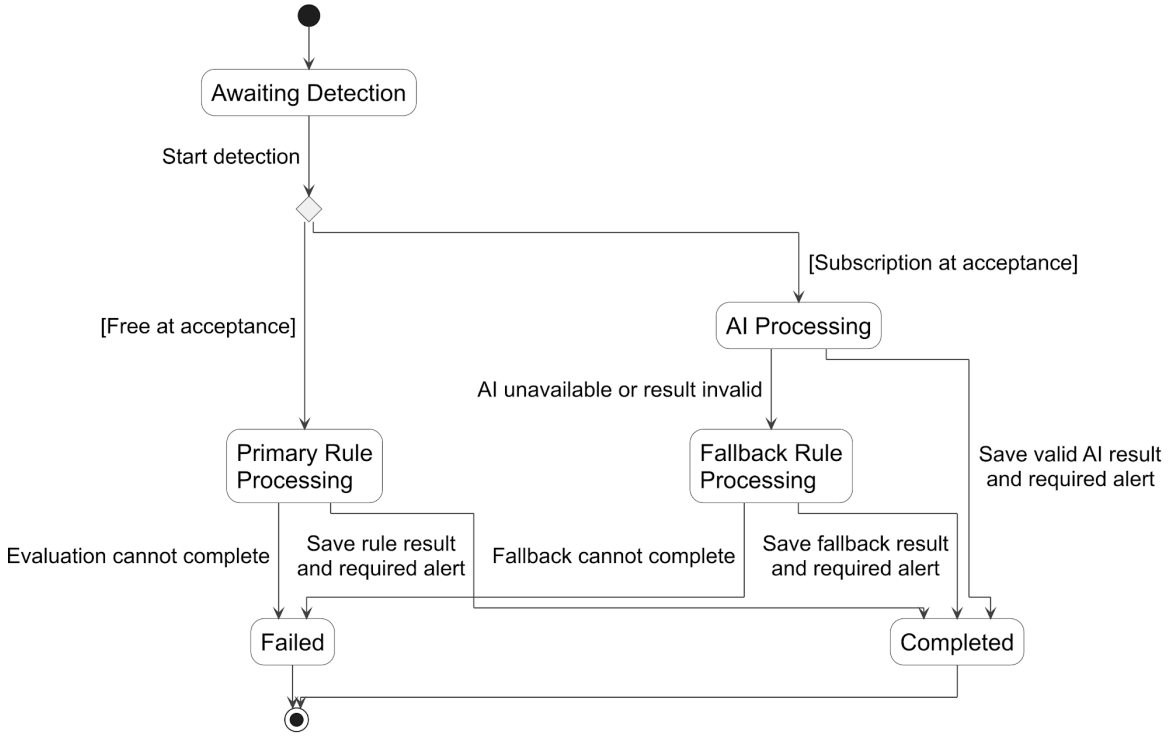


Figure 4.3.9.4.1. Initial Transaction Detection State Machine Diagram.

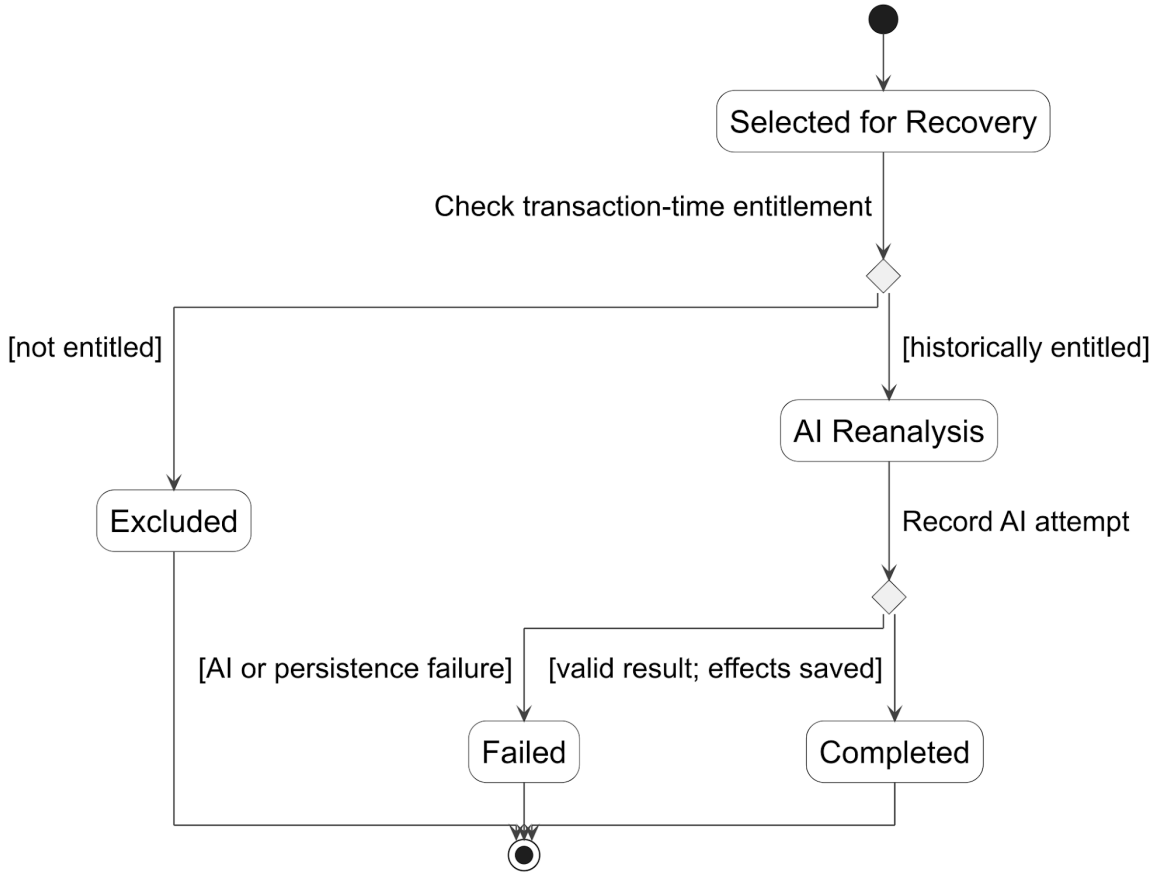


Figure 4.3.9.4.2. AI Recovery Operation State Machine Diagram.

SRS 3.10.3: one recovery operation for an existing fallback-processed transaction. Eligibility uses the transaction event time and historical Subscription validity, even if the current plan expired. Recovery is not new ingestion and cannot open a case automatically. Store a separate attempt/model version, preserve prior fallback and Rule Alert history, and reconcile effects idempotently. Anomaly creates/reuses the transaction's single AI Alert; a later Normal result never deletes an existing alert or case. Repeated execution of the same operation must not duplicate effects. Failed means this operation cannot complete; a later controlled recovery operation keeps its own identity/history instead of rewriting this outcome. Excluded ends selection without AI execution. No timeout, retry schedule or external wire schema is invented.

3.10 Transaction and Alert Review

3.10.1 Class Diagram

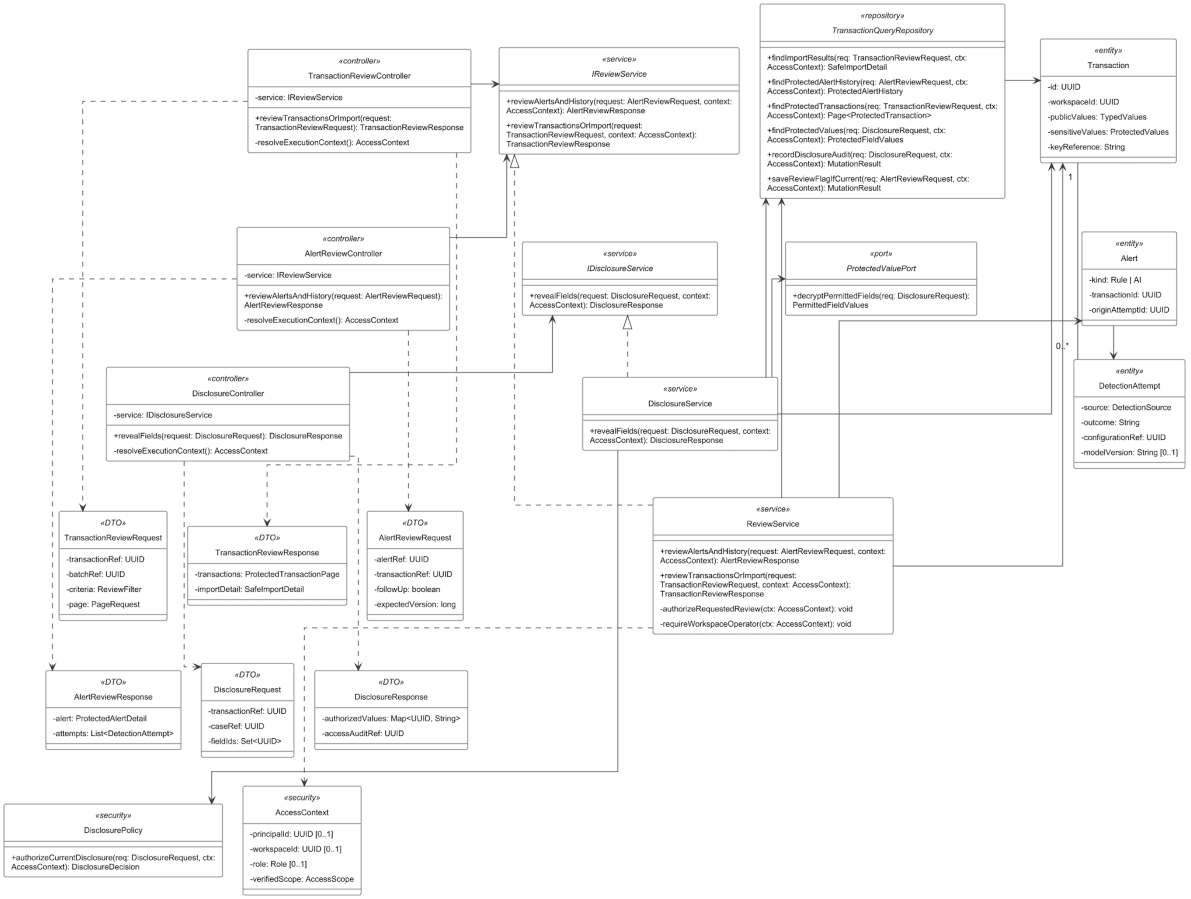


Figure 4.3.10.1.1. Transaction and Alert Review Class Diagram.

3.10.2 Sequence Diagrams

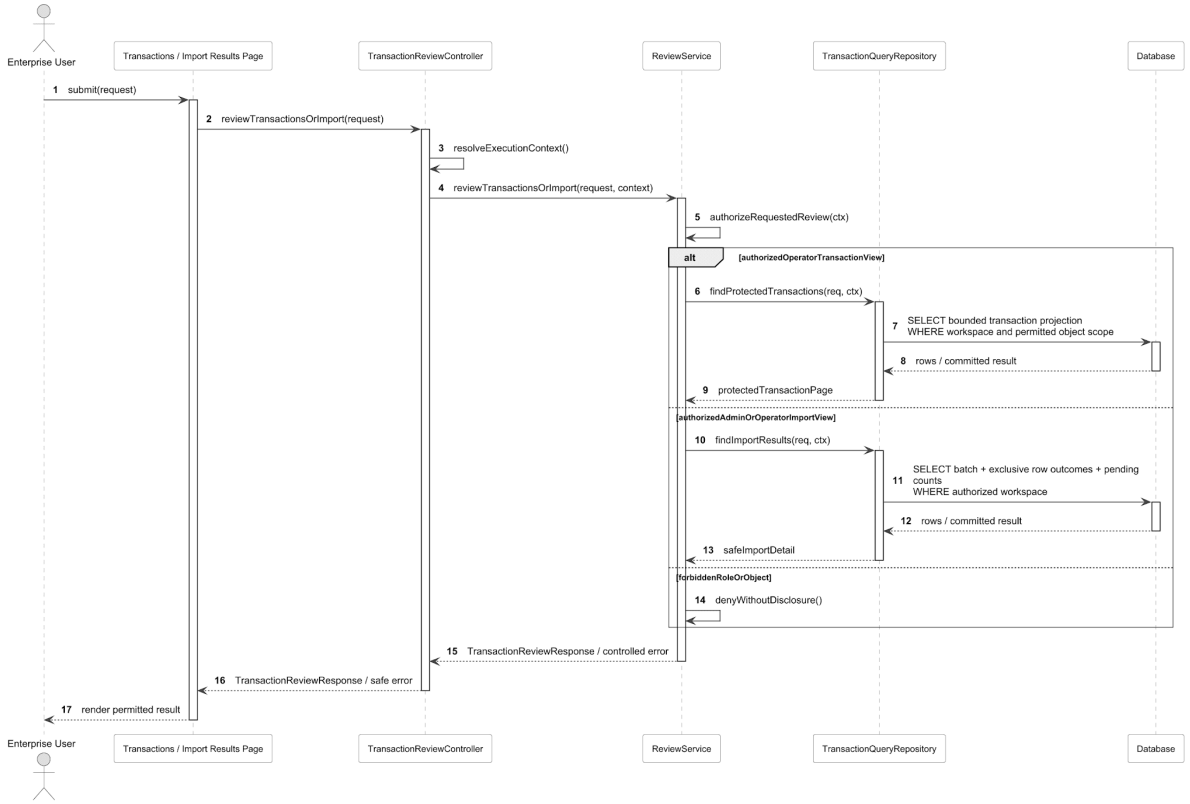


Figure 4.3.10.2.1. Review Transactions and Role-Scoped Import Results Sequence Diagram.

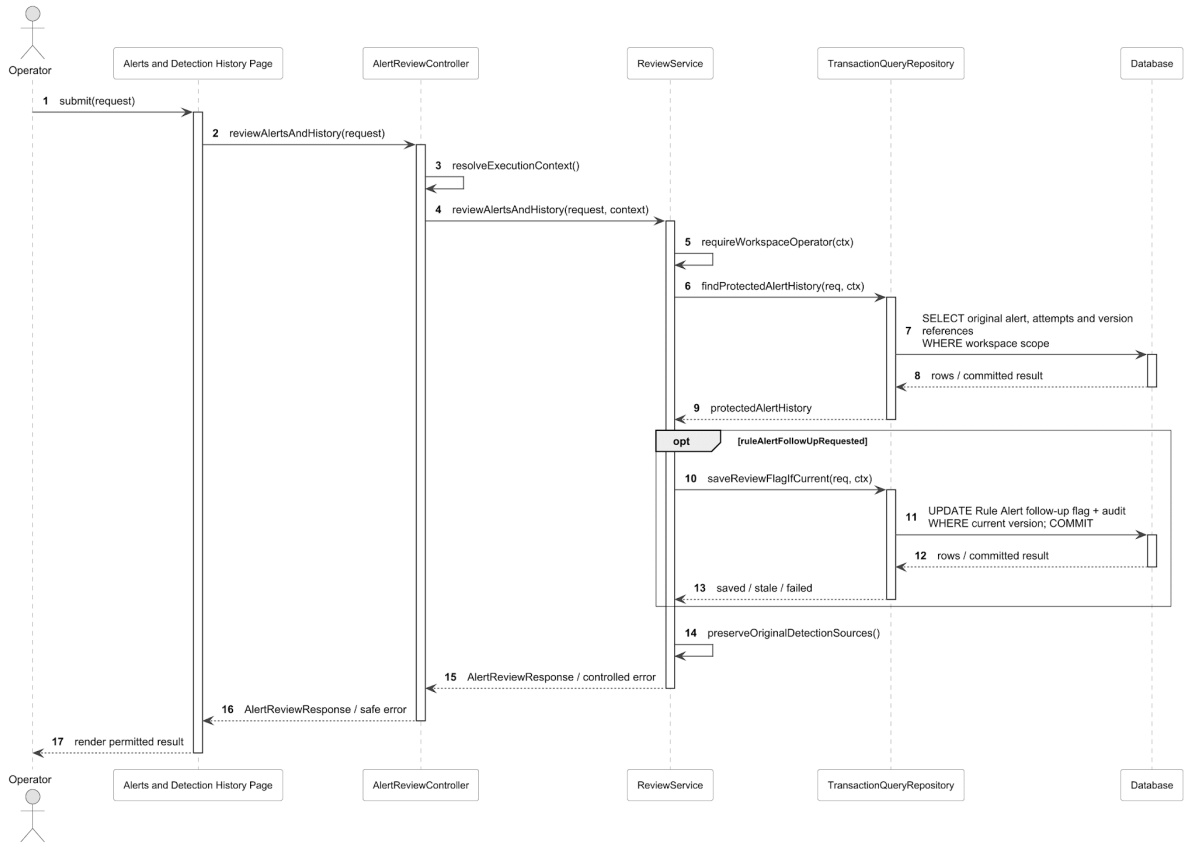


Figure 4.3.10.2.2. Review Rule/AI Alerts, Follow-Up Flags and Attempt History Sequence Diagram.

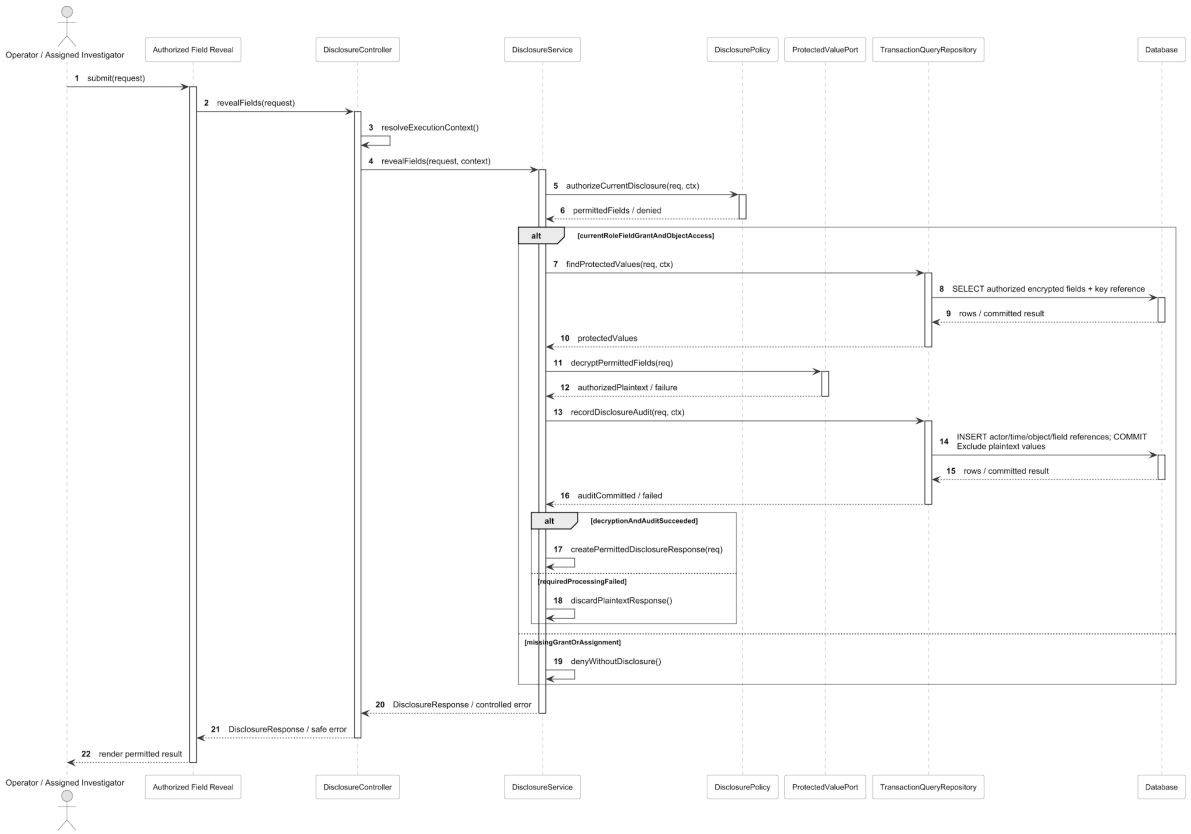


Figure 4.3.10.2.3. Explicitly Reveal Only Currently Authorized Sensitive Fields Sequence Diagram.

3.11 Case and Investigation Management

3.11.1 Class Diagram

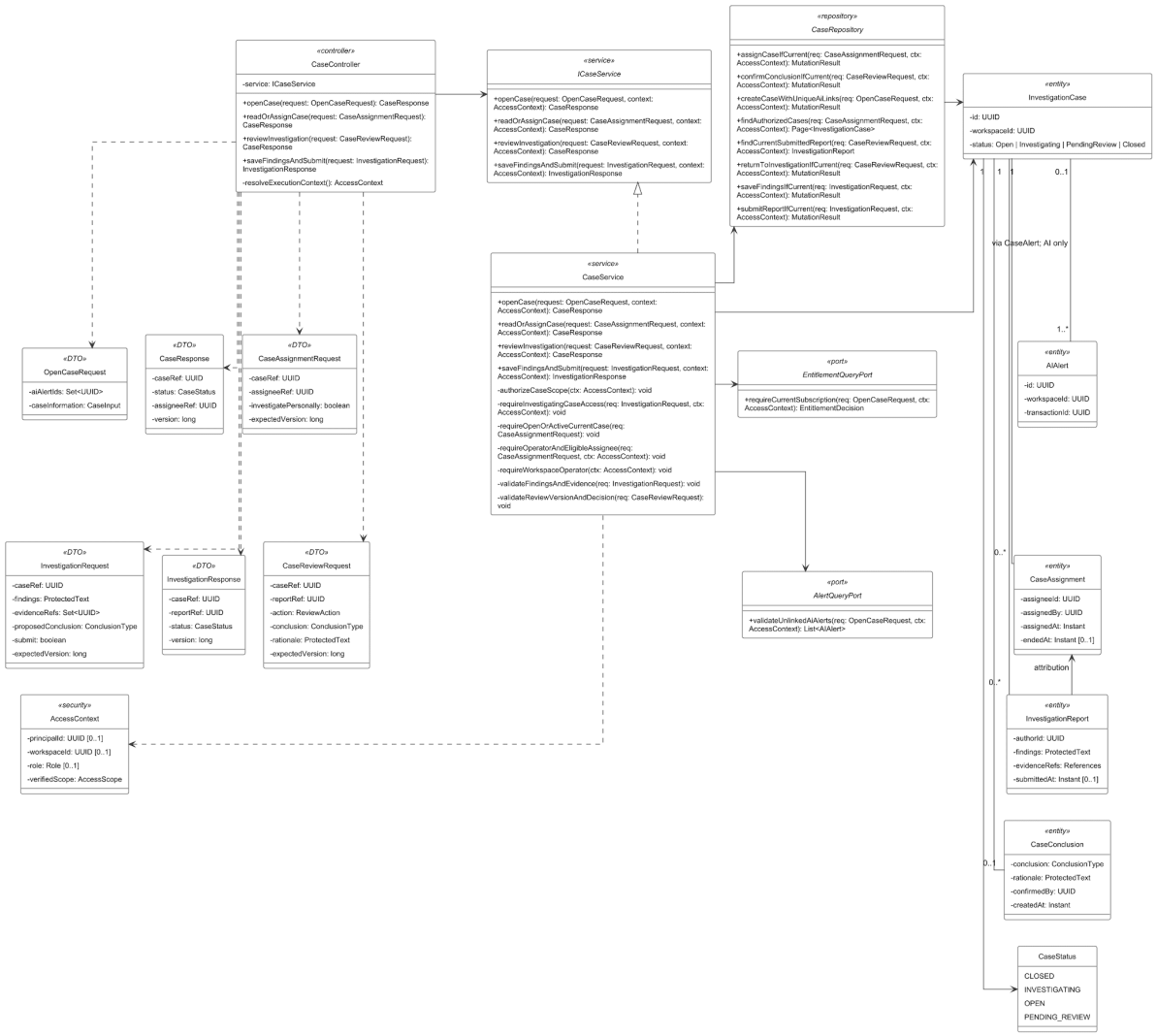


Figure 4.3.11.1.1. Case and Investigation Management Class Diagram.

3.11.2 Sequence Diagrams

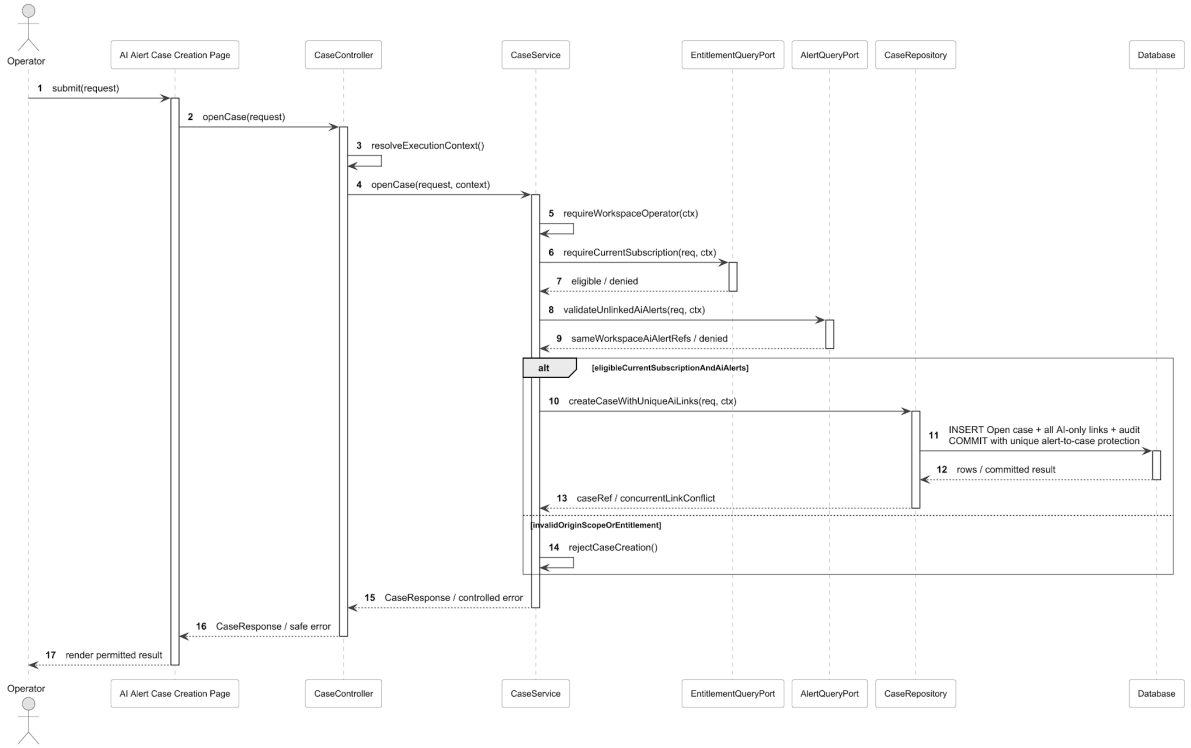


Figure 4.3.11.2.1. Open One Case from Eligible AI Alert References Sequence Diagram.

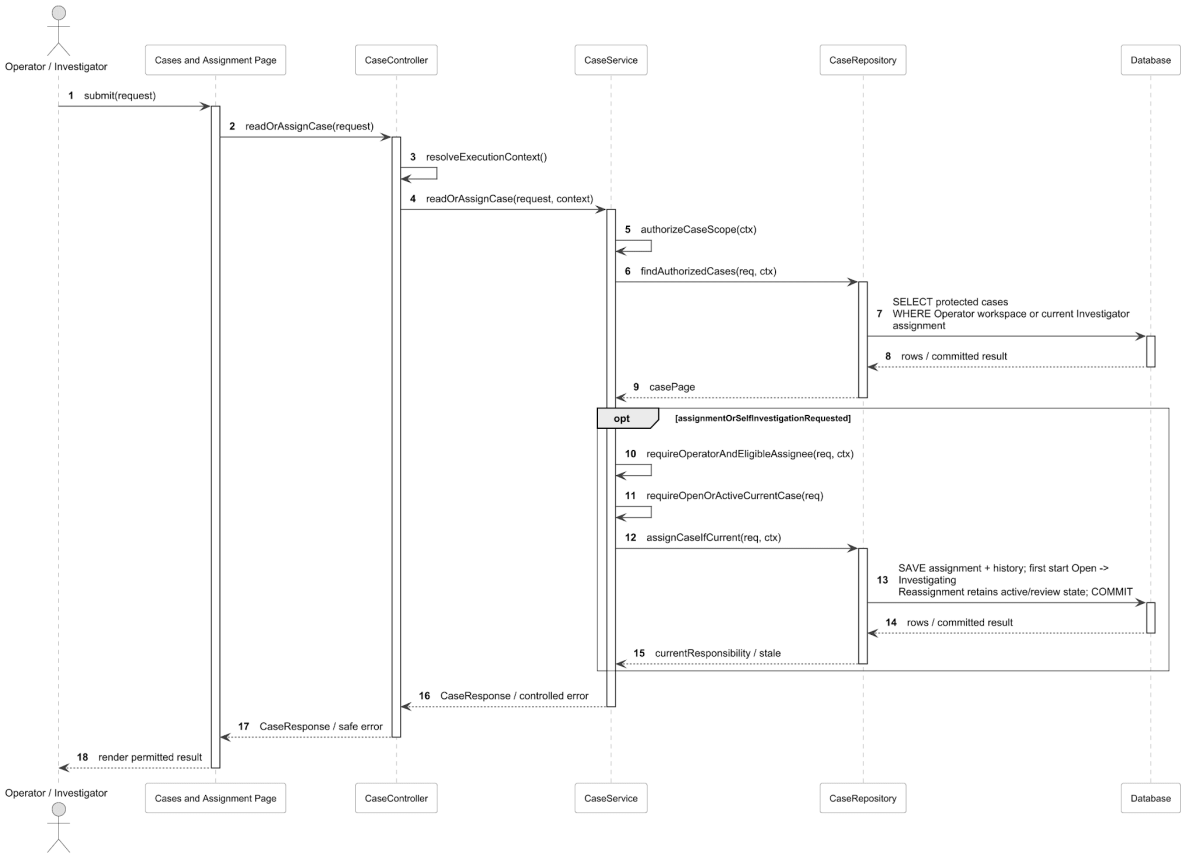


Figure 4.3.11.2.2. View Cases and Assign or Start Self-Investigation Sequence Diagram.

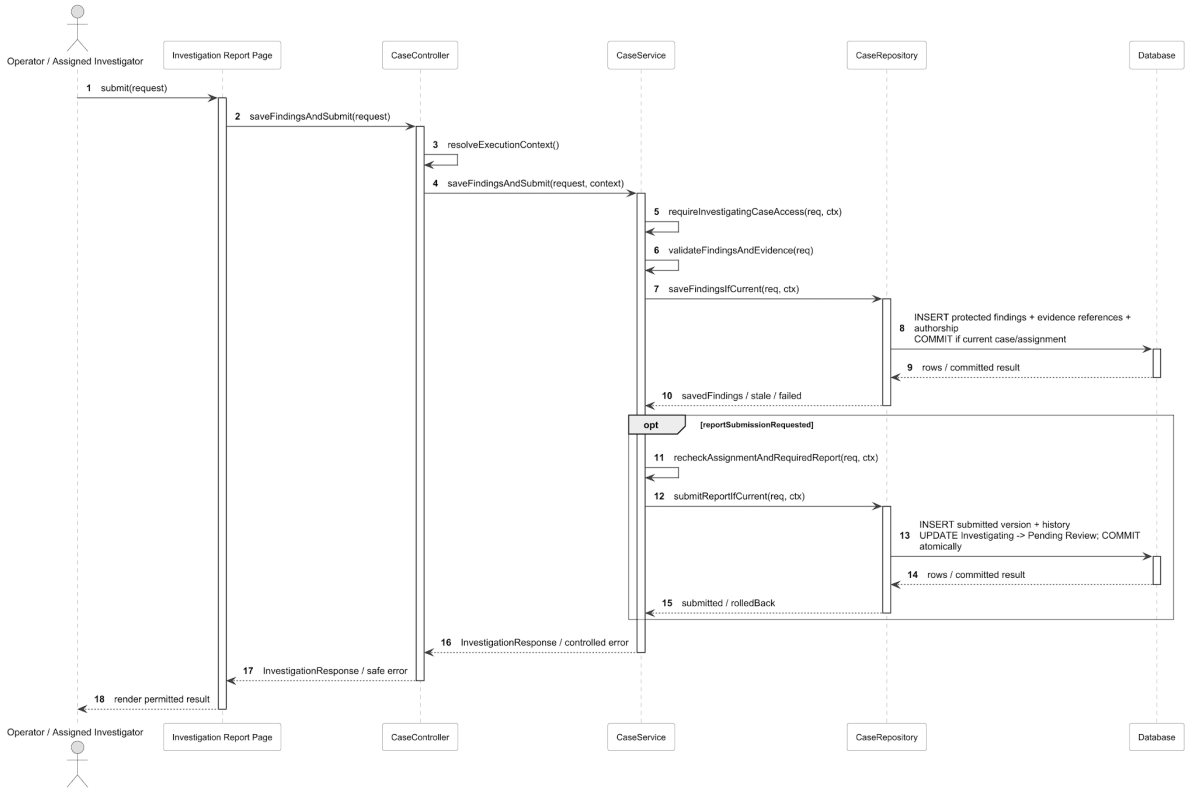


Figure 4.3.11.2.3. Record Findings and Submit an Investigation Report Sequence Diagram.

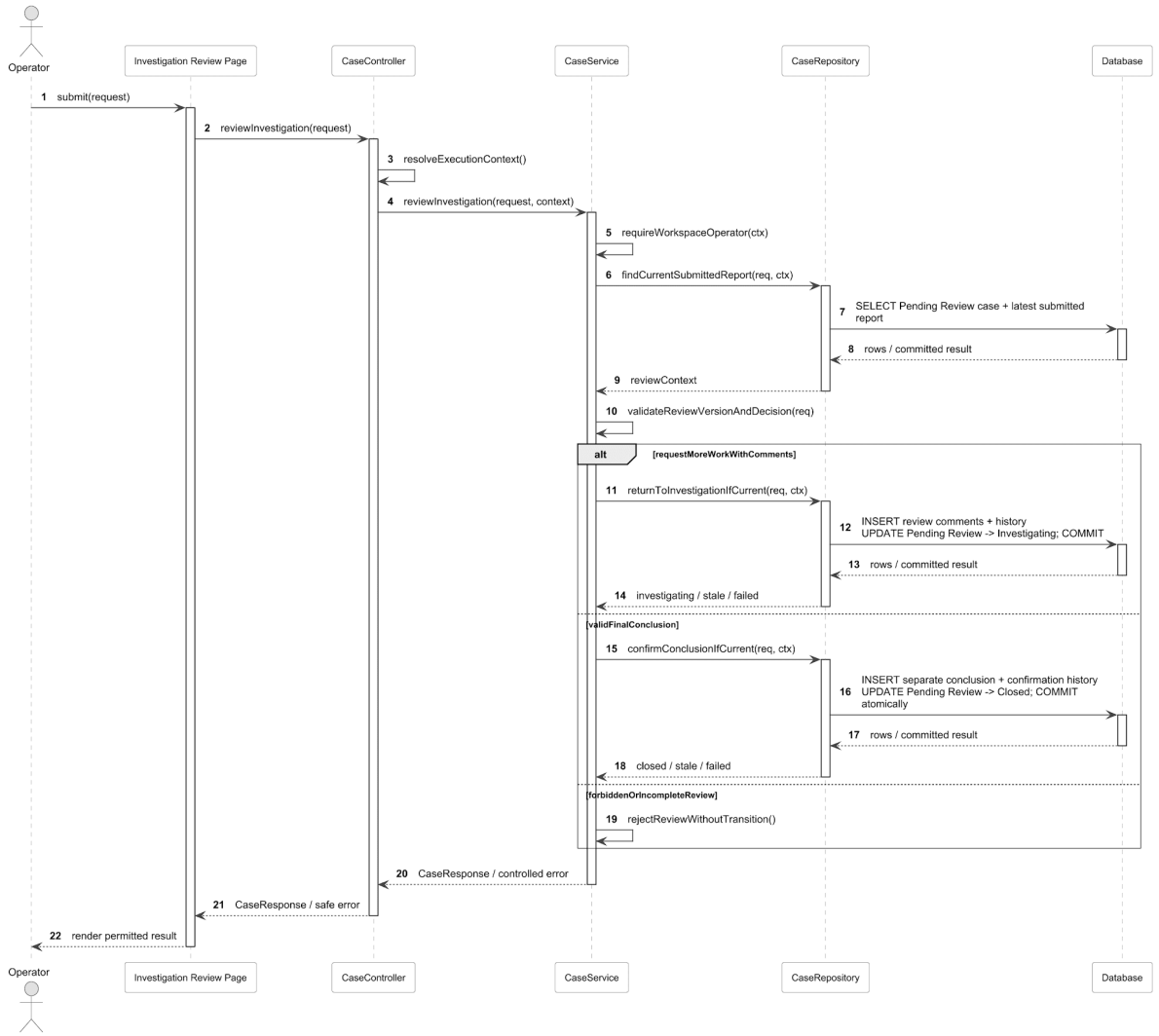


Figure 4.3.11.2.4. Request Additional Investigation or Confirm the Final Conclusion Sequence Diagram.

3.11.3 Activity Diagrams

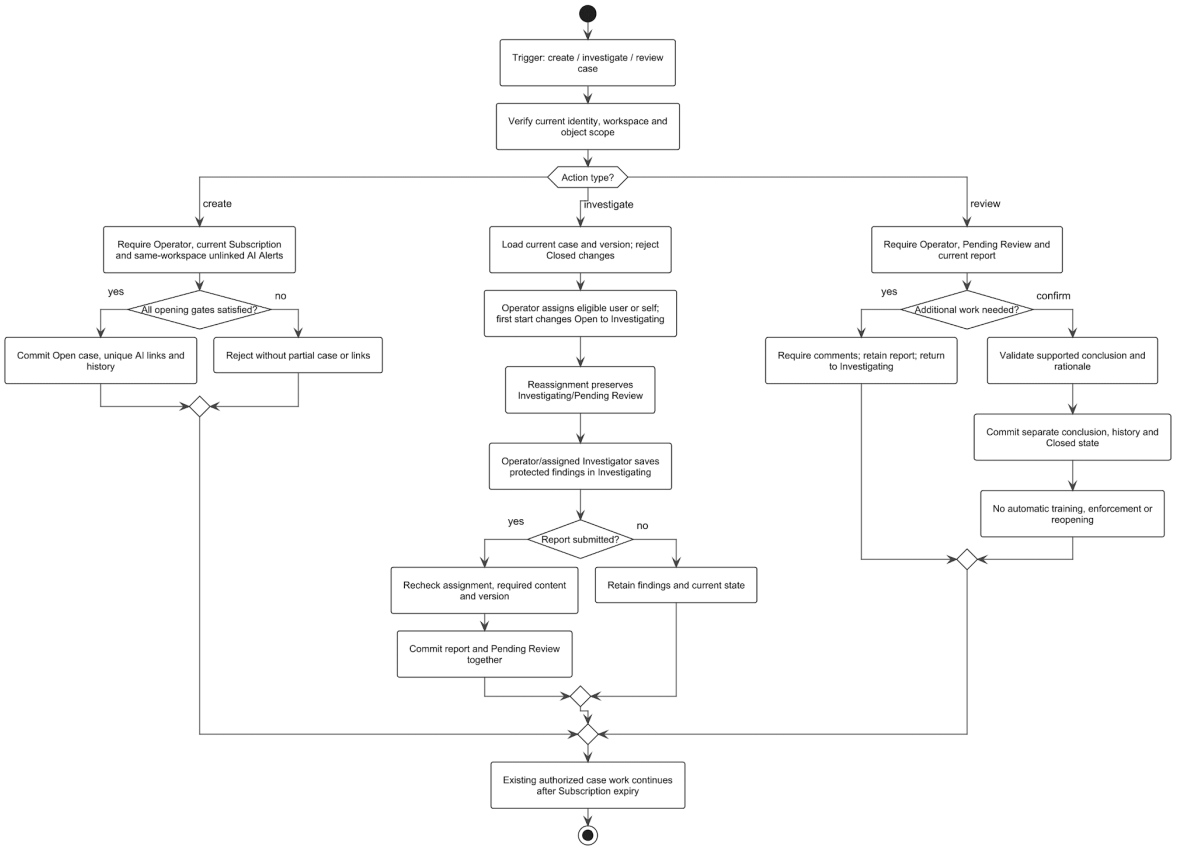


Figure 4.3.11.3.1. Investigation Case Lifecycle Activity Diagram.

3.11.4 State Machine Diagrams

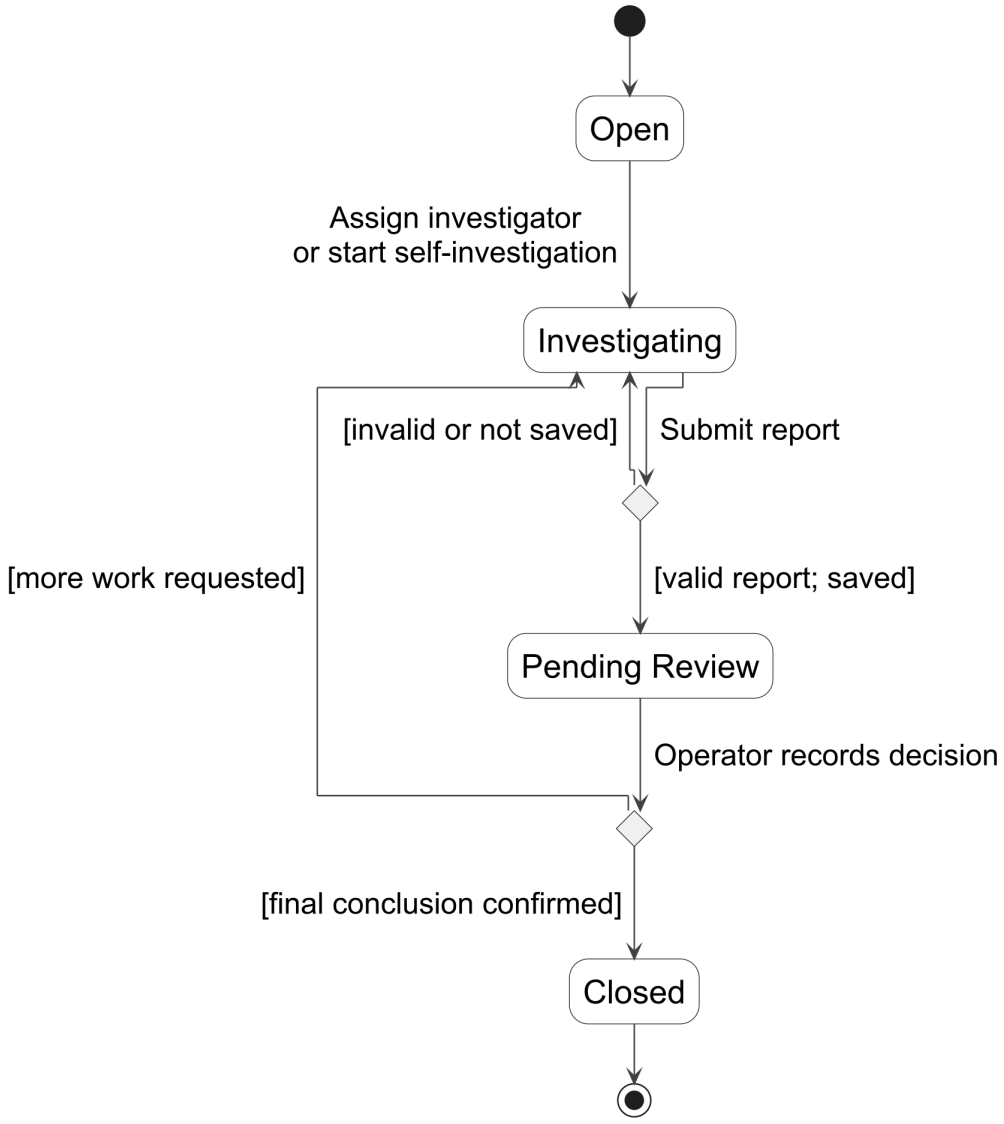


Figure 4.3.11.4.1. Investigation Case State Machine Diagram.

Operator creates an Open case under active Subscription, using eligible, unlinked same-workspace AI Alerts. Operator may assign an eligible Investigator or investigate personally. Operator investigating directly or assigned Investigator submits findings, evidence and a proposed conclusion. Recheck content, assignment and version, then commit report/status/history together. Invalid or failed submissions keep Investigating. Only Operator requests more work with comments or confirms a supported conclusion with its rationale. Invalid review actions or failed persistence keep Pending Review. Findings and reassignment retain current status and attribution; a second reviewer is not required. Conclusion is separate from Closed: Confirmed Fraud, Anomalous but Legitimate, False Alert or Insufficient Evidence. Existing cases survive Subscription expiry. The final node completes handling, retaining history without reopening, automatic AI training or customer-system enforcement.

3.12 Dashboard and Reporting

3.12.1 Class Diagram

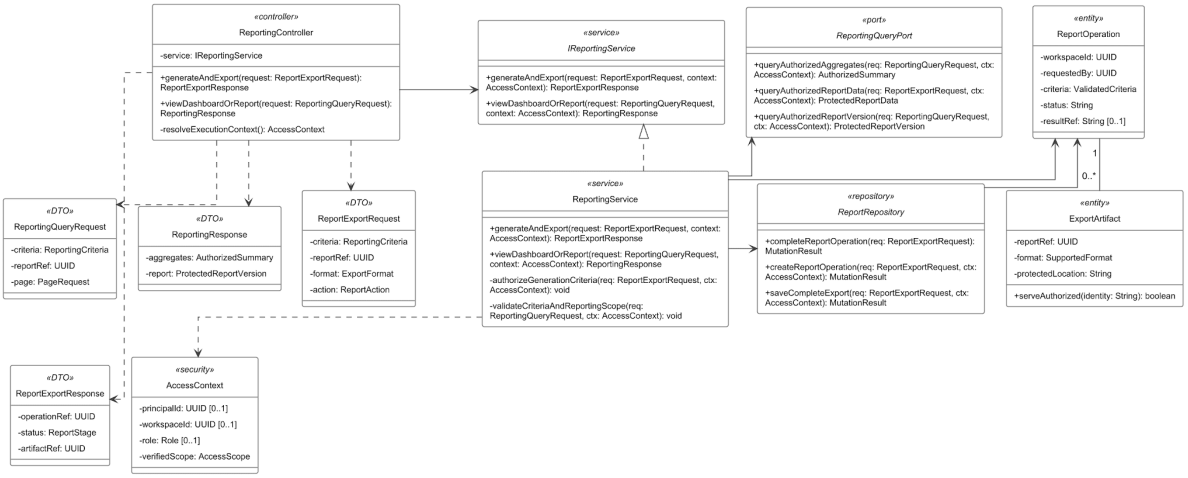


Figure 4.3.12.1.1. Dashboard and Reporting Class Diagram.

3.12.2 Sequence Diagrams

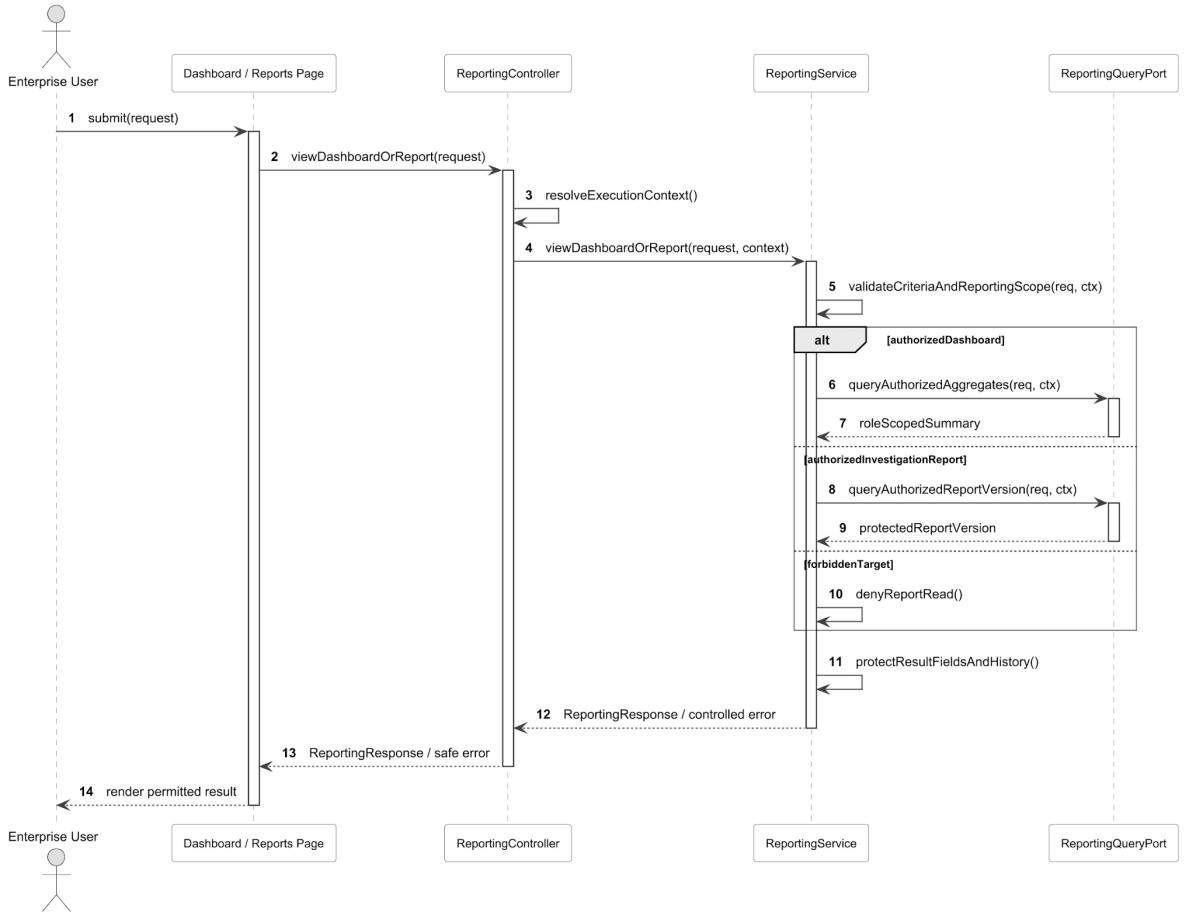


Figure 4.3.12.2.1. View Authorized Dashboard and Investigation Reports Sequence Diagram.

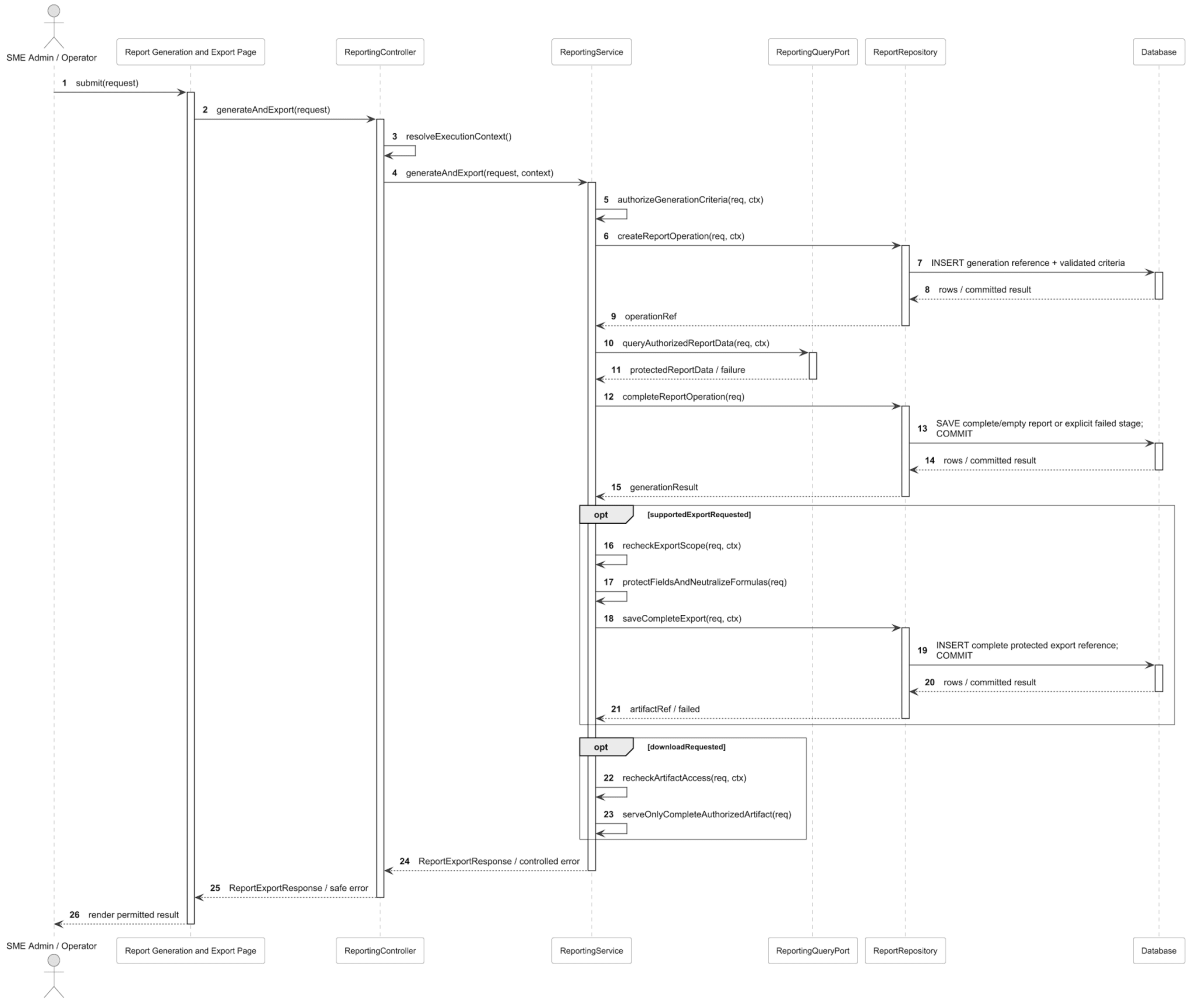


Figure 4.3.12.2.2. Generate and Export a Complete Authorized Enterprise Report Sequence Diagram.

3.12.3 Activity Diagrams

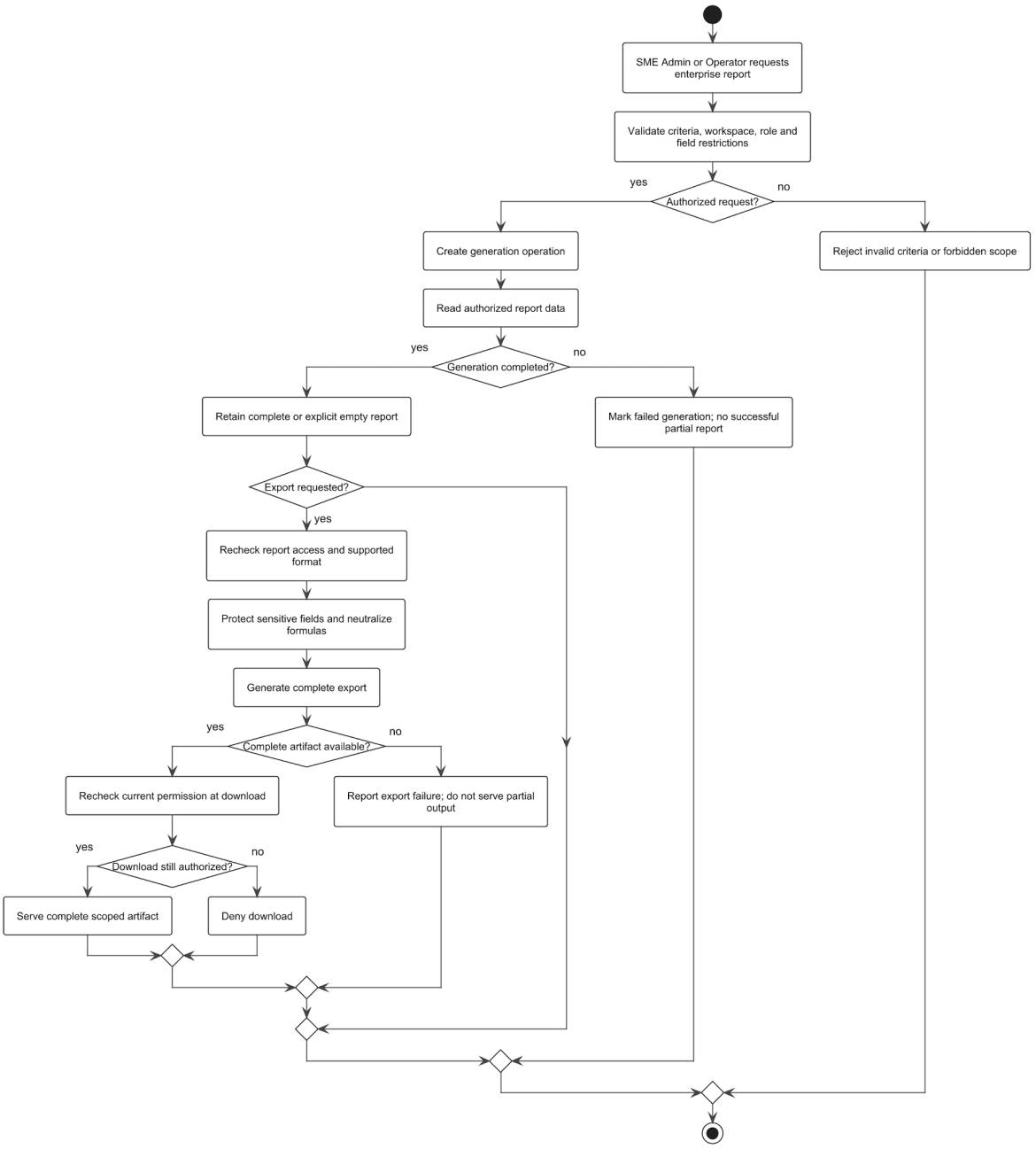


Figure 4.3.12.3.1. Enterprise Report Generation Lifecycle Activity Diagram.

3.12.4 State Machine Diagrams

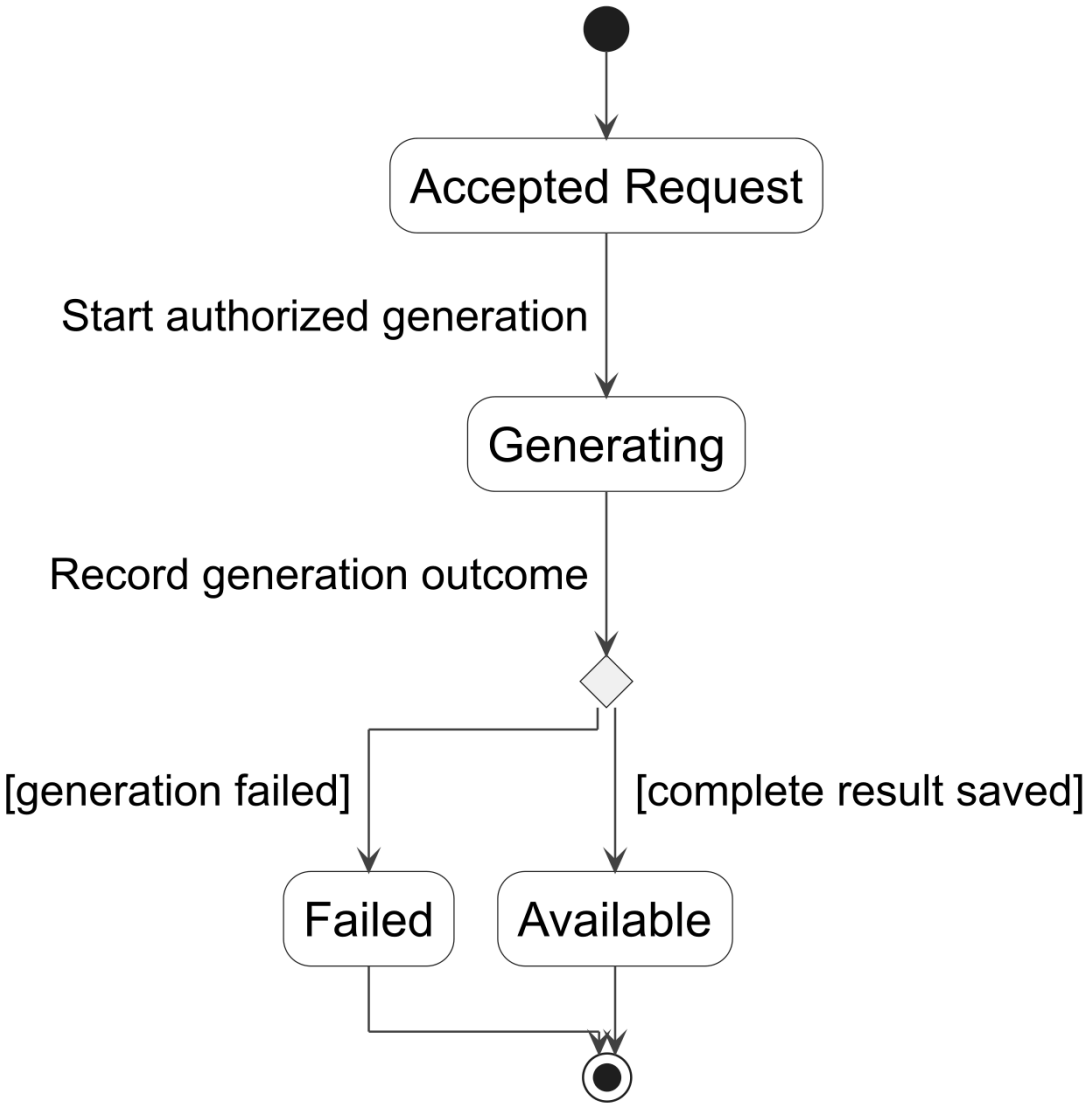


Figure 4.3.12.4.1. Enterprise Report Generation State Machine Diagram.

3.13 Notifications

3.13.1 Class Diagram

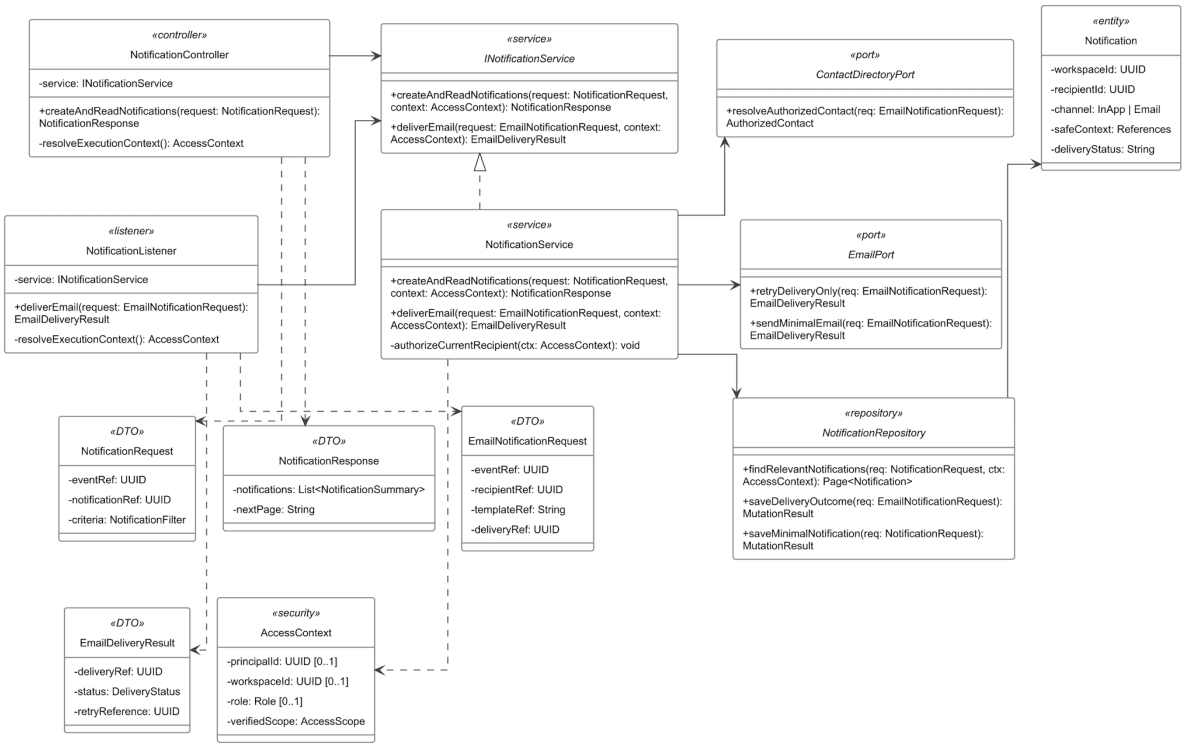


Figure 4.3.13.1.1. Notifications Class Diagram.

3.13.2 Sequence Diagrams

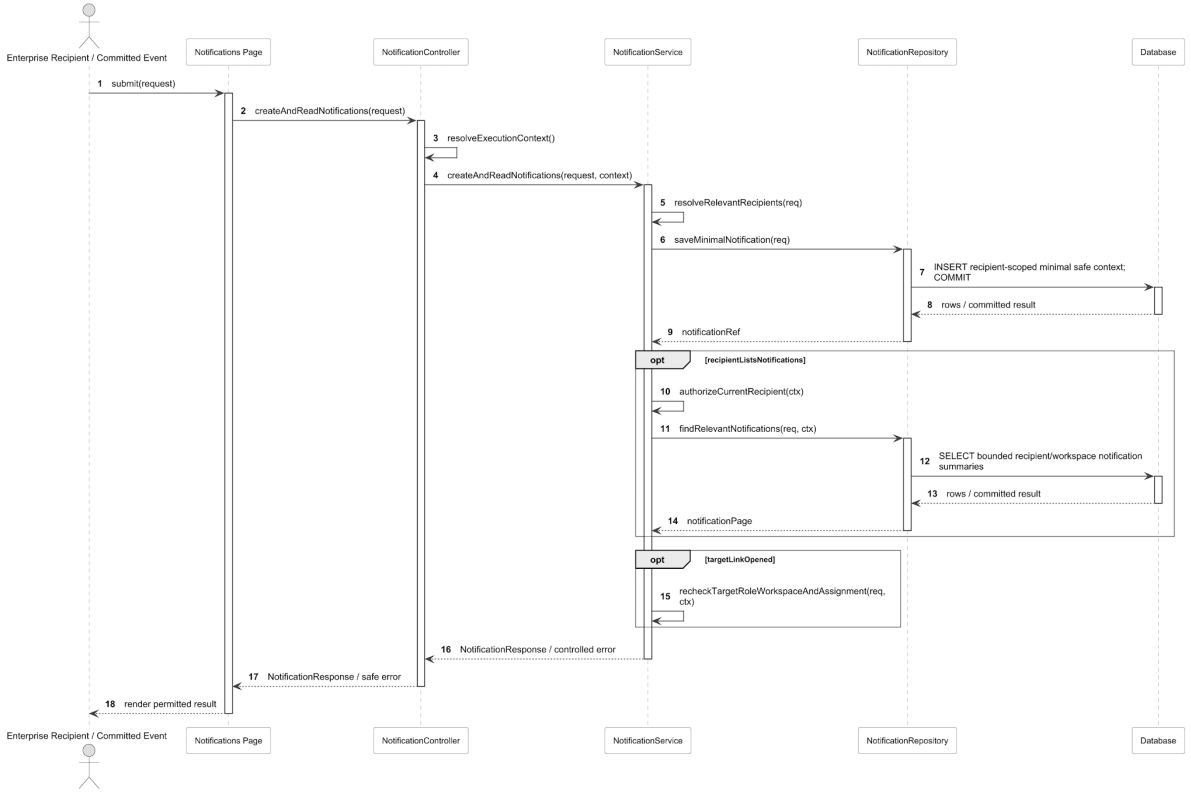


Figure 4.3.13.2.1. Create and View Relevant In-App Notifications Sequence Diagram.

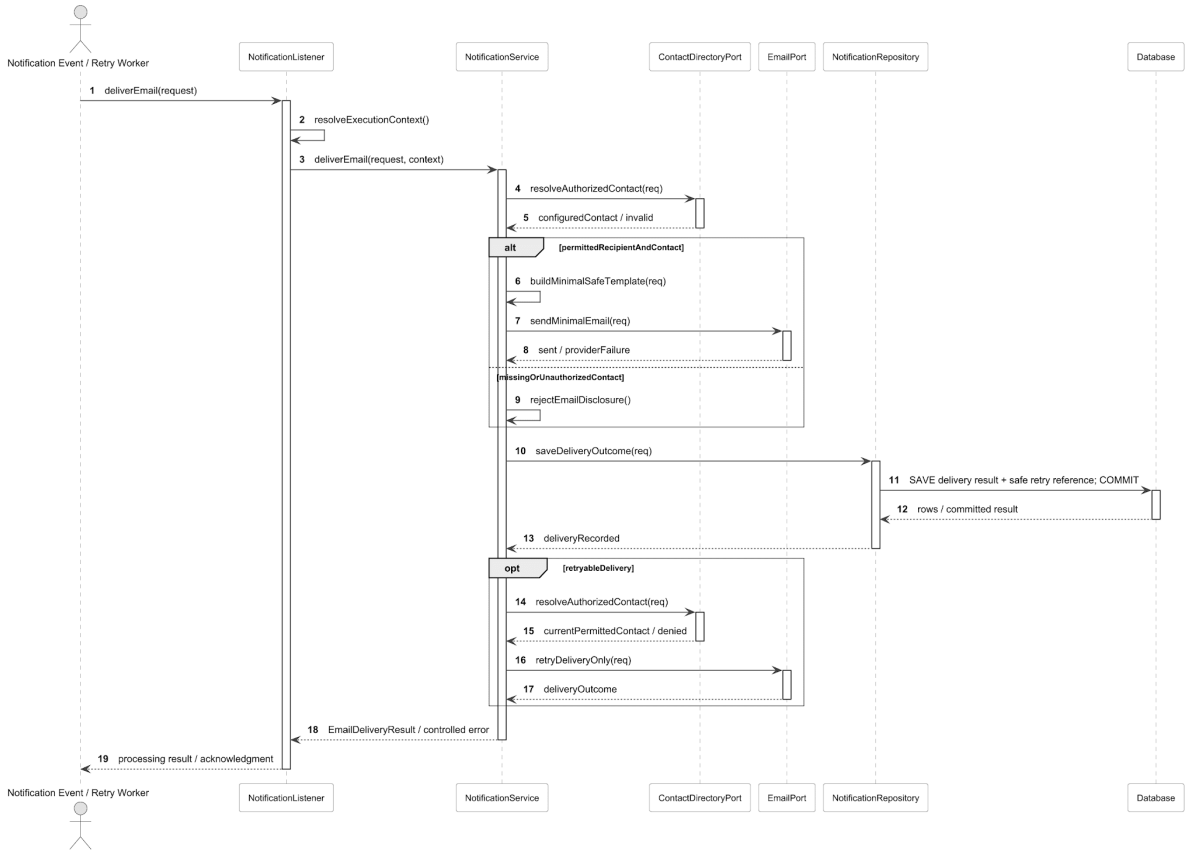


Figure 4.3.13.2.2. Send Minimal Email and Retry Delivery Independently Sequence Diagram.

3.13.4 State Machine Diagrams

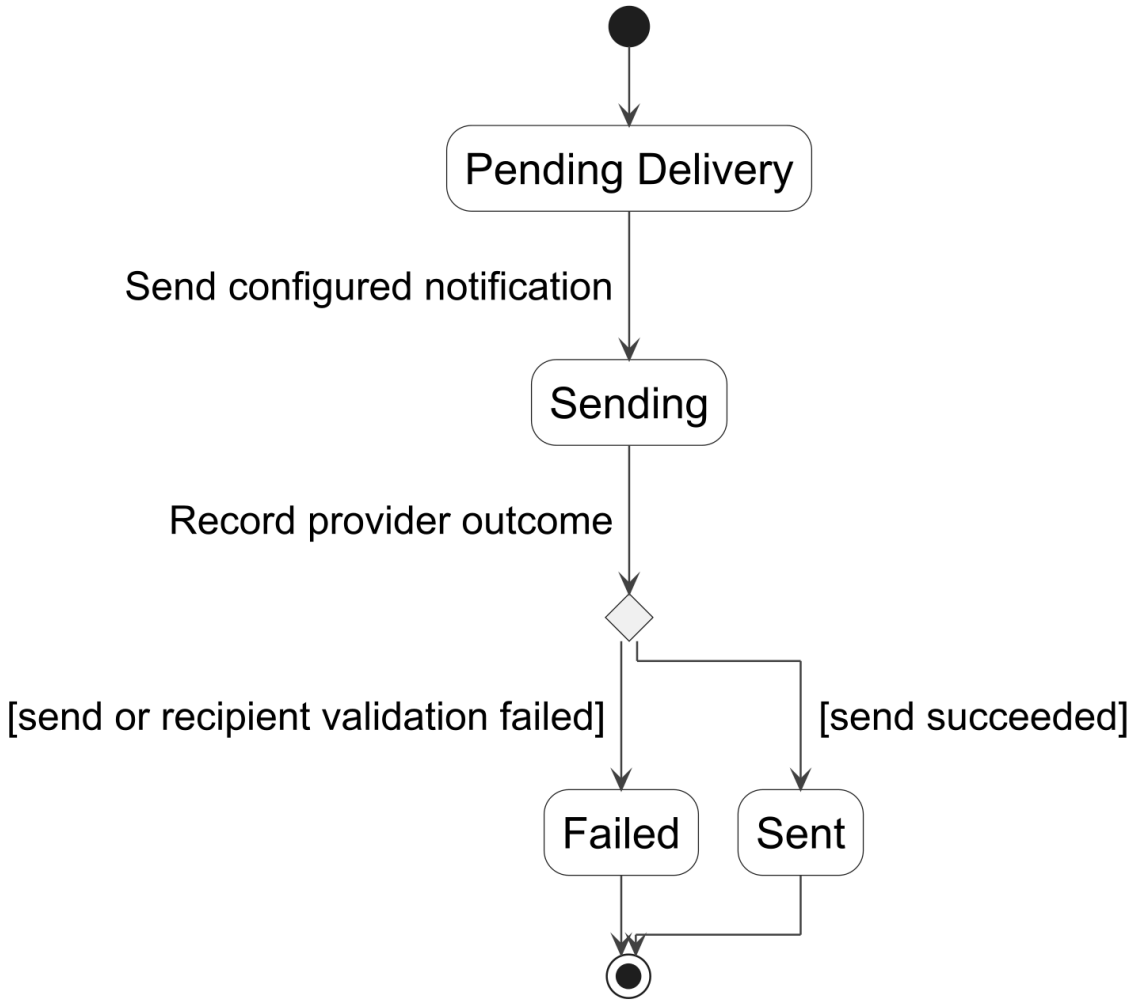


Figure 4.3.13.4.1. Email Delivery Attempt State Machine Diagram.

3.14 Platform Monitoring and Audit

3.14.1 Class Diagram

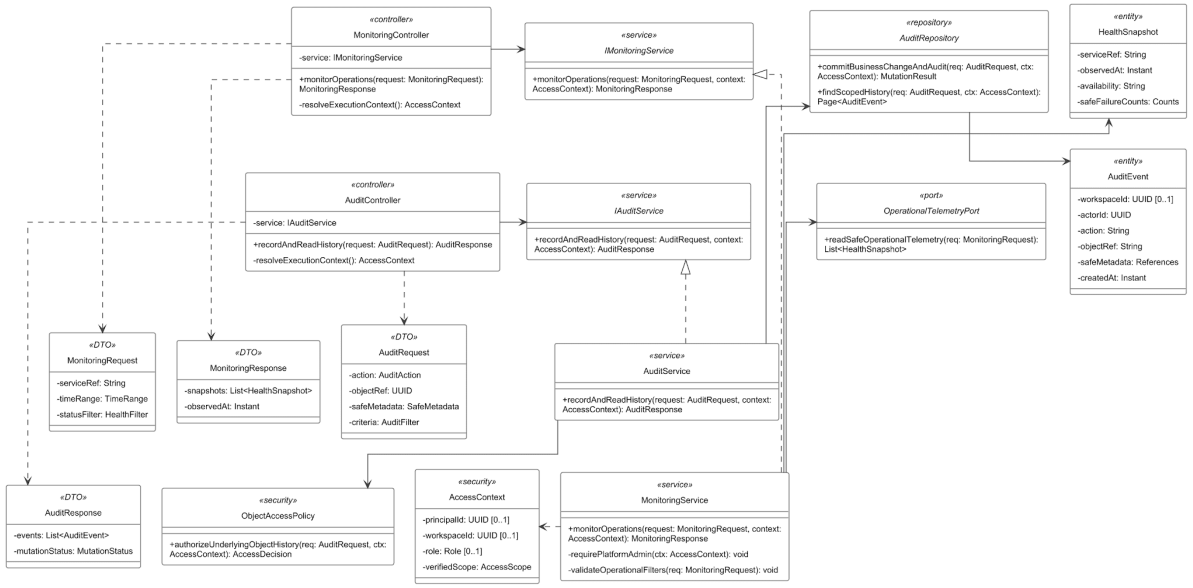


Figure 4.3.14.1.1. Platform Monitoring and Audit Class Diagram.

3.14.2 Sequence Diagrams

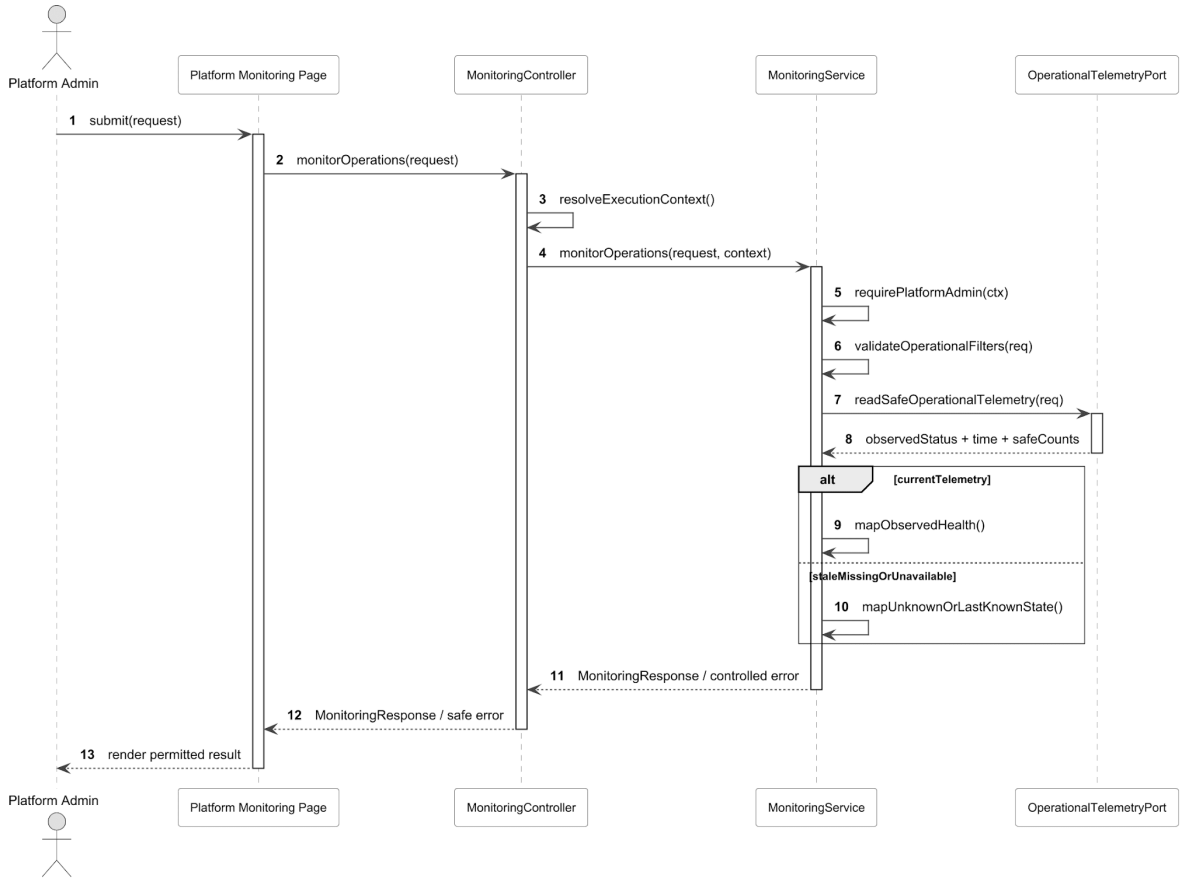


Figure 4.3.14.2.1. Monitor Platform Operations without Production Content Sequence Diagram.

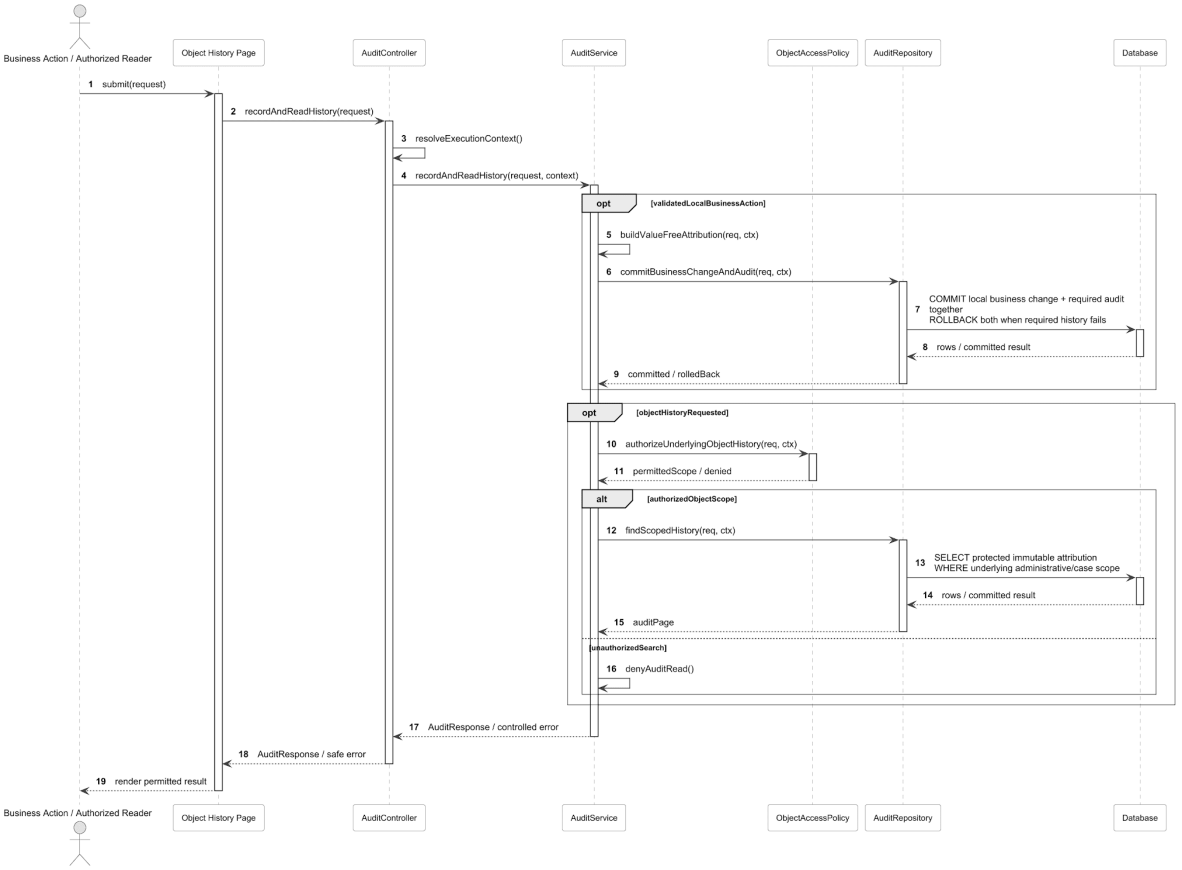


Figure 4.3.14.2.2. Commit Required Audit Attribution and Read Scoped Object History Sequence Diagram.